

AI-Enhanced Cloud-Edge-Terminal Collaborative Network: Survey, Applications, and Future Directions

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Abstract—The cloud-edge-terminal collaborative network (CETCN) is considered as a novel paradigm for emerging applications owing to its huge potential in providing low-latency and ultra-reliable computing services. However, achieving such benefits is very challenging due to the heterogeneous computing power of terminal devices and the complex environment faced by the CETCN. In particular, the high-dimensional and dynamic environment states cause difficulties for the CETCN to make efficient decisions in terms of task offloading, collaborative caching and mobility management. To this end, artificial intelligence (AI), especially deep reinforcement learning (DRL) has been proven effective in solving sequential decision-making problems in various domains, and offers a promising solution for the above-mentioned issues due to several reasons. Firstly, accurate modelling of the CETCN, which is difficult to obtain for real-world applications, is not required for the DRL-based method. Secondly, DRL can effectively respond to high-dimensional and dynamic tasks through iterative interactions with the environment. Thirdly, due to the complexity of tasks and the differences

in resource supply among different vendors, collaboration is required between different vendors to complete tasks. The multi-agent DRL (MADRL) methods are very effective in solving collaborative tasks, where the collaborative tasks can be jointly completed by cloud, edge and terminal devices which provided by different vendors. This survey provides a comprehensive overview regarding the applications of DRL and MADRL in the context of CETCN. The first part of this survey provides a depth overview of the key concepts of the CETCN and the mathematical underpinnings of both DRL and MADRL. Then, we highlight the applications of RL algorithms in solving various challenges within CETCN, such as task offloading, resource allocation, caching and mobility management. In addition, we extend discussion to explore how DRL and MADRL are making inroads into emerging CETCN scenarios like intelligent transportation system (ITS), the industrial Internet of Things (IIoT), smart health and digital agriculture. Furthermore, security considerations related to the application of DRL within CETCN are addressed, along with an overview of existing standards that pertain to edge intelligence. Finally, we list several lessons learned in this evolving field and outline future research opportunities and challenges that are critical for the development of the CETCN. We hope this survey will attract more researchers to investigate scalable and decentralized AI algorithms for the design of CETCN.

Index Terms—Cloud-edge-terminal collaborative network (CETCN), single-agent reinforcement learning (SARL), multi-agent reinforcement learning (MARL), network architecture, task offloading, resource allocation, collaborative caching, mobility management, applications, security, standards.

I. INTRODUCTION

IN RECENT years, in the fields of industry [1], transportation [2], Internet of Things (IoT) [3] and network security [4], intelligent cloud-edge-terminal collaborative network (CETCN) attracts more and more researchers' attention since it can complete complex networking operations under limited human participation. The advantages of intelligent cloud-edge-terminal collaborative networks not only can reduce network deployment costs [5], but also can improve network reliability and efficiency [6], [7]. Meanwhile, the rapid growth of the Internet has brought up new demands for the multi-dimensional performance indexes, such as extremely low latency requirements, more reasonable resource allocation, uninterrupted service demand, and reliable security guarantees [8] and traditional centralized cloud computing architecture can no longer fully meet these requirements [9], [10]. Therefore, the capabilities of edge computing nodes (base stations, wireless access point, etc.) and terminal

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devices (laptops, intelligent wearable devices, vehicles, etc.) should be utilized to support emerging applications [11].

Reasonable allocation of network resources is an issue that cannot be ignored to perform task offloading due to the inherent shortcomings of edge computing nodes and terminal devices, such as limited resources (computing and storage resources) and mobile characteristics of terminal devices. Traditionally, many computation-intensive applications are offloaded to cloud servers with powerful computing capabilities [12], [13], [14]. However, long distance transmission consumes a large amount of bandwidth resources of backhaul networks and results in a substantial increase in energy consumption and delay. For example, Google Docs can achieve effective and real-time online collaborative document sharing and editing, and such services can be executed in cloud servers, users only need to send their requirements to the cloud server and then computation resource will be allocated to process the tasks automatically. However, the long distance between users and cloud servers cannot be ignored when processing some real-time applications, such as remote surgery and smart transportation, which have also been applied in practical system, millisecond level delays could cause serious accidents. Fortunately, with the aid of edge computing nodes and terminal devices, some tasks can be offloaded to the idle edge servers or terminal devices to execute, and in this manner, the bandwidth of backbone network, energy consumption and delay can be reduced when processing real-time and delay sensitive applications. By offloading part of tasks to the network edge [15], [16], [17], the problem of resource occupancy and energy consumption can be successfully solved [18].

In CETCNs, resource allocation is another key issue concurrence with task offloading. Through reasonable resource allocation, edge devices can better handle various tasks. Many researchers have proposed various methods to solve multidimensional resource allocation problems in the cloud-edge collaborative environment, including methods based on convex optimization [19], [20], [21], game theory [22], [23], [24] and artificial intelligence [25], [26], [27]. Generally, resource allocation refers to the effective allocation of limited resources to tasks through some methods, so as to achieve users' objectives. Specifically, for the CETCN, the key terms of resource allocation can be divided as follows: 1) Resources: there are various available resources at the edge of the network, but these resources are often precious and limited, such as computing resources, storage resources and communication resources, 2) Tasks: Tasks often are generated from users, such as Ultra High Definition (UHD) broadcast and video processing. 3) Objectives: Different task have different objectives during task processing. For example, UHD live broadcast often requires extremely low transmission delay, and video processing often requires abundant computing resources. The objectives are often refers to various performance indexes. However, tens of billions of heterogeneous devices may generate tens of billions of heterogeneous tasks, it is difficult to allocate reasonable resources to these tasks through traditional methods due to its NP-hard property.

Caching content and service at edge devices can reduce the delay and increase users' Quality of Service (QoS). In addition

to offloading the generated task to the edge device, users can also retrieve popular documents from edge devices [28], [29]. Extensive research has indicated that a large part of traffic on the Internet came from the repeated viewing and downloading of popular content [30], [31]. For content providers, such as YouTube and TikTok, they need to cache the hot content in edge devices to maximize their revenue. For hardware device providers, such as Google Cloud Platform (GCP), they need to attract content providers to rent or buy hardware equipments by setting a reasonable price of their equipments. For users, they would like to be able to get the content they need at minimal cost. On the one hand, duplicate content acquisition puts huge traffic pressure on the network. On the other hand, various applications and service caching have additional computation requirements and also generate delay and energy consumption. When users request application service or content from the cloud servers, users may suffer unbearable delays due to the long distance transmission, and operators will consume more unnecessary bandwidth resources of the core network and the content providers will spend more money on purchasing cloud servers. At the same time, if users request content or service from the edge layer, the limited caching capacity should be considered when designing appropriate caching strategy. In this context, content caching and service caching based on the CETCN can reduce latency, save cost and take advantage of the rich bandwidth resources at the edge. Due to the heterogeneously distribution and the limitation of edge resources, the content and the service placement problem should be reviewed comprehensively in the CETCN.

Mobility management is another research topic that should not be ignored in the CETCN. User's mobility in real-world scenarios makes task offloading, resource allocation, and caching decision problems more difficult [32]. The obstacle mainly comes from the uncertainty and dynamics of the channel due to the user's mobility [33], [34] which is often ignored by researchers [35], [36], [37]. Therefore, in order to provide users for seamless access services, efficient mobility management solutions need to be proposed to improve the users' QoS [38]. In particular, mobility management has the following advantages:

- **High energy efficiency:** The mobility of the devices allows the distance between the user and the device to be adjusted in order to consume less transmission power and improve transmission rate, so that the user can optimally upload the task to the corresponding mobile devices.
- **Low latency:** Shortening the distance between mobile devices and users can also reduce the processing delay of user-generated tasks. Therefore, users can offload tasks with higher transmission rates to mobile devices to reduce the time for the task to reach the mobile device.
- **Low interference:** Another problem that fixed mobile devices may face is that they cannot control the interference faced by the signal in the process of transmission. However, this problem can be solved by using mobile devices, which can reduce interference by moving the location.

The above analysis indicates that task offloading, resource allocation, caching and mobility management issues play a vital role in the CETCN. However, there are a few drawbacks when we adopt traditional methods or technologies to solve these issues in the CETCN. For example, traditional methods require an accurate environment model and optimize the problem in one shot, but accurate environment model is difficult for the complex system to obtain. Fortunately, as a broad frontier subject, DRL and MADRL plays an increasingly important role in academia when solving sequential decision-making problems and we summarize the advantages of DRL and MADRL as follows:

- Traditional methods require accurate models of the problem, but accurate models for real-world problems are often difficult to obtain. On the contrary, DRL methods often make little assumptions about the environment, and formulate the problem as an Markov decision process (MDP). Due to the little assumption of the environment, the accurate model is not required for DRL methods.
- The problems of task offloading, resource allocation, and caching based on DRL focus on long-term performance optimization, and in this manner, the system can achieve stable performance compared with traditional methods, which are based on one-shot optimization.
- DRL can effectively responds to high-dimensional and dynamic environment states through iterative interactions with the environment, thus its trained agent can obtain the near-optimal policy intelligently.
- MADRL can provide centralized training and decentralized execution framework for the network vendors to make optimal decisions. Thus, the network vendors only require local observation information to make decisions without exchanging information with other vendors. In this manner, the signaling overhead can be significantly reduced.

In summary, integration of the CETCN with existing DRL approaches lays the foundation for creating a new network paradigm. Since this new paradigm introduces DRL and MADRL in all aspects of next-generation mobile communications, it is called AI-enhanced CETCN in this paper. Table I is the abbreviations used in this survey.

A. Existing Surveys and Tutorials on Cloud-Edge-Terminal Collaboration

In the following, we will investigate the relevant research topics that have been summarized in cloud-edge-terminal collaboration network. Specifically, it includes network architecture, computation offloading and scheduling issue, caching issue, mobility management issue, applications and security issues which related to DRL in the CETCN system. It is worth noting that the term “AI-enabled” mentioned in Table II refers to whether the surveys utilize DRL or MADRL to make decisions for computation offloading and scheduling issue, caching issue and mobility management issue.

Plenty of literature investigated the research findings and related challenges of cloud and edge computing, as shown in Table II Mach and Becvar [39] provided an overview of

TABLE I
LIST OF ABBREVIATIONS USED IN THIS SURVEY

CETCN	Cloud-Edge-Terminal Collaborative Network
ITS	Intelligent transportation System
IIoT	Industrial IoT
DRL	Deep Reinforcement Learning
FL	Federated Learning
SDN	Software Defined Network
MAR	Mobile Augmented Reality
ESTI	European Telecommunications Standards Institute
IaaS	Infrastructure as a Service
PaaS	Platform as a Service
SaaS	Software as a Service
MEC	Multi-access Edge Computing
ML	Machine Learning
MDP	Markov Decision Process
POMDP	Partial Observable Markov Decision Process
MARL	Multi-agent Reinforcement Learning
NE	Nash Equilibrium
Dec-POMDP	Decentralized Partial Observable Markov Decision Process
GNN	Graph Neural Network
DQN	Deep Q-Network
DDQN	Double DQN
AC	Actor-Critic
DDPG	Deep Deterministic Policy Gradient
MADDPG	Multi-agent Deep Deterministic Policy Gradient
COMA	Counterfactual Multi-Agent Policy Gradients
MF-Q	Mean-Field Q-Learning
MF-DDPG	Mean-Field Deep Deterministic Policy Gradient
HIF	High Impedance Fault
UAV	Unmanned Aerial Vehicle
RSU	Road Side Unit
AR	Argument Reality
VR	Virtual Reality
D2D	Device-to-Device
AI	Artificial Intelligence
IoT	Internet of Things
5G	5-Generation
CaaS	Cache as a Service
V2I	Vehicle to Infrastructure
V2V	Vehicle to Vehicle
IoCVs	Internet of Connected Vehicles
MNO	Mobile Network Operator
MINLP	Mixed Integer Non-Linear Programming
NOMA	Non-Orthogonal Multiple Access
NFV	Network Function Virtualization
SFC	Service Function Chain
RAN	Radio Access Network
eNB	Macro Base Station
RNC	Radio Network Controller
DNS	Domain Name System
ML	Machine Learning
DL	Deep Learning
DNN	Deep Neural Network
VANET	Vehicle Ad-hoc Network
QoE	Quality of Experience
QoS	Quality of Service
Massive MIMO	Massive Multiple-Input Multiple-Output
RIS	Reflected Intelligent Surface
MARL	Multi-Agent Reinforcement Learning

the computation offloading of mobile edge computing (MEC). Ni et al. [40] comprehensively reviewed the architecture and characteristics of IoT applications in fog computing.

TABLE II
EXISTING SURVEYS AND TUTORIALS ON CLOUD-EDGE NETWORKS

survey	network architecture	computation offloading & scheduling	caching	mobility management	applications	security	AI-enabled
2017 [39]	✓	✓		✓			
2018 [40]						✓	
2019 [41]					✓		
2020 [42]	✓						
2020 [43]					✓	✓	✓
2020 [44]	✓	✓	✓			✓	
2020 [45]		✓					✓
2020 [46]		✓	✓		✓		✓
2021 [47]	✓						
2021 [48]	✓	✓		✓	✓	✓	
2021 [49]		✓					✓
2021 [50]					✓	✓	✓
2022 [51]		✓	✓		✓	✓	✓
This survey	✓	✓	✓	✓	✓	✓	✓

Moura and Hutchison [41] reviewed the literature on applying theoretical game theory, including classical and evolutionary game theory, to wireless networks, emphasizing the upcoming use cases of MEC. Wang et al. [42] provided a systematic survey of the architecture design of the connected vehicles based on edge computing and provided a new computation and communication-assisted architecture based on edge computing. In [43], the authors analyzed the application scenarios of edge computing and practical implementation methods of deep learning, namely DL training and reasoning in the custom edge computing architecture, and discussed the challenges and future trends of more general and fine-grained intelligence. Qiu et al. [44] summarized the scenarios of edge computing in IIoT, then, they proposed a future architecture of the IIoT from the perspective of edge computing, and finally, they analyzed the technological progress of edge computing in communication, load balancing, shared security and application scenarios. In [45], machine learning applications in MEC systems were reviewed, and the latest advances in this field prospected. In [46], the authors propose a novelty concept-autonomous IoT (AIoT), which combines the IoT and Autonomous Control System (ACS). Moreover, apart from basic DRL approaches, such as DQN, DDPG, A3C, the authors also introduced the multi-agent DRL and its applications in network layer, application layer and perception layer in ACS. Sarmiento et al. [47] investigated and analyzed the characteristics and limitations of existing technologies which provide connectivity between sites in the domain of software-defined networking (SDN). They also introduced Kubernetes and discussed some research directions of SDN applications in the distributed cloud edge infrastructure management field. Siriwardhana et al. [48] provided an analysis of current and future mobile augmented reality (MAR) systems from the perspective of cloud, edge, terminal, and hybrid architecture options. The requirements and existing issues of the MAR system were discussed, such as communication, mobility and energy management, service offloading, and privacy protection. In [49], the authors first introduced a

unified resource scheduling model and then summarized the current research progress of resource allocation based on limited resources at the edge. The authors in [50] provided a brief review of the advanced DRL algorithms including actor-critic architecture based DRL algorithms and multi-agent DRL algorithms. Furthermore, they analyzed the cooperation and confliction of multi-agent system. Finally, they reviewed the literature on applying DRL and MADRL algorithms to address issues in IoT scenarios. In [51], the authors reviewed the applications of MARL in emerging networks. On the one hand, they introduced the cooperation and confliction relationships in the future network, and specifically, they classified the MARL into fully cooperative setting, fully competitively setting and mixed setting. On the other hand, the authors utilized the MADRL approaches to address the problems in the emerging network, such as network access and control problem, offloading and resource allocation problem and so on.

In this survey, we provide a comprehensive review of the convergence of the CETCN and AI. Firstly, edge computing is a supplement rather than a substitute for cloud computing, which cannot be discussed separately and independently. Secondly, AI has gained more and more attention in academia in solving decision-making problems. Thirdly, the further enhancement of the computing power of the intelligent terminal enables users to handle decision-making tasks on the terminal devices directly. Therefore, based on the above three reasons, we believe it is necessary to make a full and comprehensive survey of AI-enabled CETCN. In addition, based on the review of existing literature, we summarize some experiences and lessons of the CETCN and point out the application prospects of AI-enabled CETCN.

B. Our Contributions

This paper focuses on a specific type of AI technology, i.e., single-agent DRL based on MDP and partial observable MDP (POMDP) and explains that they are insufficient to model

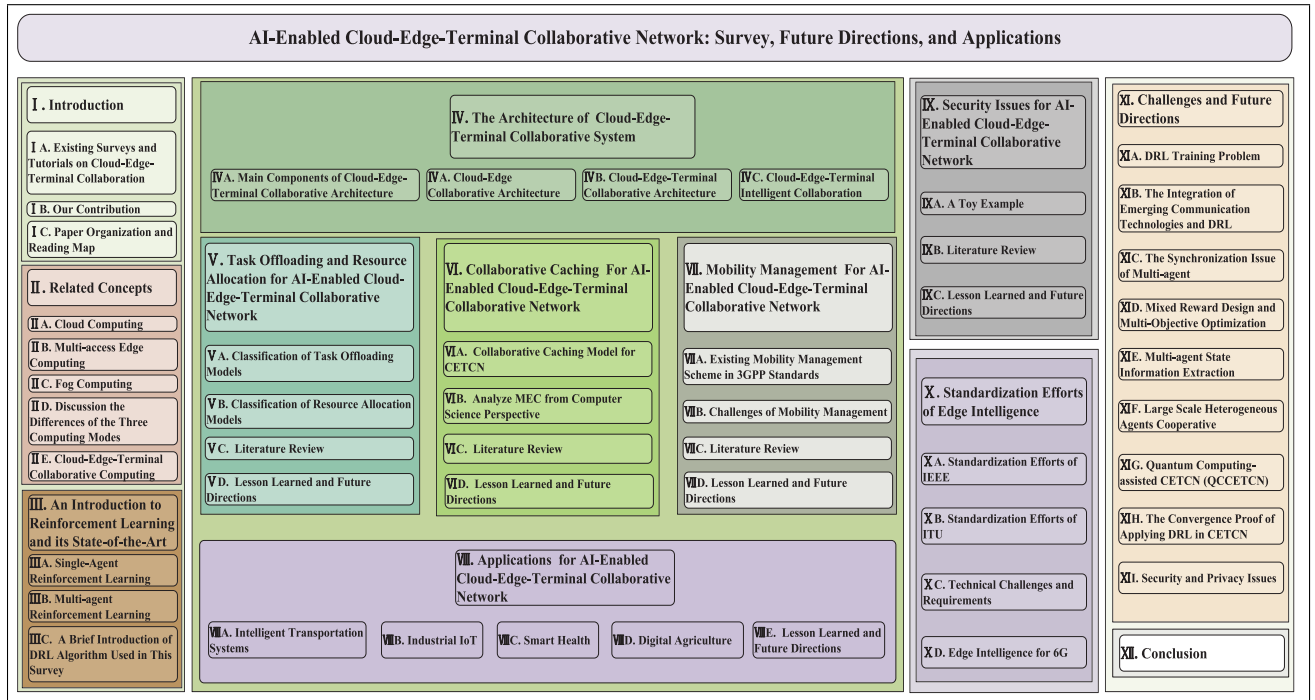


Fig. 1. Reading Map of the Survey.

the sequential decision-making problems in the CETCN when involving multiple vendors, i.e., terminal devices, edge devices and cloud. Furthermore, we present MADRL based on Markov Game (MG) and Dec-POMDP to guide different vendors to make optimal decisions in the CETCN. In the following, we present our major contributions of this survey.

- The main issues faced by the CETCN, including task offloading, resource allocation, caching and mobility management issues, these issues that should be addressed to guarantee the high quality of services, such as minimizing the delay and energy consumption. It is worth noting that this is not only helpful for those who focus on AI, but also for those who are interested in solving the CETCN problems.
- Additionally, we present a brief explanation of AI algorithms which is the mainstream in the current industry and academia to enable future researchers to use this paradigm as an important tool for future studies of the CETCN.
- We comprehensively review the recent works and use cases in the literature that utilize AI to improve the network performance of the CETCN.
- We point out the further development directions for the convergence of deep reinforcement learning and CETCN.

C. Paper Organization and Reading Map

In the following, we give a brief introduction to the structure of this article. The first part of this paper mainly includes Section I which is the introduction, Section II which introduces the related concepts of the CETCN system, such as cloud computing, fog computing and MEC, and Section III introduces the related concepts of reinforcement learning, several

state-of-the-art reinforcement learning algorithms and their links with the CETCN. The second part adopts a deduction and summary structure. In Section IV, it first introduces various cloud edge collaboration architectures, and then gives a comprehensive overview of the main issues in the cloud edge collaboration architecture, mainly including task offloading, resource allocation, collaborative caching and mobility management in Section V, Sections VI and VII. Finally, we listed two scenarios of cloud edge collaborative architecture application, namely intelligent transportation system and industrial Internet of Things in Section VIII. The last part lists the challenges and possible development directions of applying AI in the CETCN in Section XI, and finally summarizes the full text in Section XII. The reading map is shown in Fig. 1.

II. RELATED CONCEPTS

In this section, we introduce and compare several key concepts in the CETCN. The following subsections introduce the main concepts in cloud computing, fog computing and edge computing in Sections II-A–II-C, and then we integrate these concepts in Fig. 2. We also discuss their similarities and differences in Section II-D with a Venn diagram in Fig. 3. Finally, we introduce the cloud-edge-terminal collaborative computing, which can take full advantages of the above mentioned computing paradigms in Section II-E.

A. Cloud Computing

Cloud computing evolved from traditional distributed computing and has become a relatively popular Internet infrastructure service. Network virtualization [52], [53] is one of the essential supporting technologies of cloud computing which is based on virtual machine (VM). It connects

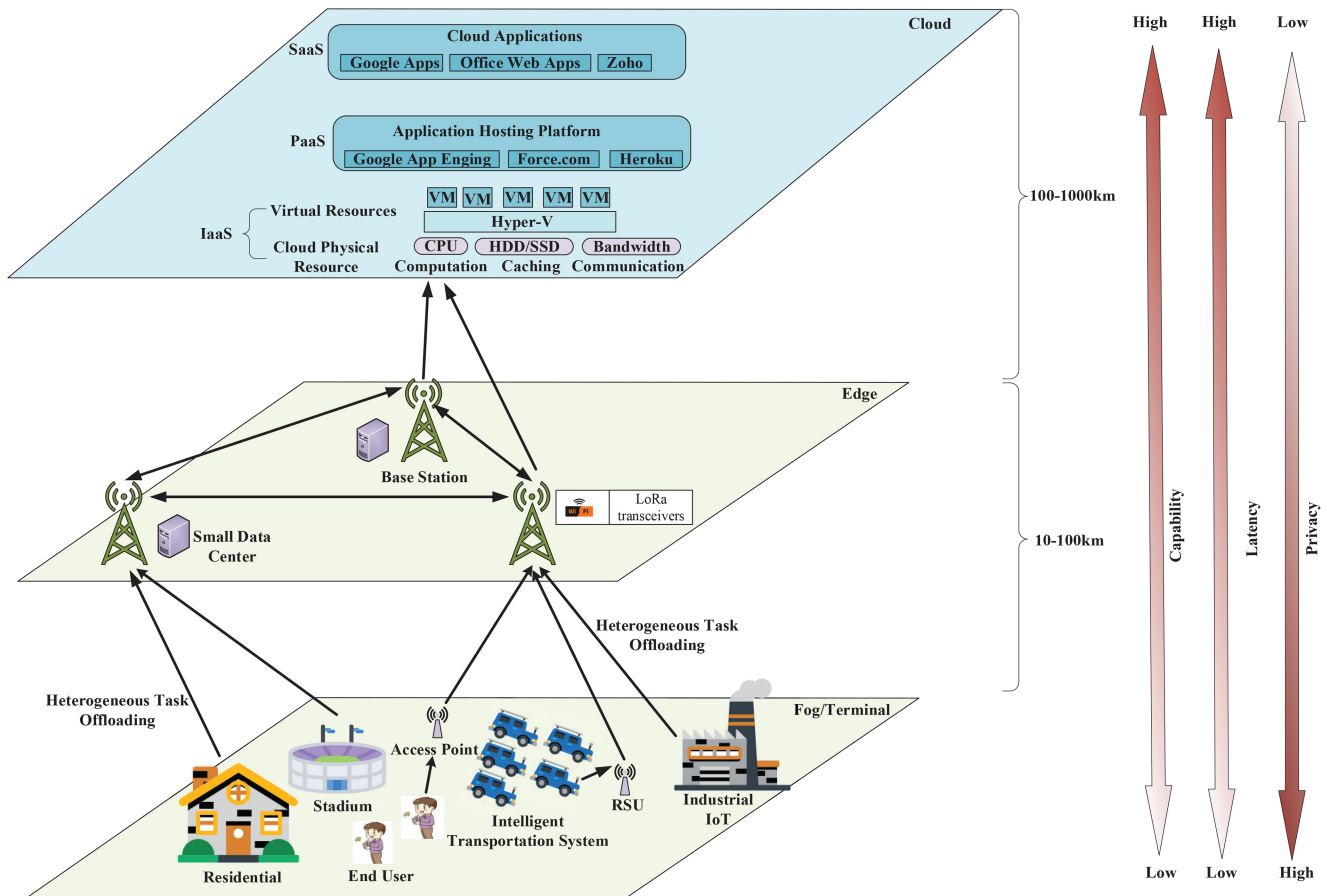


Fig. 2. The main components of cloud-edge-fog/terminal system.

and combines various resources as a whole at the bottom in a particular way, and provides them for individuals and businesses on demand [54], [55], [56]. It is universally acknowledged that the service type of cloud computing can be divided into Infrastructure-as-a-Service (IaaS) [57], [58], [59], Platform-as-a-Service (PaaS) [60], [61], [62], and Software-as-a-Service (SaaS) [63], [64], [65] according to the different service levels. However, with the emergence of cloud-edge-device collaborative networks, cloud computing has been unable to meet the requirements of the emerging applications. In the following, we summarize the problems and challenges faced by cloud computing.

1) *Challenges of Cloud Computing*: Cloud computing has revolutionized the way organizations and individuals access and manage computing resources. However, it also brings considerable challenges [66], [67]. Therefore, we list several key challenges of cloud computing in the following:

- **Data Availability and Reliability** [68], [69]: Cloud service outages can result in temporary unavailability of services and data. Operators need to carefully consider service-level agreements (SLAs) to ensure adequate uptime and reliability.
- **Data Transfer and Latency** [70]: Uploading and downloading large volumes of data from the cloud servers can be time-consuming and bandwidth-intensive. High latency can impact the performance of real-time applications. Specifically, on the one hand, for real-time

applications that require immediate response, connecting devices to remote cloud servers can bring intolerable delays. On the other hand, for mobile users in remote areas, relying on cloud services may result in the inability to access certain services.

- **Cost Management** [71], [72]: Cloud services typically involve pay-as-you-go pricing models. While this model can offer cost savings for some users, it can also lead to unexpected costs if not carefully monitored and managed. For example, using service level agreements (SLAs) that exceed the requirements can result in premium fees. In addition, complex pricing models can also lead to additional costs.
- **Performance and Scalability** [73]: Although scalability allows programs to accommodate to varying workloads by allocating or deallocating resources, the scalability of the cloud computing can be affected by various factors, including server load, network congestion, the existence of multi-tenant environment and cost.
- **Security and Privacy issue** [74]: Data security and privacy are major concerns in cloud computing. Storing sensitive data and applications on third-party servers raise questions about data breaches. In addition, due to the large amount of information stored on cloud devices through the network, there are many vulnerabilities that can be exploited by malicious actors.

Serverless computing, also known as Function-as-a-Service (FaaS), is a popular cloud computing execution model in which the cloud provider can dynamically manage server resources [75]. In a serverless architecture, although developers write and deploy individual functions or code snippets, and the cloud provider takes care of automatically scaling, managing, and maintaining the underlying infrastructure to execute those functions, serverless computing also faces the following challenges [76], [77]:

- **Cold Start Latency:** When a serverless function is invoked after being idle for a period, there can be a delay known as a “cold start”. This occurs because the cloud provider needs to allocate and initialize the resources required to execute the function. Cold start latency can impact real-time and highly responsive applications.
- **Resource Limitations:** Serverless platforms impose resource limitations on individual functions, such as maximum execution time and memory allocation. This can be challenging for tasks that require extensive computation or have specific resource requirements.
- **Vendor Lock-In:** Adopting a serverless platform often means relying heavily on the features and APIs provided by the specific cloud provider. This can lead to vendor lock-in, making it difficult to switch to a different provider without significant code modifications.
- **Stateless Execution:** Serverless functions are typically stateless, which means they do not retain data between invocations. Handling state and maintaining session information across function calls can be challenging and may require additional mechanisms, such as external databases or caching solutions.
- **Testing and Deployment:** Testing serverless applications can be more complex due to the event-driven nature and potential interactions between different functions. Additionally, deploying and managing a set of interrelated functions can require careful coordination.
- **Scalability and Performance:** Although serverless platforms offer automatic scaling, they may not handle sudden or extreme spikes in demand as effectively as anticipated, and thus resulting in a decrease in service quality due to insufficient instance resources.

2) *Challenges of Emerging Applications to the Traditional Cloud Computing:* With the increasing number of applications and their demands for wider bandwidth and lower latency, these emerging applications and demands also bring great challenges to cloud computing which can be summarized as follows:

- **Efficient distribution of content:** Media content distribution, occupies the vast majority of the network resources. Due to the repeated request of the popular media content, the centralized distribution mode causes a considerable waste of bandwidth. Also, it brings difficulties in supporting larger-scale users and large-scale data distribution services.
- **Task offloading:** Since smart mobile terminals are limited in computing and storage capabilities, how to exploit the resources of cloud computing or other adjacent devices to assist in completing tasks is becoming

increasingly important. Task offloading is an excellent way to overcome the resource limitation problem of mobile terminals. However, since the user experience of these applications relies heavily on latency, these applications are not suitable for offloading to remote cloud computing centers for execution.

- **Mobile big data analysis:** A large number of applications of smart mobile devices generate massive data, and the analysis and processing of these data can accurately locate the enterprise market and product applications. Traditional batch or streaming data processing that relies on data centers will bring high bandwidth usage and latency.
- **Intelligent Transportation/Government/Medical applications:** If these intelligent field applications, such as traffic information release, traffic scheduling, accident and emergency handling, are all centrally processed in remote data centers, in that case, it is likely to miss the best processing opportunity due to computing or transmission delays.

It can be seen from the above analysis that centralized cloud computing is challenging to satisfy the emerging applications and services due to the heterogeneity and intelligence of terminal equipment, especially for the IoT and various real-time interactive applications. Therefore, it has become urgent to explore new network computing models to make up for the shortcomings of cloud computing.

B. Multi-Access Edge Computing

The centralized processing mode of cloud computing has become the obstacle of the evolution of ever-changing ubiquitous terminals [78], [79], ubiquitous network technologies, and new networked applications and services. Therefore, the industry and academia have begun exploring and researching the other computing models [80]. Since Cisco proposed fog computing in 2011, the concept of similar computing models in the post-cloud computing era has been proposed successively in different noun forms, such as edge computing [81].

Specifically, the data centers were initially deployed centrally far away from terminals and have been extended to the network edge [82]. In specific technical implementation, the network edge can be not only edge routers, communication base stations, servers, etc., but also specially deployed servers, small data centers, or even other terminal devices with idle resources. In this manner, the impact of delays caused by long-distance networks and heterogeneous wireless networks can be solved to a certain extent. These new network computing models are a natural continuation of cloud computing from a centralized processing model to a decentralized and small centralized model [83].

The concept of mobile edge computing (MEC) was first proposed by ETSI in September 2014 [84]. It is a near-end cloud computing model proposed by communication equipment manufacturers, hence its purpose focuses on the support for communication and applications and does not focus on too much technical content.

Multi-access Edge Computing was evolved and renamed from Mobile Edge Computing in September 2016. MEC aims to overcome the challenges faced by cloud computing in cellular scenarios, especially in emerging 5G/6G application scenarios [85], [86], [87]. With the development of access technology, access paths are not limited to mobile cellular networks but also include Wi-Fi networks and wired networks. Therefore, ETSI renamed its concept to multi-access edge computing based on maintaining the MEC abbreviation. In the following, when we mention MEC, we do not distinguish between mobile edge computing and multi-access edge computing unless otherwise stated.

It is possible to deploy MEC servers at the LTE Macro Base Station (eNB) [88], [89], 3G Radio Network Controller (RNC) sites [90], [91], [92]. Since cellular aggregation sites can be installed both inside enterprises (indoors) [93], and in specific public places (outdoors) [94] to cover specific scenarios, they can provide local fast services to users directly from base station clusters. The MEC server platform provided by ETSI ISG MEC includes a MEC hosting infrastructure [95] and a MEC application platform [96], [97].

Of course, the server platform provided by MEC can support and provide a variety of mobile communication value-added services and applications, such as augmented reality content transmission [98], video analysis [99], content optimization [100] and content caching [101], etc.

In recent years, because MEC is proximity to terminal devices, combining the decentralized and centralized computing models and supporting end collaboration, MEC shows great development opportunities in many application areas [102] such as the IoT, 5G, smart grid, virtual/augmented reality, and drones. Compared with cloud computing, MEC can achieve lower latency, save energy and enhance security for users. The advantages are described briefly as follows.

Low latency: Service delay consists of propagation delay, computation delay, and communication delay, which depends on propagation distance, computing power, and data rate, respectively. First of all, for the MEC network, the information propagation distance is usually tens of meters, generally no more than 1 kilometer. In contrast, cloud computing needs to transmit information from end users to the core network, with a distance ranging from tens of kilometers to thousands of kilometers. In this manner, the propagation delay of the MEC is shorter than cloud computing. Second, the transmission of user information from the terminal device to the cloud center needs to go through the wireless access network, the backhaul network and the Internet which leads to increased delays. Last but not least, the computing resources on cloud servers are richer than those on edge devices. But at the same time, computing resources on the cloud will be shared by more users. These two factors narrow the gap between cloud computing and MEC in computing latency. Experiments demonstrate that the delay caused by cloud computing is in the range 30-100ms which is intolerable for latency sensitive services, such as online games which require processing latency within 1ms [103]. In contrast, MEC is expected to be a promising technology for implementing delay-sensitive applications due to its shorter transmission distance and simpler protocol.

Energy saving: Because IoT devices are usually powered by batteries, IoT devices cannot cooperate to complete large and complex tasks [104]. Powering massive IoT devices is still a critical problem in designing IoT systems, as it is impossible and impractical to replace the batteries of the IoT devices frequently. Therefore, by offloading part of the tasks to MEC servers, the battery life of IoT devices can be prolonged. Some energy saving experiments have been carried out to demonstrate the effectiveness of computing offloading, such as in [105], [106].

Enhanced privacy and security: The traditional cloud computing system is consisted of the remote public data centers, which are highly vulnerable to intruders. In addition, since the management permissions of users' data are separated in cloud computing, it will cause the problem of privacy data leakage [107] and this issue can be handled by using MEC servers. On the one hand, distributed deployment reduces the centralization of valuable information. On the other hand, MEC servers can be private cloudlets, which will reduce the risk of privacy leakage. For example, the enterprise manages data by themselves through deploying MEC servers, thereby the data security can be ensured [108].

C. Fog Computing

Fog Computing is introduced by Cisco [109] to address the problems faced by cloud computing. It is a novel computing paradigm that can sink the various resources to the network edge. However, unlike MEC which mentioned above, fog is connected to the cloud closely [110], in other words, it cannot running in an independent way. With fog computing, the processing of some applications can take place at the network edge and thus can provide low latency services to the users. It also provides densely distributed fog nodes for collecting data generated by terminal devices. These fog nodes include access points and routers which are closer to the users.

The fog computing system is composed of three parts, including cloud servers, fog servers and IoT/end users [111]. The fog servers consist of one or more fog regions which can be owned and controlled by different providers. Each fog region is consisted of many fog nodes, including gateways, access points, etc. The IoT/end user is consisted by two entities, one is user devices, the other is IoT devices. It is noteworthy that one of these two entities can be absented in the real-world system.

Fog computing has several critical characteristics [112] that are summarized as follows. (1) Heterogeneity, fog computing nodes can include access points (APs), edge routers, RSUs, and even mobile terminals (such as smart phones). (2) Decentralized management, different from cloud computing, fog computing nodes are deployed distributed at the network edge, and in this manner, the fog nodes can manage its resources in a distributed way. (3) Support for interacting with the cloud. The delay-sensitive applications can be executed at the fog nodes, while computation-intensive applications and delay-tolerant applications can be executed at the cloud servers due to its powerful computing capability.

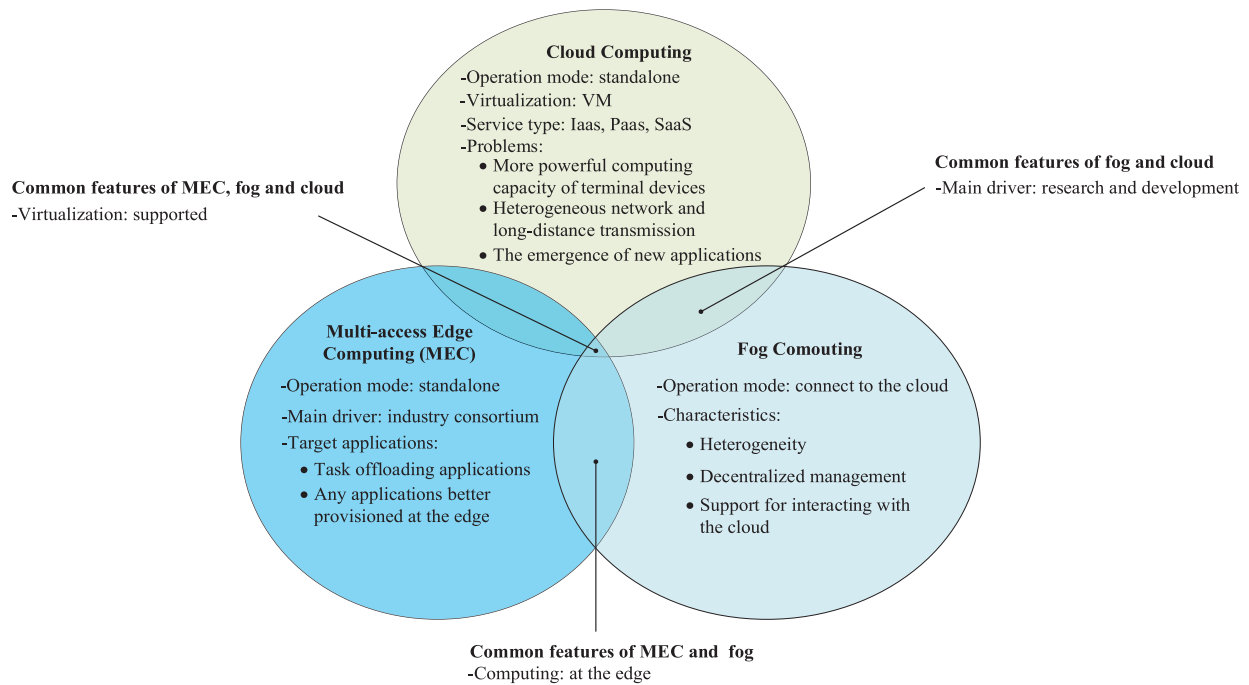


Fig. 3. The relationship between MEC, Fog Computing and Cloud Computing.

D. Discussion the Differences of the Three Computing Modes

This section focuses on the similarities and differences among cloud computing, fog computing, and edge computing. From a geographical perspective, these three computing modes range from core networks to edge networks. Therefore, it is necessary to point out several differences of these three computing modes.

The first difference is that cloud computing and fog computing are proposed by Google and Cisco, respectively. For example, the OpenFog consortium, it aims to accelerate the deployment of Fog technology by developing core technologies such as open architecture, distributed computing, networking and storage. Only MEC is proposed by the industrial consortium and the initial purpose of MEC focuses on business and interests. It aims to combine information industries such as communication and IT, and further increase IT and cloud computing service capabilities in the Radio Access Network (RAN). The second difference is that MEC operates in a stand-alone mode and fog computing operates in a stand-alone mode or connects to cloud.

The third difference is that the applications targeted by these three computing modes are different. Cloud computing is responsible for big data analysis of long-period data, and can operate in areas such as periodic maintenance and business decision-making. The principle of fog computing is the same as cloud computing, which uploads data to remote centers for analysis, storage and processing. However, fog computing sets up many scattered central nodes, namely the so-called “fog nodes” to process, which can make the calculation process faster and more efficient to obtain the calculation results. Edge computing focuses on the analysis of real-time and short-term data to better support the timely processing and execution

of local businesses. Fig. 3 provides a summary of MEC, fog computing and cloud computing.

E. Cloud-Edge-Terminal Collaborative Computing

With the further enhancement of the computation and caching capabilities of terminal devices, the use of terminal devices for task offloading, resource allocation, cooperative caching, and mobility management decisions can further reduce energy consumption and transmission delay in the CETCN. In fact, there is currently no internationally recognized definition for CETCN. In order to solve this problem, the industrial and academic communities have made many efforts [113], [114], [115], [116]. We briefly depict the CETCN in Fig. 2 and give a definition of CETCN in the following. *In this paper, we define the CETCN as a distributed computing architecture that consists of three layer, including the terminal layer, the edge layer and the cloud layer. This architecture is designed to optimize the computing, storage, and communication resources of terminal devices, edge devices and cloud devices to cater to the demands of various emerging applications and services.* Furthermore, In a broader sense, the collaborative computing can be the terminal-edge collaboration, edge-edge collaboration, edge-cloud collaboration, and terminal-edge-cloud collaboration. In this collaborative system, the terminal layer includes a large amount of terminal devices, such as sensors, vehicles, AR/VR devices, intelligent wearable devices with very limited computing capability, and wireless access points used for communication between terminal layer and edge layer. The edge layer consists of not only edge devices, such as edge servers used for processing and storing data, but also consists of base stations, WiFi and LoRa used for managing network traffic and acting as intermediaries between the terminal layer

and the cloud layer. These edge devices are widely deployed at urban centers, factory area and densely populated areas. The cloud layer is composed of various cloud servers, the number of deployed cloud servers depends on population density, business landscape, data-intensive applications and so on. Moreover, the authors in [113], [114], [115], [116] proposed a CETCN framework, which is different from the one proposed in this work. The main difference is that we have added some emerging scenarios with potential applications for CETCN at the terminal layer, such as industrial IoT and smart transportation. At the edge layer, we have also added other radio access methods such as WiFi and LoRa. In the cloud layer, we have provided a detailed description of the components of the cloud server which does not mentioned in [113], [114], [115], [116].

Moreover, Fig. 2 also shows a comprehensive picture of the data life cycle management in future wireless networks [117], including data generation and analysing at the terminal layer, cross layer transmission of data and data processing at edge/cloud layer. By integrating AI into the data life cycle of wireless networks, it becomes possible to enhance efficiency, improve performance, and address challenges associated with massive data volumes and complex network operations. Therefore, in the following, we summary several key aspects of wireless AI for the data life cycle management.

- **Data Collection and Preprocessing:** AI techniques, such as machine learning and deep learning algorithms, can be used to preprocess the data and clean the redundant data collected from wireless network devices and sensors [118]. This process can ensure that the data used for analysis and decision-making is accurate and reliable [119].
- **Data Transmission and Communication:** AI can optimize the wireless communication process by managing data traffic, allocating resources, and predicting network congestion [120], [121]. For example, AI-driven adaptive modulation and coding schemes can dynamically adjust transmission parameters to improve data transfer rates based on the channel conditions [122].
- **Data Processing and Analytics:** AI algorithms can perform real-time data analysis [123], pattern recognition [124], and abnormal detection [125]. This enables intelligent decision-making, predictive maintenance, and data-driven optimizations for various applications, such as network management and IoT applications.
- **Data Storage and Management:** AI can be employed to optimize data storage and management strategies. For instance, AI-driven data de-duplication and compression techniques can reduce storage requirements and enhance data retrieval speed [126].
- **Data Utilization and Application:** AI can enhance the utilization of data collected from wireless networks by enabling advanced applications like predictive maintenance [127], intelligent traffic management [128], [129], and personalized services.
- **Network Resource Management:** AI-based techniques can be used for dynamic spectrum allocation [130], interference mitigation [131], [132], and energy-efficient management of network resources [133].

- **Security and Privacy:** AI can be employed to detect and mitigate security threats in wireless networks. Additionally, AI can help enforce privacy policies and ensure compliance with data protection regulations [134].

However, in the CETCN system, there are still several issues need to be addressed. We summarize the issues as follows:

- **Heterogenous capacity of edge servers and terminal devices:** The further development of wireless network technology makes the emerging network services generate extra energy consumption and delay problems. At the same time, due to the size limitations of edge and terminal devices and the differences in computing and caching capabilities among different devices, it is a challenging problem to reasonably utilize limited edge resources to provide differentiated services to users.
- **Large-dimensional and multi-objective problems:** Due to the coupling of various resources, the optimization problem often have multiple optimization objectives with different constraints. At the same time, with the rapid growth of terminal devices, the dimension of the optimization problem will be further expanded, which resulting in NP-hard joint optimization problems.
- **Mixed-integer programming:** Mixed integer programming often appears in CETCN optimization problems. As an example, for the typical problem of task offloading in the CETCN, the discrete caching decision variable and the continuous task offloading variable should be jointly considered.

From the above analysis, it can be seen that due to the heterogeneity and limitations of edge devices, including edge servers and terminal devices, as well as the dynamic nature of resource requirements of various applications, it is still challenging to achieve efficient resource allocation while performing tasks [135]. Most of the existing algorithms can achieve efficient task offloading, resource allocation, collaboratice caching and mobility management in specific scenarios, however, they lack a unified framework to study these decision-making problems in practical scenarios, which consists of multiple criteria, such as heterogeneity, scalability, and mobility [136].

In principle, algorithms based on reinforcement learning obtain optimal solutions on the basis of long-term optimization by continuously learning the changes of the dynamic environment [137], [138]. For example, a RL-based task offloading algorithm was developed to reduce the average transmission delay of tasks [139]. In recent years, the combination of deep learning and reinforcement learning which called DRL has further enhanced the decision-making ability when facing complex and uncertain environments [140], [141]. In this regards, We summarize some popular AI algorithms to set the stage for our selected issues in CETCN.

III. AN INTRODUCTION TO REINFORCEMENT LEARNING AND ITS STATE-OF-THE-ART

The aim of this section is to introduce the mathematical background of SARL and MARL in Sections III-A and III-B, and briefly reveal its connection with the CETCN through

several examples. We also introduce several DRL algorithms used in this survey in Section III-C.

A. Single-Agent Reinforcement Learning

In this subsection, we summarize the MDP and POMDP which is useful in modelling the RL problems. For more details, the interested readers can refer to the following literature [142], [143], [144].

1) *Markov Decision Process*: We introduce the MDP through reinforcement learning model. Reinforcement learning maximizes its reward by interacting with the environment and taking corresponding actions. RL is generally regarded as an MDP and defined by a tuple $\langle \mathcal{S}, \mathcal{A}, R, T, \gamma \rangle$. At each episode, the agent receives a state $s \in \mathcal{S}$, selects an action $a \in \mathcal{A}$ and gets a reward $r \in R$ according to the policy, and then transits to the next state [145]. We often use a discount factor γ to express the impact of future rewards on the present.

In the following, we describe the key components of reinforcement learning in a more mathematical perspective.

- $\text{state}(s_t)$: The state is generally discrete in reinforcement learning, and s_t generally represents the state of the agent at time step t .
- $\text{action}(a_t)$: This is the output of the agent, when an agent observes the environment state, it will take an action a_t as the feedback of s_t .
- $\text{policy}(\pi)$: The policy π represents the probability that the agent chooses an action a based on the state s and $\pi(a|s) = P(a_t = a|s_t = s)$.
- $\text{state transition probability}(P_{ss'}^a)$: state transition probability depicts the probability when an agent takes action a and transits to the next state s' .
- $\text{reward}(r_t)$: When the agent takes an action, the environment will give the agent a reward. In general, the actions taken by the agent will have an impact on subsequent rewards. Therefore, to measure this effect, we use a discount factor γ to accumulate the rewards that the agent subsequently receives and can be represented as $R_t = \sum_{t=1}^{\infty} \gamma^t r_t$.
- value function : The purpose of RL is to learn a policy that maximizes the expected return. We define the state-value function as

$$V^{\pi}(s) = \mathbf{E}_{\pi}[R_t | s_t = s]$$

we can also define a state-action value function and can be represented as

$$Q^{\pi}(s, a) = \mathbf{E}_{\pi}[R_t | s_t = s, a_t = a]$$

Example: Several CETCN problems can be formulated as an MDP. For example, the authors in [146] model the task offloading and resource allocation problem as an MDP. The agent is the set of N mobile users. The state space is consisted of the computing resource. The action space is defined as the new task offloading decision and resource allocation decision. For each time step, the agent chooses an action and receives a reward, then transfers to the next state. Due to the continuity of the state and action space, the RL algorithm which called DDPG can be adopted to solve the problem under continuous action and state space.

2) *Partial Observable Markov Decision Process*: MDP usually assumes that the agent can obtain all the information in the environment. However, this assumption is difficult to realize in most practical systems. For example, the edge server can only obtain the tasks which generated from its connected terminal users. Thus, the agent only has part information of the whole system to make decisions. Hence, the problem describes above can be modelled as the POMDP [147], which extends the MDP to the partial observable environment. Specifically, a POMDP is denoted by a 7-tuple $\langle \mathcal{S}, \mathcal{A}, R, T, \gamma, \mathcal{O}, Z \rangle$, where $\mathcal{S}, \mathcal{A}, R, T$ and γ denotes the state, action, reward, state transition probability and discount factor, respectively, which is similar to the MDP, the observation is represented by $\mathcal{O}$, and $Z : \mathcal{S} \times \mathcal{A} \times \mathcal{O} \rightarrow [0, 1]$ denotes the transition probability based on a state $s \in \mathcal{S}$, an action $a \in \mathcal{A}$ and an observation $o \in \mathcal{O}$. In other words, the agent will receive a new observation when the agent chooses action a and the environment transmits from s to the next state s' .

Example: Several problems in the wireless communication system usually can be modelled as a POMDP. For example, the terminal equipments cannot obtain all the channel state information of the whole system, and can only obtain the estimation of the channel state information through interaction with the environment. Hence, in [148], the authors propose a dynamic task offloading problem in fog computing system based on POMDP. The agent is the set of IoT devices and has a sequence of computation tasks that need to be processed. Due to the limited computing capacity of IoT devices, the IoT devices need to decide whether tasks require to be processed at the local or offloaded to the fog servers to process through wireless link. If the task is decided to offload to the fog server to execute, the IoT devices need to consider which sub-channel to connect based on the estimated channel state information. Since the agent can not obtain the full information about the available sub-channels, this task offloading problem is considered as a POMDP. A finite state Markov chain is adopted due to the sub-channels fading is block fading. Based on the model, the IoT devices can estimate the channel state information (CSI) about the sub-channels and then choose a subset to connect for transmission. Hence, the state space is consisted of quantized CSI and the queue length, and the observation space of the agents is queue length. Given the observation of the agent, the agents need to minimize the task processing energy consumption under the maximum tolerable delay.

B. Multi-Agent Reinforcement Learning (MARL)

Multi-agent reinforcement learning solves the sequence decision problem involving the interaction of multiple players. Therefore, the dynamic characteristics of the system are mainly determined by the joint actions of players. In other words, the reward of an agent is no longer a function of its own actions, but depends on the joint actions of all players involved in the interaction. Therefore, in order to maximize long-term benefits, one player needs to consider the actions of other players when taking actions. In the following, we give a brief introduction of MARL from a mathematical view

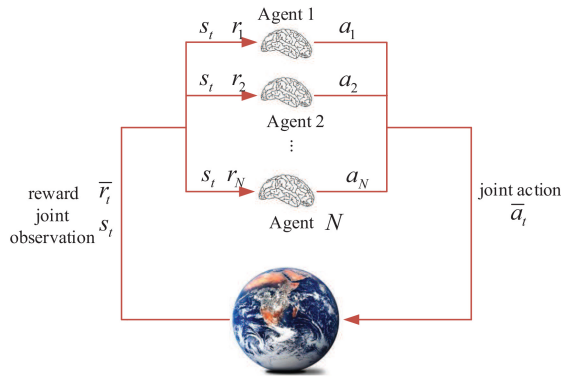


Fig. 4. Multi agents acting in the environment.

and give several example to illustrate the benefits of using MARL.

1) *Markov Game*: The communication network such as the CETCN involves multiple network entities such as edge nodes or mobile users. These entities not only interact with the environment, but also with other entities. Therefore, considering the interaction between entities, it is necessary to develop effective MARL algorithms for future Internet problems.

The multi-agent MDP can be defined by Markov Game [149] or Stochastic Game [150], as can be seen in Fig. 4, joint action is the result of each agent's independent choice of action. As shown in Fig. 4, the MG is defined by 5-tuple $(\mathcal{S}, \mathcal{N}, \{\mathcal{A}_i\}_{i \in \mathcal{N}}, \mathcal{T}, \{R_i\}_{i \in \mathcal{N}}, \gamma)$, where $\mathcal{S}$ is the state space, also called joint observation. $\mathcal{N} = \{1, 2, \dots, N\}$ denotes the set of the agents, $\mathcal{A} = \mathcal{A}_1 \times \mathcal{A}_2 \times \dots \times \mathcal{A}_N$ is the joint action space. In the area of the CETCN, the states of each agent include the various indicators of devices. The action state can be the allocated bandwidth, the allocated computing cycles and the caching status. The reward functions are usually related to the optimization objectives, such as minimizing the delay or maximizing the transmission rate.

At time step t , each agent chooses an action a_t^i based on the current system state s_t . Then, the state s_t transmits to the next state s_{t+1} , and receives the reward $R_i(s_t, a_t, s_{t+1})$. The aim of each agent is to find a policy $\pi_i^* : \mathcal{S} \rightarrow \mathcal{A}_i$ to maximize its own reward. Specifically, in multi-agent systems, the value function $V^i : \mathcal{S} \rightarrow \mathbb{R}$ of agent i is represented by all the agents' policies,

$$V_{\pi^i, \pi^{-i}}^i(s) := \mathbb{E} \left[\sum_{t=0}^T \gamma^t R_i(s_t, a_t, s_{t+1}) \mid a_t^i \sim \pi^i(\cdot | s_t), s_0 = s \right]$$

where $-i$ represents the agents that expect agent i . The above equation indicates the common solution of the MG is to find an optimal solution for each agent that no agent will receive a better solution under the situation that other agents do not change their solutions, and this statement is true for all agents. We call this optimal solution as Nash equilibrium (NE) [151], such that

$$V_{\pi^i, *, \pi^{-i}, *}^i(s) \geq V_{\pi^i, \pi^{-i}, *}^i(s).$$

The NE represents that the optimal policy $\pi^{i,*}$ is the best response of $\pi^{-i,*}$ for all the agents. Generally speaking, it

always exists in finite-space infinite discounted MGs [152] and NE is often more than one.

Example: In particular, the relationship among agents can be divided into cooperative and competition when involves multi-agent environment. Both of these two situations are common in the CETCN. For example, the terminal devices can cooperative with each other to achieve a higher transmission rate. However, when it comes to task offloading issues, the terminal devices may compete with each other to obtain limited edge resources to reduce the task processing delay as much as possible. The MGs can handle the following MARL settings: full cooperative setting, full competitive setting and mixed setting. In cooperative setting, in order to avoid low efficiency (avoid lazy agents), the authors in [153] propose team-average reward, and the goal is to maximize the long-term average reward. When in the full competitive setting, the MGs can be seen as zero-sum MGs. In the mixed setting, the MG consists of the cooperative and competitive agents [154]. The most popular MARL algorithm which called MADDPG [155], have been developed for mixed setting to calculate the Nash equilibrium solution of the MGs.

2) *Dec-POMDP*: In particular, the agents cannot receive the whole system information timely in the CETCN and the uncertainty of other agents and the environment has a great influence on learning an optimal solution. In this case, the Partial Observable Markov Game (POMG) under cooperative setting can be modelled as Dec-POMDP [156], [157] and it is a directly extension of POMDP from single-agent to multi-agent. The Dec-POMDP is defined by 7-tuple $(\mathcal{S}, \mathcal{N}, \{\mathcal{A}^i\}_{i \in \mathcal{N}}, P, R, \gamma, (O_i)_{i \in \mathcal{N}}, Z)$, the first six are the same as the previous definitions of MG, O_i is the observation space of agent i . In the Dec-POMDP, at each time step, each agent i makes its own decision based on its own observation o^i instead of system state s . Then, the action a^i of each agent forms an action set $\mathcal{A}_i = \{a^i | i \in \mathcal{N}\}$ and executes in the environment. The objective of each agent is to learn a policy $\pi_i : O_i \mapsto \mathcal{A}_i$ to maximize its cooperative long-term reward.

Example: Task offloading [158] and cooperative caching [159] among multiple base stations can be modelled as Dec-POMDP. For example, in [158], the author considers the problem of multi-channel access and task offloading in an industry 4.0 scenario based on MEC. In the proposed scenario, the edge device layer includes N machine type agents (MTAs), such as smart phones, robots, among which the MTAs can better utilize the spectrum resources, save bandwidth and avoid core network congestion through D2D communication. Each MTA can upload and download historical information, including observations and actions from the MTA administrator by D2D communication. Each MTA represents an agent, at each time step, each agent selects an action to minimize the total calculation delay of all the offloaded tasks. The actions of each agent include the state of each task and the selected channel. If the agent i decides to execute the task at the MEC sever, it needs to select the free channel to transmit. The environment is partial observable since each agent can only obtain its own selected channel. Thus, the agent's observation space includes a set of history information about channel selection. Based on the agent's

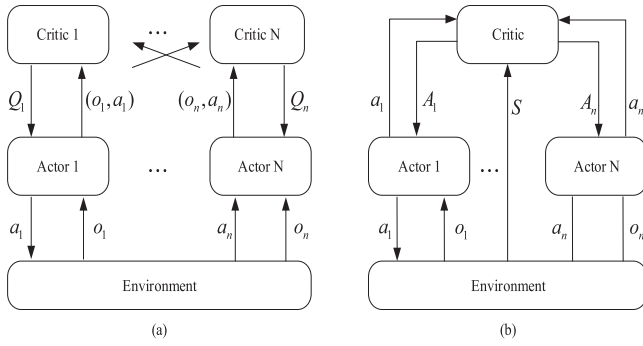


Fig. 5. Learning framework of MADDPG and COMA.

observation, each agent selects an action and receives a reward quantified by the total delay.

C. A Brief Introduction of DRL Algorithm Used in This Survey

- **DQN**: DQN solves the problem of unstable convergence [160] of approximate learning state-action value functions by combining Q-learning and deep learning. The first key technique is called **replay buffer**. At each time step t , DQN first stores the experience gained by the agent in the replay buffer, and then uniformly samples mini-batches of samples from this buffer for Q-learning updates. The second is called the **target network**, it performs as an independent network to replace the required Q network to generate the Q-learning objective, therefore, further improving the stability of the training process.
- **DDQN**: Although the DQN algorithm performs well in many games, it has been found that DQN algorithm have the problem of overestimating the Q value. In order to solve the problem, DDQN was proposed by Google DeepMind in 2016 [161]. The core of Double DQN is to try to avoid the overestimation problem by separating the selection action from the evaluation action.
- **Actor-Critic**: This algorithm simultaneously learns the agent's policy function and state value function [162], [163], in addition, the algorithm also applies bootstrapping to estimate the Q-value function. Actor-Critic algorithm through minimizing the square of temporal difference error to learn the critic function.
- **DDPG**: DDPG can be seen as an extension of DQN in continuous action space and can also be seen as a combination of Deterministic Policy Gradient and deep neural network [164]. The DDPG algorithm includes the Q-value function and the policy function at the same time. The Q-value function is updated by the TD method. The policy function is updated by the policy gradient method.
- **Multi-agent Deep Deterministic Policy Gradient (MADDPG)** [155]: It is an extension of DDPG algorithm to multi-agent scenario. As shown in Fig. 5(a), the critic of each agent can evaluate the impact of the joint strategy on the expected reward through obtain other agents' actions. MADDPG can be utilized to solve various issues in the CETCN. For example, MADDPG can be adopted

to learn the optimal task offloading strategy for each network entity, and in this manner, the computation complexity can also be reduced.

- **Counterfactual Multi-Agent Policy Gradients (COMA)** [165]: As shown in Fig. 5(b) COMA has a local actor to make actions which is similar to MADDPG. However, unlike the MADDPG, the COMA uses the centralized critic to estimates the action-value function. In a multi-agent environment, there exists a credit allocation problem, that is, the multi-agent algorithm uses the status and behavior of all agents to estimate the expected reward, so the agent may not be able to calculate the contribution of its behavior to the total reward. The COMA address the issue through using the counterfactual baseline method, which represented as follows,

$$A^i(s, \mathbf{a}) = Q(s, \mathbf{a}) - \sum_{a'^i} \pi^i(a'^i | \tau^i) Q(s, (\mathbf{a}_{-i}, a'^i)),$$

where $Q(s, \mathbf{a})$ is the Q-value of the current joint action $\mathbf{a}$, the second term $\sum_{a'^i} \pi^i(a'^i | \tau^i) Q(s, (\mathbf{a}_{-i}, a'^i))$ is the counterfactual baseline. In addition, COMA uses a single forward pass to compute the counterfactual advantage function for each agent in actor and critic network. Due to COMA uses only one centralized critic, it is only can be used in cooperative scenarios. Moreover, it is impossible to applying COMA in large scale system due to it need to know the global state to calculate the advantage function.

- **Mean-Field Q-Learning (MF-Q)**: When the joint action space and the number of the agents grows exponentially, MADDPG and COMA may be intractable. To address this issue, mean-field game (MFG) has been proposed to address it [166], [167]. Recently, by combining the MFGs and RL, we can solve MFGs at a very large scale both in terms of population size and environment complexity. Therefore, MF-Q [168] has been proposed to address the above issues. The Q-function is defined as follows

$$Q_i(s, \mathbf{a}) = \frac{1}{N_i} \sum_{k \in \mathcal{N}_i} Q_i(s, a_i, a_k) \approx Q_i(s, a_i, \hat{a}_i)$$

where $\mathcal{N}_i$ is the set of neighboring agents of agent i , $N_i = |\mathcal{N}(i)|$. The average value of $Q_i(s, a_i, a_k)$ can be approximate by $Q_i(s, a_i, \hat{a}_i)$. In this manner, the effect of neighboring agents can be converted to the mean-field value $\hat{a}_i$ and the representation agent can interact with the mean-field term instead of the whole population.

- **Mean-Field Deep Deterministic Policy Gradient (MFDDPG)**: In [169], the authors showed the connection with MDP and they explained how RL can be used to learn MFG solutions in a model-free way. Hence, the last algorithm we discuss is MFDDPG [170] which combine the MFG and DDPG algorithm. In the MFDDPG framework, the MFG can deal with continuous spaces. Similar to DDPG, the key of the algorithm consists in updating the critic network by minimizing an empirical square error and the actor network by making one step of gradient descent.

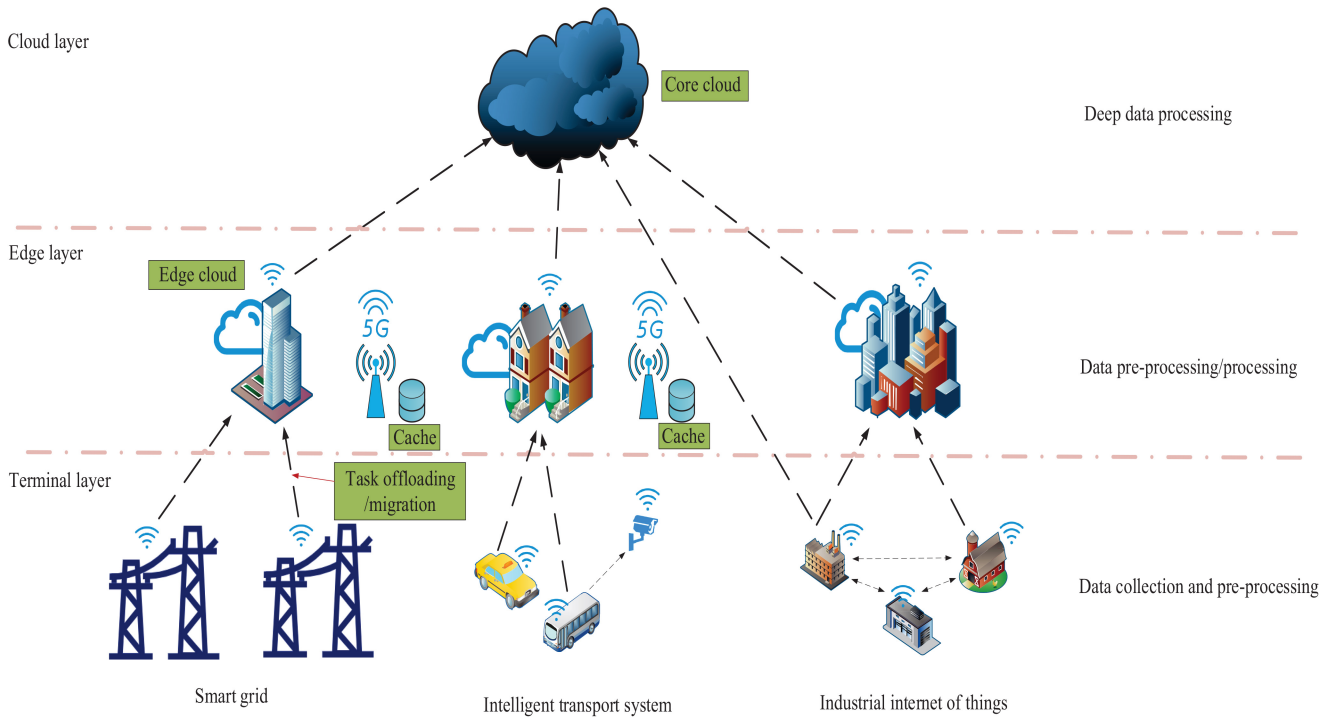


Fig. 6. The cloud-edge-terminal cooperation network architecture.

IV. THE ARCHITECTURE OF CLOUD-EDGE-TERMINAL COLLABORATIVE NETWORK

This section proposes the role of each component in the CETCN and introduces several potential network architectures of the CETCN in Section IV-A. The strict requirements of current and emerging applications and the improvement of computing power of edge devices jointly realize these architecture options, including cloud-edge based architecture and cloud-edge-terminal based architecture. This section also summarizes the work under each architecture in Sections IV-B and IV-C. Finally, we briefly introduce the connotation of cloud-edge-terminal intelligent collaboration, and explain its characteristics and advantages in emerging applications in Section IV-D.

A. Main Components of Cloud-Edge-Terminal Collaborative Architecture

There are various cloud-edge-terminal architecture for different applications. As shown in Fig. 6, we depict a simplified cloud-edge-terminal collaborative network architecture, which consists of cloud layer, edge layer and terminal layer. The terminal devices are deployed in the terminal layer to collect data for pre-processing and then transmit to the edge layer to computing and processing. Finally, some tasks that edge layer is not able to process will be sent to the cloud layer for execution. The specific roles of each layer are introduced as follows:

- **Terminal layer:** The first layer is terminal layer, which including sensors and intelligent devices to collect data from the physical environment or generate tasks from users. In some specific cloud edge collaboration systems,

such as the Internet of Things, terminal devices can also be various sensors, such as infrared sensors, environmental sensors, etc.

- **Edge layer:** Then, the collected data or the task will be sent to edge layer through communication links. If the computing resources installed in the edge nodes are sufficient to complete the task, then the task will be processed at the edge layer. Due to the distance between terminal layer and edge layer is short and the power for terminal devices are usually supplied by batteries, the communication methods between terminal layer and edge layer should be short distance and low power. Certainly, wireless local area network (WLAN) can provide a stable communication link between terminal layer and edge layer [171]. However, considering the requirements mentioned above, many other communication methods are more suitable for such transmission, such as ZigBee, Wi-Fi, and Bluetooth [172], [173].
- **Cloud layer:** Finally, the tasks that the edge layer can not able to process will be sent to cloud layer. The communication distance between edge layer and cloud layer is long and above local transmission methods are not suitable for such transmission. After receiving the tasks from edge layer, the cloud layer will process the task and transmit the results to the terminals through edge layer.

B. Cloud-Edge Collaborative Architecture

The cloud layer mainly comprises cloud computing architecture, such as computing units, storage units, and data centers. The cloud layer and the edge layer are connected through the core network. Although the edge layer can process

tasks from the terminal layer, Those tasks that need in-depth processing still need to be uploaded to the cloud center for processing.

The edge layer is the core layer of the cloud-edge-terminal collaboration architecture. From a hardware perspective, one the one hand, the edge layer contains various network and computing devices, such as edge servers, roadside units, gateways, and edge controllers. On the other hand, the edge layer provides wireless access to a wide variety of smart devices through wireless channels, fiber, and LTE. From a software point of view, the edge layer can manage and orchestrate services. The edge layer can collect and process data from the terminal layer and perform data analysis and real-time control. The edge layer can use its storage capacity to store related data and services for terminal devices. In addition, the cloud layer and the edge layer are complementary to each other and the task of the edge layer can be offloaded to the cloud layer for processing when the resources in the edge layer is not sufficient.

In [174], the authors propose an infrared intelligent service platform which can support cloud-edge cooperation and provide intelligent infrared services. The actual case study showed that the framework simplifies the development of intelligent services and ensures the security of industrial data. In [175], to solve the problem of infrastructure switching when the IoT system runs multiple applications for different tasks, this paper proposes a cloud edge cooperation framework based on attention enhancement and verifies the effectiveness of the framework through several most advanced convolutional neural networks. In [176], this paper proposes a HIF detection method based on migration learning under the cloud-edge cooperation framework of the IoT system. The framework can integrate all the features in different networks to form a basic cloud convolution neural network model for HIF detection and analyze the performance of the framework through seven different distribution networks. In [177], the authors introduce a new edge cloud collaboration architecture to solve resource allocation and decision-making problems and introduces the implementation method of equipment defect detection based on AI to show the effectiveness of the cloud edge collaboration architecture.

Qing [178] propose a cloud-edge collaborative power system that integrates big data and artificial intelligence platforms to realize global virtualization and global perception in the 5G scenario. At the same time, the platform has a certain reference value for cloud-edge collaborative management and control upgrading of a power system. In [179], the authors propose a collaborative cloud edge service cognitive framework to deploy the deep learning models, ensuring the reasoning quality and improving the operational efficiency and service reliability. In [180], the authors propose an intelligent distributed edge computing system architecture which supports the training and reasoning of deep learning models across heterogeneous resource-constrained IoT devices. In [181], to solve the privacy problems faced by many applications, this paper proposes an edge cooperation framework and verifies the effectiveness of the framework through a case study in a real scene. In [182], Xu et al. combine deep reinforcement learning

and cloud edge collaborative caching to build an intelligent edge caching framework to reduce content redundancy and transmission delay.

In [183], the authors propose a edge-cloud collaborative computing architecture for smart grid which consists four layers. The four layers including perceptual layer, network layer 1, application layer (CC node) and network layer 2. Then, they introduce the applications of smart grid, such as energy generation and transportation. Finally, they point out the future direction of cloud-edge architecture when applying to the smart grid. The authors in [184] develop a big data analysis architecture for edgeCcloud computing. In the proposed architecture, an intelligence module is introduced to the edge layer to process the task efficiently. Finally, the experiment results show the efficiency of the proposed architecture.

C. Cloud-Edge-Terminal Collaborative Architecture

The terminal layer consists of various terminal devices, such as mobile phones, different automobiles in intelligent transportation systems, and various applications in industrial IoT. These devices all have specific computing and storage capabilities and can continuously generate and collect large amounts of data. Based on the user's service quality requirements, these data can be processed at the terminal layer or offloaded to the edge layer and cloud layer for processing. For example, some applications which require powerful computing ability need to rely on the mighty computing power of the cloud layer. In contrast, low-latency services, such as unmanned driving, may need to process the task nearby and return the processing result to the vehicle in time to avoid traffic accidents.

To solve the problem of the insufficient computing power of edge devices and the mismatch between edge platform and AI algorithm, Zhang et al. [185] construct a lightweight software platform to enable edge intelligent devices to process data and make data sharing between edge devices possible. In [186], authors analyze the problem of how to build multiple mobile devices into a group to run artificial intelligence algorithms together. Through this framework, devices can reduce the burden of performing resource-intensive tasks. Nguyen et al. [187] propose a task offloading framework in which the parked vehicles are used as the expansion of infrastructure. In the peak period of the network, the edge server is deployed as the primary node, and the parked vehicles are used as subnodes for task offloading. The simulation results show that the framework can effectively and flexibly process tasks. In [188], a UAV-driven mobile edge computing framework is proposed. Under this framework, UAVs can help other UAVs perform computing tasks to solve the lack of computing services in remote areas. On this basis, DRL is used to solve the problems of task offloading and resource allocation, which we will discuss in later sections. The authors in [189] propose a cache-assisted collaborative intelligence framework between edge devices, which completes the storage and computing tasks of the deep neural network model through the cooperation between edge devices. Due to the limited computing capacity of terminal devices, the authors in [190]

propose a terminal-edge-cloud collaborative task offloading scheme to achieve the tradeoff between computation and communication capacity. The authors in [191] propose a terminal-edge-cloud collaborative architecture to make task offloading decisions for IIoT scenario. Then, they propose a novel task offloading method to obtain the balance between efficiency and delay.

D. Cloud-Edge-Terminal Intelligent Collaboration

Cloud-edge-terminal intelligent collaboration is not a simple combination of cloud computing, edge computing, and artificial intelligence technologies. Its core value lies in the realization of cloud-edge-terminal endogenous intelligence. At present, there is no internationally recognized definition of cloud-edge-terminal intelligent collaboration. To solve this problem, industry and academia have given different definitions [192], [193], [194]. *In this paper, we define cloud-edge-terminal intelligent collaboration as the realization of low-cost through and high-efficient collaborative optimization among devices, edges, clouds, and networks.*

Obviously, in a broad sense, cloud-edge-terminal collaborative intelligence includes different combinations of AI services based on terminals, edge, and cloud, such as terminal-edge-terminal collaboration, edge-cloud collaboration, edge-to-edge collaboration, and terminal-edge-cloud collaboration, as well as cloud intelligence (data are on the cloud, all intelligent analysis and decision-making are completed in the cloud) and terminal intelligence (data acquisition, processing, and intelligent analysis are completed on the terminal device).

It can be seen that cloud-edge-terminal collaborative intelligence in a broad sense is a natural extension of cloud intelligence and terminal intelligence. In some particular scenarios, the optimal collaborative method of cloud-edge-terminal collaborative intelligence may be cloud intelligence, due to the limited resources of terminal and edge device. However, edge-cloud-terminal intelligent collaboration generally finds the optimal collaborative method among all possible collaborative methods to obtain a better collaborative method than pure cloud intelligence and terminal intelligence.

The features and advantages of cloud-edge-terminal intelligent collaboration can be summarized at least from the following four aspects:

- **Lower bandwidth cost.** According to the prediction of the Internet Data Center, by 2025, the total amount of data in the world will reach 180 ZB [195], and the current network bandwidth is hard to meet the demand for such data transmission, which will become the bottleneck of cloud computing. The most significant advantage of cloud-edge-terminal intelligent collaboration is that data analysis is processed at the edge, avoiding the need for large amounts of data to be uploaded to the cloud and greatly saving bandwidth costs.
- **Higher level of intelligence.** Cloud-edge-terminal collaborative intelligence can efficiently realize various AI services and applications through the collaboration between devices, edges, and clouds. Deep reinforcement learning is the core support for realizing

cloud-edge-terminal collaborative intelligence. These AI algorithms are also essential for quickly analyzing large amounts of data. In addition, the perfect and unified cloud-edge-terminal collaborative intelligent system architecture guarantees for the further intelligence of AI algorithms.

- **Higher real-time performance.** As more and more sensors are deployed on network terminals, an astonishing amount of real-time data will be generated every second. Since end users and cloud computing centers often need to go through multi-level networks, if the acquired information is uploaded to the cloud, the end-to-end latency is difficult to satisfy the performance requirements of applications such as virtual reality and autonomous driving. By sinking computing power from the cloud to the network edge, cloud-edge-terminal intelligent collaboration can complete data processing with low latency which can ensure real-time data analysis.
- **Stronger privacy protection.** Privacy protection has become one of the hottest topics recently as people become more aware of the importance of preventing public data breaches. In traditional cloud computing, users must upload all data to the cloud for encryption. Cloud-edge collaborative intelligence ensures that the original data can be processed at the network edge, thus avoiding the risk of user privacy exposure.

Next, we mainly introduce and summarize five aspects of the cloud-edge-terminal collaboration network combining AI technologies, especially DRL and MADRL methods. The five aspects include task offloading, resource allocation, caching, mobility management, and intelligent emerging applications.

V. TASK OFFLOADING AND RESOURCE ALLOCATION FOR AI-ENABLED CLOUD-EDGE-TERMINAL COLLABORATIVE NETWORK

Task offloading and resource allocation techniques are promising methods used in the CETCN that can significantly improve the efficiency of task offloading and reduce the task execution delay. This is beneficial for meeting the growing performance requirements of emerging applications, such as low latency and high reliability. In addition, SARL and MARL can provide a good solution for task offloading and resource allocation issues due to the complex and decentralized nature of the CETCN. Hence, in this section, we firstly introduce the models of task offloading and resource allocation for emerging applications in Sections V-A and V-B. These models provide the background for abstracting various operations into optimization problems. Then, we review the works that have been done in task offloading and resource allocation domain in Section V-C. Finally, we point out the possible research directions on task offloading and resource allocation issues for the CETCN in the future in Section V-D.

A. Classification of Task Offloading Models

In this subsection, we introduce two popular task offloading models, corresponding to binary offloading and partial offloading respectively. A toy example can be seen in Fig. 7.

TABLE III
THE ADVANTAGES AND DISADVANTAGES OF DIFFERENT COMPUTING ENTITIES

Computing entity	advantage	disadvantage
terminal device	(1) provide extremely low transmission latency (2) solve the congestion of the core network	(1) the computing capacity and battery life is limited
edge server	(1) provide low transmission latency and abundant bandwidth resource (2) solve the congestion of the core network	(1) the computing and storage capacity is limited (2) randomness of geographical distribution of edge devices
cloud server	(1) powerful computing and storage capacity	(2) high transmission latency

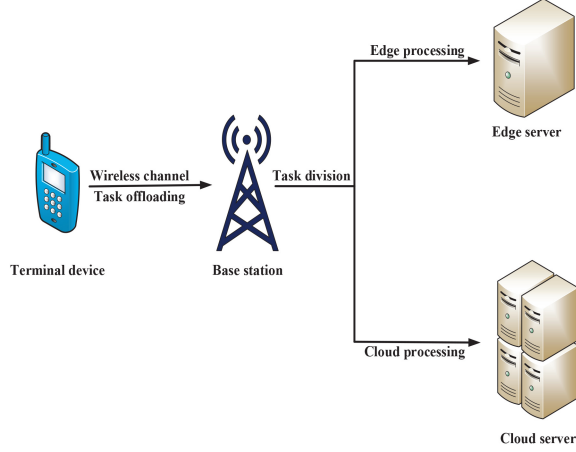


Fig. 7. A toy example of task offloading. The architecture of task offloading is composed of terminal device, base station, edge servers and cloud server. The terminal device uploads the generated tasks to the edge server through the wireless channel. Then the edge server chooses whether to process the task at the edge server or process the task at the cloud server according to the offloading mode of the task.

1) *Task Model for Binary Offloading*: Binary offloading refers to that the task cannot be divided into multiple parts and only can be processed at terminal devices, edge servers or cloud server as a whole. This task offloading model is suited for highly integrated and simple tasks. Such tasks can be represented by a four-tuple $\{s, f, \tau, c\}$, where s represents the size of the task (in bits), f represents the CPU cycles required for processing one bit task, τ represents the time constraint for processing the task (in seconds), c is the storage space required for a typical task. The four parameters can not only capture the basic properties of various tasks, but also can evaluate the time delay and energy consumption [196]. In general, there are more than one edge server at the edge layer, hence, we can use x_{ij} to denote the task i which processed at the j th edge server. Similarly, we can use x_{ic} to denote whether the task i processed at the cloud server and x_{il} to denote the task that processed at the terminal devices. In the binary offloading mode, x_{il}, x_{ij}, x_{ic} must satisfy $x_{il}, x_{ij}, x_{ic} \in \{0, 1\}$ and $x_{il} + \sum_{j=1}^J x_{ij} + x_{ic} = 1$, where J denotes the amount of edge servers.

2) *Task Model for Partial Offloading*: In the contrary, if the task can be divided into multiple parts, and different part can be processed at different devices, this task offloading model is refer to partial offloading. Similarly, in the partial offloading mode, x_{ij}, x_{ic} must satisfy $x_{il}, x_{ij}, x_{ic} \in [0, 1]$ and $x_{il} + \sum_{j=1}^J x_{ij} + x_{ic} = 1$. $x_{il} = 0$ / $x_{ij} = 0$ / $x_{ic} = 0$ refers to

the situation which terminal device / j th edge server / cloud server do not process the i th task.

Although terminal devices, edge servers, and cloud servers can process tasks generated by different users, these three entities have their own advantages and disadvantages in processing tasks. We summarize the advantages and disadvantages in the Table III.

B. Classification of Resource Allocation Models

The joint computing-communication resource allocation plays a vital role in realizing delay-sensitive and energy-efficient CETCN.

1) *Computing Model for Task Offloading*: The CPU is the core computing component of various computing entities. In general, we adopt CPU-cycle frequency f_m to evaluate the performance of CPU. The advanced CPU architecture adopts the DVFS technique which can adjust the CPU cycle frequency (voltage), which usually has a maximum value f_m^{max} . Based on the computing model introduced in Section V-A, the execution time for a particular task $\{s, f, \tau, c\}$ can be calculated as follows:

$$t_m = \frac{sf}{f_m} \quad (1)$$

The energy consumption is another critical aspect of resource allocation scheme according to the circuit theory [197], [198], [199], [200]. In particular, the authors in [199] have demonstrated that the power consumption is related to the production of $V_{cir}^2 f_m$, where V_{cir} is the voltage supplied by circuit. Furthermore, in [197], [200], the clock frequency of CPU chip is approximately linear to the voltage. Thus, the energy consumption is given by κf_m^2 , where κ is the constant. From the above analysis, for a typical task $\{s, f, \tau, c\}$, the energy consumption can be represented by

$$E_m = \kappa s f f_m^2 \quad (2)$$

When we adopt partial offloading model, the task handled by different computing entity is decided by x_{im} , where x_{im} is referred to x_{il}, x_{ij}, x_{ic} . For example, if part of the task handled by j th edge server is x_{ij} , then the energy consumption can be calculated by $\kappa x_{ij} s f f_j^2$, where f_j denotes the CPU-cycle frequency of j th edge server. One can observe from (1) and (2), when allocate computing resources, we need to consider the time deadline and the energy consumption.

2) *Communication Model for Task Offloading*: In CETCN, an efficient communication resource allocation scheme is

TABLE IV
CHARACTERISTICS OF TYPICAL WIRELESS COMMUNICATION TECHNOLOGIES AND 6G PERFORMANCE REQUIREMENTS

	NFC	RFID	Bluetooth	WiFi	LTE	5G	6G
Max. Coverage	10cm	3m	100m	100m	up to 5km	Excellent coverage	Full coverage
Operation Freq.	13.56MHz	LF:120-134kHz HF:13.56MHz UHF:850-960MHz	2.4GHz	2.4GHz,5GHz	TDD:1.85-3.8GHz FDD:0.7-2.6GHz	6-100GHz	0.1-10THz
Data Rate	106,212, 414kbps	Low (LF) to High (UHF)	22Mbps	135Mbps	Downlink:300Mbps Uplink:75Mbps	Indoor/Dense outdoor: up to 10Gbps Urban/Suburban: >hundreds of Mbps	100-1000Gbps

critic to delay-sensitive applications. To be specific, the wireless channel is different from wired ones in the following aspects [201]:

- Due to reflection and refraction from buildings, walls, and trees, wireless signals face multipath fading during propagation, which makes the channel height time-varying and lead to heavy interference.
- The broadcast nature of wireless transmission causes interference between different signals resulting in a probability of error in detection. To tackle this problem, interference management has become one of the most active research directions in wireless communication systems [202], [203], [204].
- Spectrum scarcity has been a major problem for high-speed radio access, which has prompted the design of new transceiver architectures [205], [206] and network architectures [207], [208] to improve spectral efficiency. At the same time, the development of spectrum sharing and aggregation technologies [209], [210], [211] is also conducive to promoting the effective use of fragmented and under utilized spectrum resources.

The uncertainty of wireless channels is a critical factor in designing efficient resource allocation schemes in CETCN. For example, when the wireless channel is in deep fading, the delay of task processing reduced by the powerful computing power may not be enough to compensate the increased transmission delay generated by wireless channel. Furthermore, increasing data transmission rate through improving the transmission power may lead to high energy consumption.

Nowadays, various commercialized technologies for modern mobile communication systems have emerged, including *near-field communication* (NFC), *radio frequency identification* (RFID), Bluetooth, WiFi and 5G. These technologies can support task offloading from terminal devices to edge layer or cloud layer.

Furthermore, in addition to the huge improvement in maximum coverage, operation frequency and data rate of 6G, the IMT2030 promotion group said that the air port delay of the sixth generation mobile communication system would reach. 1 to 1 millisecond, supporting high-speed mobile communication, the spectrum efficiency will be 1.5 to 3 times higher than 5G, and the energy efficiency will be 20 times higher than 5G. The 6G requirements are also pointed out in terms of maximum coverage, operation frequency and data rate in [212]. We list the existing communication technologies and 6G requirements in Table IV.

3) *Caching Model for Task Offloading*: On the one hand, the storage capacity of terminal devices and edge servers are limited, on the other hand, most of the traffic on the network is generated by the repeated requests for popular content and applications. Hence, designing reasonable caching placement scheme can efficiently realise the traffic pressure of the core network and reduce the delay of content acquisition [213], [214], [215]. Detailed analysis of cooperative caching can be found in Section VI.

Future mobile communication systems will carry a large number of emerging services, such as AR/VR and driverless driving which require restrict delay guarantee. Therefore, only using edge servers or cloud servers cannot meet the delay requirements of such applications. Besides, advanced communication technologies will further improve the QoS of users.

From another perspective, due to the heterogeneity of tasks (different tasks have different requirements for delay and bandwidth) and the complexity of edge environments (such as interference between wireless channels), it is difficult to obtain the accurate system models for task offloading which is required for traditional methods. But lucky, the development of DRL can effectively solve the problem. In the following, we will summarize the recent progress of using DRL to solve task offloading in CETCN.

C. Literature Review

From the user's perspective, a key benefit of CETCN is that it can leverage the capabilities of edge devices for computation offloading, thereby speeding up the computation process and saving energy. Another important aspect of computing offloading is how to use the limited resources at the edge to make the optimal task offloading decisions, such as how to use the available bandwidth, the size of the data volume that needs to be offloaded, and the user's latency requirements to determine how to offload. When tasks are offloaded to CETCN, it is necessary to properly allocate computing resources. Since edge computing nodes are numerous and scattered, we investigate papers that discuss how to efficiently allocate resources to applications that need to be offloaded.

Plenty of efforts have been put into task offloading and resource allocation, such as in [39], authors provided a systematic review of task offloading in edge computing scenarios. Because the dimension of the parameters of the problem is very high, the traditional optimization schemes are often

TABLE V
SUMMARY OF CLOUD-EDGE-TERMINAL COLLABORATIVE TASK OFFLOADING BASED ON AI

reference	Proposed solution	Cooperation type	Objective	Evaluation method	Model
[216]	Q-Learning	Terminal-edge	Minimize the long-term energy consumption	Numerical analysis	MDP
[217]	Q-Learning	Terminal-edge-cloud	Minimize the transmission and computation delay	Simulation analysis	MDP
[218]	Q-Learning	Terminal-edge-cloud	Minimize the computation and communication delay	Simulation result	MDP
[219]	DQN	Terminal-edge-cloud	Minimize the energy consumption	Simulation analysis	MDP
[220]	DQN	Edge-cloud	Minimize the latency	Simulation analysis	MDP
[221]	DQN	Edge-cloud	Minimize the energy consumption and delay	Simulation analysis	MDP
[222]	DQN	Terminal-edge	Minimize the system-wide data processing cost	Simulation Analysis	MDP
[223]	DDQN	Terminal-edge-cloud	Minimize the computing energy consumption and task processing delay	Simulation analysis	MDP
[188]	DDQN	Terminal-edge	Maximize the total utility of the system	Numerical analysis	MDP
[224]	DDQN	Terminal-edge	Minimize the total delay and waiting time	Simulation analysis	MDP
[225]	AC	Terminal-edge-cloud	Minimize the delay	Simulation analysis	MDP
[226]	AC	Terminal-edge-cloud	Minimize the weighted task delay and energy consumption	Simulation analysis	MDP
[227]	AC	Edge-cloud	Minimize the long-term cost of cloud server	Simulation analysis	MDP
[228]	DDPG	Terminal-edge-cloud	Minimize the maximal task latency	Simulation analysis	MDP
[229]	DDPG	Edge-cloud	Minimize the total cost of mobile users	Simulation analysis	MDP
[230]	DQN	Terminal-edge	Minimize the energy consumption	Simulation analysis	POMDP
[231]	AC	Terminal-edge-cloud	Minimize the end-to-end delay	Simulation analysis	POMDP
[232]	A3C	Terminal-edge	Minimize the energy consumption	Simulation analysis	POMDP
[233]	PPO	Terminal-edge	Maximize the benefit	Simulation analysis	POMDP
[234]	DDPG	Terminal-edge-cloud	Minimize the total energy consumption	Simulation analysis	POMDP
[235]	MADDPG	Terminal-edge-cloud	Minimize the task execution delay	Simulation analysis	POMDP
[236]	A3C	Terminal-edge	Minimize the latency	Simulation analysis	MG
[237]	A3C	Terminal-edge	Minimize the computational delay	Simulation analysis	Dec-POMDP
[238]	PPO	Terminal-edge-cloud	Minimize the latency and energy consumption	Simulation analysis	Dec-POMDP
[239]	DQN	Terminal-edge-cloud	Minimize the latency	Simulation analysis	Dec-POMDP

limited. Hence, machine learning algorithms can reduce the difficulty of solving the optimization problem by finding the approximate optimal solution. Therefore, in [45], the author made a review of the application of machine learning in the MEC system and pointed out the problems faced by machine learning in solving these issues. Based on different offloading schemes and different optimization objectives, such as delay minimization and energy efficiency maximization, different optimization problems can be obtained. Due to the challenges faced by CETCN, DRL becomes a promising method in solving these issues. Therefore, in this section, we review the papers that utilize the DRL to solve task offloading and resource allocation issues in CETCN in recent years. We summarize this section in Tables V and VI respectively.

1) *Task Offloading*: In [216], the authors study the realization of an efficient multi-user distributed offloading mechanism in a time-varying channel. First, with the help of game theory, the edge nodes need to determine the task offloading decisions. Second, the author extends the problem to scenarios where the channel information is uncertain and proposes a RL-based offloading mechanism to help edge nodes learn optimal offloading strategies. Wang et al. [217] discuss the task offloading problem in MEC system. In the

small cell scenarios, the computing tasks requested by the user are divided into computing tasks performed by the user and the MEC server, tasks performed by the MEC server alone, and tasks performed by the users themselves. Due to the different data sizes of tasks requested by users, the authors describe the problem as a transmission and calculation delays minimization problem and propose a multi-stack Q-learning method to improve learning efficiency. To address the stringent requirements of MEC systems for offloading scalability, [218] presents a collaborative computing system consisting of mobile devices, edge clouds, and central clouds. Based on Q-learning, a computational offloading strategy is designed, and the numerical results show that the scheme can effectively improve the user's service experience.

In [219], in order to minimize the energy consumption in mobile Internet, the authors formulate a power minimization problem and propose a novel DRL based green task offloading scheme in end-edge-cloud collaborative network to make task offloading decisions, extensive experimental results indicate the proposed scheme is superior to other baseline solutions. The movement of terminal devices poses difficulties for cloud edge collaborative task offloading. At the same time, maintaining continuous migration of mobile device's tasks is

TABLE VI
SUMMARY OF CLOUD-EDGE-TERMINAL COLLABORATIVE RESOURCE ALLOCATION BASED ON AI

reference	Algorithm	Cooperation type	Resource type	Objective	Evaluation method	Model
[249]	Q-Learning	Terminal-edge-cloud	Communication Computation	Minimize the processing delay	Simulation analysis	MDP
[250]	Q-Learning	Terminal-edge	Communication	Maximize the utility and maintain allocation fairness	Numerical analysis	MDP
[251]	Q-Learning	Terminal-edge-cloud	Computation Caching	Minimize the state loss function	Numerical analysis	MDP
[252]	DQN	Edge-cloud	Communication Computation	Minimize the delay and energy consumption	Simulation analysis	MDP
[253]	DQN	Terminal-edge-cloud	Computation	Minimize the system delay	Real-world analysis	MDP
[254]	DQN	Edge-cloud	Communication Computation Caching	Minimize the system delay	Simulation Analysis	MDP
[255]	DDQN	Terminal-edge-cloud	Communication Computation	Minimize the energy consumption	Simulation analysis	MDP
[256]	DDQN	Terminal-edge	Communication Computation	Maximize the weighted sum cost	Numerical analysis	MDP
[257]	DDQN	Terminal-edge-cloud	Communication Computation	Minimize the energy consumption	Simulation analysis	MDP
[258]	AC	Terminal-edge	Communication Computation	Maximize the computation rate	Simulation analysis	MDP
[259]	AC	Terminal-edge	Communication Computation Caching	Maximize the cumulative reward	Simulation analysis	MDP
[260]	DDPG	Edge-cloud	Communication Computation	Maximize the system cost	Simulation analysis	MDP
[261]	DDPG	Terminal-edge	Computation	Minimize the delay and energy cost	Simulation analysis	MDP
[262]	DDPG	Terminal-edge	Communication Computation Caching	Minimize the energy consumption	Numerical analysis	MDP
[263]	DQN	Terminal-edge-cloud	Communication Computation Caching	Minimize the task latency	Numerical analysis	POMDP
[264]	DQN	Terminal-edge-cloud	Computation	Minimize the energy consumption	Numerical analysis	POMDP
[265]	PPO	Terminal-edge-cloud	Communication Computation	maximize the energy efficiency	Numerical analysis	POMDP
[266]	MADDPG	Terminal-edge-cloud	Communication Computation	maximize the time-average completion rate Computation	Numerical analysis	Dec-POMDP
[267]	MFDDPG	Terminal-edge	Communication Computation	minimize the energy consumption	Numerical analysis	Dec-POMDP

also a challenging issue. Therefore, different from [219], the authors in [220] propose a task offloading framework under the cloud-edge cooperative network aim of minimizing the latency of the task assignment of mobile terminal devices. Then, they propose a DQN-based algorithm to make offloading decisions, and verify the effectiveness of the proposed task offloading framework. When deploying intrusion detection systems at the edge of the network, serious packet loss issues can occur. To solve the problem, in [221], the authors propose a collaborative intrusion detection system for mobile edge computing, which can transfer some detection tasks to other edge nodes with rich resources. Then, the intrusion detection task offloading problem is described as an MDP and propose a DQN-based algorithm to tackle the problem, extensive experiments show that the proposed scheme superiors to baseline schemes in terms of energy consumption, response time and packet loss rate. Different from above works, in [222], the authors study

the issue of how to reasonably utilize network resources in order to achieve the best resource utilization under limited communication and computing capabilities. Firstly, the authors propose a novel edge-terminal collaboration model. In the proposed model, they consider the terminal device as an autonomous vehicle, and the edge devices are roadside units with edge computing capabilities and are deployed on the roadside. Roadside units can provide vehicles with the ability to task offload, and vehicles can also collaborate with roadside units to provide computing capacity. Aiming to minimize the data processing cost and reduce latency, the author further models the problem as a DRL problem, and proposes a DQN algorithm to solve the problem. Extensive simulation results indicate the effectiveness of the proposed scheme in reducing cost and latency.

In [223], the authors consider the computation offloading problem in MEC from the perspective of computational cost

and user privacy. Firstly, the authors establish a QoS model based on latency and energy consumption. Then, they propose a DDQN-based task offloading algorithm to improve the Quality of Service (QoS). At the same time, the optimal offloading strategy can be obtained based on local node state information within the coherent time of the channel, thereby solving the problem of the transmission channel changing over time and also protecting the data security of other nodes. Extensive experimental results indicate that the DDQN algorithm can achieve a higher QoS compared with existing schemes. Different from [223], the authors in [188] not only take the time-varying characteristics of communication channels into account, but also consider the randomness of user requests. In this context, they propose a two-stage DRL algorithm to tackle the task offloading issue in the UAV-assisted mobile edge computing system to maximize the long term utility. The simulation results show that the proposed DRL based scheme can achieve good performance in UAV assisted MEC system. In [224], the authors consider how to efficiently offload computationally intensive tasks in congested situations. Firstly, they establish a multi-vehicle dynamic offloading model to minimize the total waiting delay of the total moving vehicles' tasks. Then, they divide the tasks into sequential subtasks and find the optimal solution for subtask offloading with the assistance of DDQN algorithm. Experiments results demonstrate that the DDQN algorithm is effectiveness than baseline solutions.

The authors in [225] propose a deep learning task scheduling framework and formulate the task scheduling problem as a delay cost minimization problem and prove the problem is a NP-hard problem. Then, considering the limited computation capability of edge nodes, they propose an edge-cloud collaboration system to solve the problem based on an actor-critic algorithm. Numerical results indicate that the proposed scheme can reduce the task processing time by 28.3%. In [226], the author proposes a three-tier MEC network architecture. The computing tasks in this system can be processed at the edge servers and cloud servers. Meanwhile, considering that randomness of the number of tasks and environmental states, traditional optimization methods are difficult to obtain effective decisions in such dynamic network environments. Therefore, the author proposes an optimization framework based on an actor-critic algorithm to minimize the task delay. Experimental results indicate that the proposed method outperforms the baseline scheme in terms of offloading efficiency. Different from the above work, the authors in [227] not only consider the end edge collaborative computing offloading problem, but also consider the service cache problem. Specifically, on the one hand, users pay fees to the cloud server to obtain computation offloading services, and on the other hand, the cloud server provides services to users and charges service fees. Then, the offloading optimization problem is novel expressed as the problem of minimizing cloud server costs while satisfying user latency requirements. Due to the NP-hard of this problem, an A3C-based optimization algorithm is proposed and simulation results indicate that the proposed method outperforms other baseline solutions.

In [228], the authors propose a task offloading scheme in end-edge-cloud orchestrated computing framework to decide when and where to offload the task which takes the mobility of edge servers and the network conditions into account. The goal is to enhance the efficiency and responsiveness of the system by dynamically allocating resources based on the user's mobility, network connectivity, and computational requirements. Then, considering the high dimensional and coupled variables, they propose a DDPG-based offloading approach to solve the task offloading problem. Finally, extensive experiments indicate that the proposed approach is superior to other baseline solutions. In [229], the authors propose a novel approach to optimize computation offloading and service caching in MEC environments and explore the benefits of caching frequently used services at the edge servers to reduce latency and improve the overall efficiency of the system. Different from the work in [228], the authors in this work also consider the benefit of caching popular services at the network edge. Through caching popular services at the network edge, the repeated computations and traffic can be reduced. Then, in order to minimize the cost of mobile users, the authors formulate the task offloading problem as a MINLP and propose a DDPG-based offloading algorithm with intelligent caching decisions to achieve the near-optimal solution. Finally, the proposed algorithm can enhance efficiency and reduce the response time.

The authors in [230] formulate the task offloading problem as a POMDP and propose a DQN algorithm to make the optimal offloading decisions. The novelty of this paper lies in, on the one hand, in practical systems, mobile devices only can rely on partial state information to make offloading decisions, therefore, the authors model the offloading problem as a POMDP problem. On the other hand, the author uses motion prediction to predict the future location of mobile users. By using historical data and machine learning techniques, the system can predict where users may arrive at the next time slot. This prediction enables the MEC system to proactively offload tasks, and ensures that the tasks are offloaded to edge servers which are geographically closer to the user's future location, thereby further reducing service latency. The real-world experiment results show that the scheme is better than baseline scheme at the total cost, delay and energy consumption. The authors in [231] investigate the blockchain-enabled CETCN and formulate the task offloading problem as a POMDP. The key novelty of this paper lies in integrating blockchain as a decentralized and secure ledger to facilitate the task offloading process. The blockchain can enable computing nodes to transparently and tamper-resistantly record task requests, available resources, and execution results, and thus ensure trust and accountability in the offloading process. Through introducing smart contracts and token-based incentives, on the one hand, the interaction among participants can be managed and the fair resource allocation can be promised. On the other hand, the incentive mechanism can also improve the cloud-edge collaborative system's efficiency. At the same time, the proposed blockchain enabled end-edge-cloud collaborative system allows the every participant apply DRL based approach to make task offloading decisions. Finally, numerical results indicate that the algorithm can achieve a better performance

in many aspects. In [232], the authors investigate the task offloading problem in UDN-assisted MEC system. Ultra-Dense Networks (UDNs) refer to a wireless communication environment with a high density of small cells, such as urban areas or crowded locations. In such scenarios, efficient task offloading strategy is crucial to optimize resource utilization and enhance network performance. Firstly, the authors propose an intelligent and adaptive task offloading method which can handle the uncertain and incomplete information of the UDNs, including network conditions, user demands, and resource availability information. Besides, the online nature of the offloading strategy allows it to adapt to dynamic changes of the UDN environment, and thus making it suitable for real-time decision-making in dynamic and evolving scenarios. Finally, extensive numerical results show that the proposed method outperforms other baseline solutions at energy consumption. Different from above works, the authors in [233] propose a game-theoretic framework to help mobile users to decide where to offload their tasks. In this work, each mobile user acts as an independent player to maximize their own utility based on energy efficiency, task delay and cost. The game-theoretic approach allows for a competitive environment where mobile users and edge servers aim to maximize their individual utilities while considering the actions of other players. This enables the system to achieve a balance between offloading tasks to alleviate network congestion and executing locally to reduce latency and energy consumption. Then, they design an efficient task offloading model based on POMDP and propose a PPO-based algorithm to guide the network entities to obtain the optimal solutions. Simulation results indicate the proposed method outperforms other baseline solutions in terms of system efficiency, user satisfaction and resource utilization.

On the one hand, the task offloading scheme based on DDPG cannot guarantee satisfying time constraints when dealing with problems with strict deadlines. On the other hand, existing schemes assume that each decision time slot has all information about the state. However, in practical problems, many real environments involve unobservable states. Huang et al. [234] propose a multi-layer offloading scheme based on partially observable Markov models. This scheme can satisfy the time constraints by offloading a specific set of tasks to edge servers, while minimizing the total energy consumption of mobile devices. In addition, in order to minimize the energy consumption of the algorithm, they propose an overall decision-making mechanism for tasks that arrive within a fixed time interval. Under this mechanism, the number of calls of the DRL algorithm can be significantly reduced. The authors in [235] study the air access network combining satellite, UAV communication and mobile edge computing. The novelty of this article lies in the fact that the tasks of ground equipment can be offloaded to aerial platforms, and non orthogonal multiple access technology is adopted to improve transmission rates. Secondly, due to the fact that high and low altitude platforms can only observe partial state information, and the offloading problem is a mixed nonlinear integer programming problem, the author models the problem as a partially observable Markov model and designs a MADRL algorithm to solve the offloading problem. Finally, extensive

numerical results indicate that the proposed scheme can reduce the task processing delay of all the devices.

Blockchain has been used to ensure the transaction security of mobile commerce. However, the intensive computing during the mining process limits the participation of mobile devices. The authors in [236] investigate a multi-server task offloading problem with different capabilities and prices. Firstly, due to the heterogeneity of capability and price, the offloading decisions of miners are quite complex. Secondly, the long-term benefits of miners need to be maximized. Thirdly, due to the limited range of wireless communication, miners can only observe limited information about the environment. Considering the above challenges, the authors first model the task offloading problem as a Markov game to maximize the long-term benefits of miners. In addition, due to the dependence of miners' decisions on those of other miners, the decision behavior of miners is approximated as an average effect by introducing a mean field term. Based on the actor critic framework and mean field theory, each miner can be trained and executed in a distributed manner to obtain the optimal task offloading decisions. The numerical results demonstrate that the algorithm outperforms other baseline solutions at average miner utilities and convergence speeds. Wu et al. in [237] proposes a DRL algorithm to address the dynamic resource allocation problem. The key contribution of the paper is to use DRL algorithm to learn an optimal policy for resource allocation. To be specific, the agent takes the current state of the system into account, such as available resources and task demands, and takes actions to allocate computation tasks between the mobile device and the edge server to optimize latency and energy efficiency based on the system state. Considering the partial observability of the system and the high dimension of variables, the authors model the task offloading problem as a dec-POMDP problem and propose an actor-critic based algorithm to obtain the task offloading decisions. The simulation result indicates the proposed algorithm is better than PPO algorithm and DDPG algorithm in terms of minimizing computational latency and energy consumption. The authors in [238] consider privacy and security issues during the task offloading process. Unlike the above work, they adopt independent reinforcement learning to solve the task offloading problem under completely decentralized conditions, thereby avoiding mobile devices from exposing their local information. In order to solve the non-stationary problems faced by independent reinforcement learning, the author proposes a novel neural fictitious self-play architecture. This architecture solves non-stationary problems through a built-in self play mechanism, while utilizing gated recurrent unit to handle partial observability of the system under completely decentralized conditions. Through a large number of simulation experiments, it has been verified that the proposed method is superior to the original independent reinforcement learning. In [239], the distillation module is introduced to improve the learning efficiency of some observable edge computing. Firstly, the offloading problem with time constraints is modelled as a decentralized observable Markov decision process with distillation, where each edge server makes offloading decisions based on its own observation

learning strategy. This method avoids overfitting by observing historical distillation and trusting distillation. At the same time, this method can learn appropriate offloading decisions in partially observable systems to reduce communication costs. Extensive simulation results demonstrate the proposed method achieve a high offloading efficiency compared to traditional DRL methods.

2) *Resource Allocation*: After offloading tasks, the servers must decide how much computing, communication and storage resources should be allocated to the related services. Resource allocation schemes based on cloud computing or edge computing have been widely studied in academia and industry [240], [241], [242], [243], [244], [245], [246], [247], [248]. At the same time, the development of DRL has brought new opportunities for resource allocation under CETCN.

With the explosive growth of mobile data, centralized cloud computing models have become an obstacle to support the continuous exchange of information generated by large-scale IoT devices. By introducing edge computing and software defined networks, these problems can be effectively alleviated. However, cloud edge collaborative networks still face with problems such as low utilization of edge resources and terminal energy. Therefore, the authors in [249] propose a joint communication and computing resource allocation mechanism based on reinforcement learning to optimize processing latency under energy constraints. The simulation results show that this scheme can minimize long-term average energy consumption while maintaining low latency. In Edge-IoT networks, devices are interconnected and process data at the network edge and traditional resource allocation methods may struggle to provide optimal solutions. Therefore, the authors in [250] propose an enhanced Q-learning scheme to dynamically adapt resource allocation decisions over time. This adaptability enables the network to respond changing conditions and varying device requirements effectively. By employing an online Q-learning approach, the scheme can continuously learn and refine resource allocation strategies based on real-time observations and interactions within the network. This dynamic optimization approach can achieve a balance between satisfying the individual needs of devices and maintaining fairness across the network. Experimental results indicate that the proposed scheme can improve the utility of applications while also ensuring a fair distribution of resources. In [251], the authors propose a novel approach to optimize resources in a terminal-edge-cloud collaborative computing system. The key purpose of the proposed approach is to achieve efficient task offloading and resource allocation strategy. By using Q-learning, the system continuously refines its decision-making process through taking computational capabilities, network conditions, and energy constraints into account. This dynamic optimization seeks to improve overall system performance and resource utilization. Simulation results indicate the proposed method outperforms traditional methods in terms of task completion time and energy consumption.

The authors in [252] address the challenges of optimizing computation offloading and resource allocation in a heterogeneous network consisting of both cloud and edge computing

resources. In the proposed cloud-edge heterogeneous network, devices can offload computation tasks either to centralized cloud servers or to edge nodes located closer to the user. This work focuses on efficiently distributing computation tasks among available resources, and the central objective is to minimize task completion time and energy consumption. Experimental results indicate that the proposed scheme can enhance overall network performance, reduce latency, and improve resource utilization in cloud-edge computing scenarios. In [253], a network resource allocation optimization problem is proposed in a cooperative UAV-assisted edge computing framework. The key purpose of the proposed strategy is to maximize the utilization of available resources while minimizing latency and energy consumption. Additionally, the paper also outlines strategies for enhancing communication and coordination among UAVs to ensure efficient task execution and data processing. The simulation results indicate that the proposed algorithm outperforms other baseline solutions. In [254], the authors address the challenges of content distribution in F-RANs by leveraging deep reinforcement learning. The proposed approach involves training DQN algorithm to make resource allocation decisions that maximizing the efficiency of content distribution and resource utilization while minimizing latency.

With the continuous growth of devices, slicing, and various wireless access methods, the optimal offloading problem has become increasingly complex. In [255], the author considers binary offloading and partial offloading problem in MEC system. Through utilizing stochastic game theory, the dynamic task offloading problem is modelled as a game between mobile users, with the aim of maximizing bit rate while minimizing energy consumption. At the same time, in order to alleviate the problem of overly optimistic estimation of the Q function, the authors adopt the DDQN algorithm. Through utilizing the DDQN algorithm, the optimal offloading decision can be obtained. The authors in [256] propose a collaborative cloud-edge computing framework to provide low latency services and alleviate the network congestion. Firstly, in order to minimize the weighted sum cost of the system, they consider the resource allocation problem under the time-varying channel states, bandwidth constraints, and random load arrival constraints. Then, in order to minimize the latency and maximize the bandwidth utilization, the authors model the resource allocation problem as an MDP and propose a DRL based algorithm to solve the optimization problem. In addition, the novelty of this article lies in adopting circular encryption method to solve the security issue of data leakage. Finally, extensive experiment results indicate the scheme can reduce the average service latency compared to baseline solutions. Due to the severe network congestion caused by offloading a large amount of training data to cloud servers, MEC can serve as a supplement of cloud computing to reduce service latency caused by cloud computing. Therefore, the authors in [257] propose an intelligent resource allocation model based on DDQN to solve the problem of limited resources and complex and ever-changing environment of MEC. In this model, they creatively introduce a federated learning framework to reduce the amount of data uploaded over wireless channels and

protect data privacy. The experimental results show that this scheme outperforms traditional methods in minimizing system energy consumption and service delay.

Although MEC can improve the service experience of users, its computing offload still faces many problems, such as security and privacy issues, collaborative computing issues, and dynamic optimization issues. Therefore, in order to address the above issues, the author in [258] proposes a blockchain enabled computing offloading scheme, where blockchain technology is used to ensure the reliability and irreversibility of data in MEC systems. Then, through optimizing offloading decisions, power allocation, and block size, the throughput of the MEC system can be maximized. In addition, considering the fading characteristics of wireless channels, the author transforms the above optimization problem to an MDP and proposes an algorithm based on A3C to solve the MDP problem. The final results indicate that this method outperforms existing schemes in terms of convergence speed and total rewards. In order to satisfy the extreme strict demands of users for low latency services in the Internet of Vehicles (IoV), the authors in [259] propose a learning based mobile vehicle service framework. They model the dynamic changing process of the vehicle network as a Markov process, and propose a high performance A3C algorithm to solve complex resource allocation problems. This method uses SDN and ICN to dynamically orchestrate communication resources and computation resource to enhance the performance of wireless network. Finally, a large number of simulation experiments have shown that the algorithm converges quickly and improves the total compensation of network operators.

In [260], the authors consider a simple task offloading model. In this model, both task volume and time constraints need to be considered simultaneously. In order to handle these tasks reasonably, the authors propose a cloud edge collaborative multi-resource model to handle resource constrained large-scale user requests. Meanwhile, due to the inability of traditional discrete space based methods to scale well in complex environments, they propose a DDPG algorithm for continuous action space to solve the multi-resource allocation problem. In this algorithm, the authors use convolutional residual blocks, gated recursive units, and attention layers to replace fully connected networks to extract features. The simulation results show that the proposed algorithm can reduce operating costs, improve service quality, and improve the convergence speed of the algorithm. Unlike the above work, the authors in [261] consider the joint optimization of task offloading and resource allocation to minimize the long-term latency and energy consumption of mobile devices. Due to the differences in decision-making between task offloading and resource allocation, the author first used the A2C algorithm to solve the task offloading problem, and then used the DDPG algorithm to solve the resource allocation problem. By alternately using these two algorithms, the optimal solution can be effectively obtained. Finally, experiment results the performance of the DDPG algorithm is outperform other existing resource allocation schemes. The authors in [262] consider the resource allocation problem of communication and computing fusion in both single and multi-base station

scenarios. In single base station scenario, an optimization framework based on DRL is proposed to synergistically optimize task scheduling, transmission power, and computing power. In multi-base station scenario, in order to reduce interference between base stations, a MADDPG algorithm is proposed to minimize total energy consumption and latency. The simulation results show that the proposed method outperforms the baseline scheme while considering interference, transmission power, and the computing power of the base station.

The authors in [263] investigate the joint offloading and resource allocation problem of heterogeneous tasks of multiple fog nodes. The heterogeneity of tasks includes two aspects: heterogeneity of service quality requirements and heterogeneity of resource requirements. Then, this problem is described as a stochastic game with partially observable characteristics, where each fog node can cooperate to maximize rewards, and at the same time, each fog node can only access local observations. Compared with the joint action learning scheme, this scheme greatly reduces feedback and memory consumption. At the same time, in order to solve the problem of partial observability, the author adopted a deep recursive Q-network to approximate the optimal value function. Finally, the numerical results indicate that the proposed algorithm has a higher average success rate and lower average overflow compared to the baseline scheme. Due to the workload of users accessing edge networks and the uncertainty of available resources of edge nodes, the optimal resource allocation problem in edge networks faces significant challenges. The author in [264] proposes a computational offloading and resource allocation scheme to minimize computational energy consumption. Unlike the work in [263], the author considers nonoffloadable tasks and offloadable tasks. Compared to nonoffloadable tasks, offloadable tasks can be processed at multiple nodes in the network. Due to the uncertainty of edge node resources, the author models the task offloading problem as a stochastic game with the goal of maximizing the overall performance of the system. Compared with traditional methods, the method proposed by the author can effectively reduce computational energy consumption while ensuring convergence. In [265], the author considers the issue of using DRL to achieve task offloading decisions while improving energy efficiency in green MEC scenarios. Firstly, the author decouples the problem into two sub problems: subcarrier allocation problem and power control problem. In order to improve practicality, each MEC server can only obtain partial information of the heterogeneous network. Then an optimization algorithm based on the PPO framework is proposed to achieve efficient resource allocation. The final simulation results show that the algorithm outperforms the baseline scheme in terms of effectiveness, robustness, and scalability. Different from above works, the authors in [266] consider the sequential decision-making problem of user association and resource allocation in MEC scenarios to maximize the average completion rate of tasks with interdependent relationships. Due to the fact that edge servers can only observe the computing tasks, communication, and computing resources within their coverage range, the author models this problem as a Dec-POMDP. In this

model, edge servers make decentralized decisions based on user association, resource management and local information of their network status. Considering the continuous property of variables, a multi-agent DDPG algorithm is developed to solve this problem. The simulation results show that this algorithm can effectively improve the completion rate of tasks. The authors in [267] consider the problem of computing offloading in ultra dense networks, with the aim of minimizing the computational cost of the system while considering user energy consumption and latency. The novelty of this article lies in the combination of DRL and mean field game theory, which not only effectively solves the problem of traditional methods difficult to model complex environments, but also solves the decision-making behavior of dense users, thus reduces the complexity of the problem.

D. Lesson Learned and Future Directions

It has been shown that in recent years, ML has been proposed to optimize the task offloading and resource allocation decisions, especially the DRL. Since DRL can provide a better approach for the large dimension, distributed and dynamic aspect of CETCN, it plays a more and more critical role in improving the effectiveness of the CETCN. Indeed, DRL provides a better solution in a short time, which satisfies the requirements of time-intensive applications. Table III summarized in this survey indicates that a lot of efforts has been put to investigate the task offloading and resource allocation issues. Moreover, most studies focus on cost, latency, and energy consumption but have not investigated them in a unified framework. Additionally, different optimization models focus on different objectives and it is necessary to provide a solution that optimize as many objectives as possible and then provide a unified criterion to measure the performance in the CETCN system.

It is essential to note that there are still several open issues in the task offloading and resource allocation problems that need to be investigated by the research community. Indeed, it is a challenge to allocate resources at a lower cost under the constraints of delay and energy. Furthermore, mobile devices may lose their connectivity because of the users' mobility, especially in the intelligent transportation system. Therefore, the offloading scheme should design fault-tolerant mechanisms to resend the lost data or design flexible mobility management mechanisms to adapt to the occurrence of the mistakes. Last but not least, privacy protection issue is still the focus of users' attention. The privacy protection mechanism based on edge computing is not mature, mainly due to the limited computing power of edge nodes. Therefore, cloud edge collaborative privacy protection mechanism still needs to be carefully designed.

We list the several most leading research directions in the task offloading and resource allocation problems as follows:

- **Dynamic Workloads:** Plenty of researches have considered the static scenarios of offloading. Due to the dynamic of the device mobility and channel state, more sophisticated offloading strategies need to be designed to adapt to the dynamic characteristics.

- **Real-world Trace:** Most proposed task offloading and resource allocation schemes are demonstrated based on numerical and simulation analysis. Many of them simplify the models of the real world, for example, simplified wireless channel models [268], [269], [270]. Hence, it is necessary to consider more practical and realistic scenarios evaluated by real-world traces such as [253].
- **Federated Learning Offloading:** Federated learning allows devices to collaboratively train models while keeping data localized. Future research may focus on enhancing federated learning techniques for offloading and aggregating model updates efficiently across edge devices, ensuring privacy and accuracy.
- **Green and Energy-Efficient Offloading:** Given the increasing importance of sustainability, future directions might involve incorporating energy-aware algorithms that optimize offloading decisions to reduce the carbon footprint of computing tasks.
- **6G and Network Slicing:** The rollout of 6G networks and network slicing capabilities can enable more dynamic and fine-grained control over network resources. Future research might explore how to exploit these capabilities for efficient task offloading based on real-time network conditions.
- **Security and Privacy-Enhanced Offloading:** With growing concerns about data privacy and security, future research should focus on developing secure protocols for offloading sensitive tasks and data, ensuring end-to-end encryption and authentication.
- **Autonomous Resource Management:** Develop self-adaptive algorithms that enable edge devices and cloud resources to autonomously allocate and adjust resources based on real-time data, workload changes, and performance metrics.
- **Predictive Analytics:** Utilize predictive analytics and machine learning to forecast resource demands and trends, allowing for proactive resource allocation adjustments and optimization.
- **Quality of Experience (QoE) Optimization:** Shift focus from mere resource allocation to optimizing the overall user experience by considering subjective factors like user satisfaction, application responsiveness, and perceived quality.
- **Multi-Objective Resource Allocation:** Consider multiple conflicting objectives, such as performance, energy efficiency, cost, and user satisfaction, and develop algorithms that find Pareto-optimal solutions for diverse use cases.

VI. COLLABORATIVE CACHING FOR AI-ENABLED CLOUD-EDGE-TERMINAL COLLABORATIVE NETWORK

Caching at the CETCN has attracted increasing attention from academia and industry due to its advantages in reducing content or service acquisition delay and relieving the traffic pressure of the core network. Hence, in this section, we firstly introduce caching model in the CETCN in Section VI-A. In Section VI-B, we analyse the MEC from the perspective of computer science. Then, in Section VI-C, we investigate the

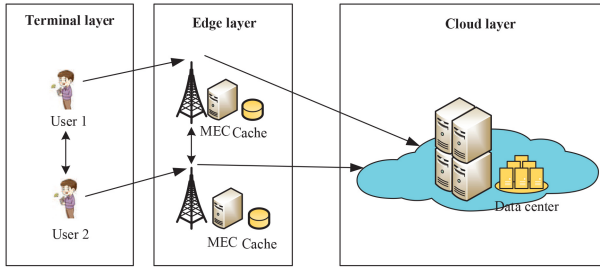


Fig. 8. A toy example of cooperative caching.

works related to content caching and service caching in the CETCN. Finally, we point out several remaining challenges and possible future directions in Section VI-D.

A. Collaborative Caching Model for CETCN

As shown in Fig. 8, the toy example of cooperative caching is consisted terminal layer with two users, edge layer with two base stations which equipped with caching capacity and cloud layer with data center. MEC servers in the edge layer can cooperative with each other to caching content and communicate with each other to realize content sharing.

As depicted in Fig. 8, when User 1 requests a content of size c or generates a task (the task should be processed by a service or an application which cached at MEC server or cloud center) which denoted as $\{s, f, \tau, c\}$ in Section IV, due to the limited caching capacity of MEC servers, the process of content retrieving can be described as follows:

- The first step: The user 1 generate a content or service request to the local MEC server, if the local MEC server has the content or can process the service at the local MEC server, then, the local MEC server transmits the content or the computing result to the user 1.
- The second step: If the local MEC server does not cache the target content or the service, then the local MEC server requests the content from other MEC servers in the cooperative domain, and then the local MEC server transmits the content to the user.
- The third step: If the edge layer does not have the content or the service, the local MEC will request the content from the data center, and then transmits the content to the user through backhaul link.

It is worth noting that collaboration between users and between edge layers is not a necessary option in the actual cloud edge collaboration system.

Due to the limited caching capacity of MEC servers, for content caching which size is c or service placement which denoted by $\{s, f, \tau, c\}$, it must satisfy the caching capacity constraint. If we assume the caching capacity of MEC server i is C_i , then the maximum number of cached content or service should satisfy $\sum_i^N c_i \leq C_i$. On the one hand, the caching placement strategy need to be designed reasonably to minimize the content retrieving latency or minimize the energy consumption. On the other hand, due to caching placement problem is often NP-hard, and the traditional method often time consuming, but lucky, the DRL method can overcome this shortcoming of the traditional methods. In the following,

we will do a literature review about cooperative caching using DRL in recent years.

B. Analyse MEC From Computer Science Perspective

Due to the fact that content caching works in application layer, therefore, in the following, we will analyze MEC from the perspective of application layer, including MEC fundamentals, architecture, edge algorithms, access latency and response time, SDN/NFV and edge AI.

1) *MEC Fundamentals*: The core components of the MEC architecture include edge servers, access network, mobile and IoT devices and backhaul network. Edge servers are the central processing units of MEC architecture. They are located at the edge of the network, closer to end-user devices compared with cloud server. These servers host virtual machines (VMs) or containers that run applications, services, and functions [271]. Edge servers enable real-time data processing, computation, and storage, and have quicker response time and shorter transmission distance [272]. The access network refers to the connectivity infrastructure that connects end-user devices (e.g., mobile phones, IoT devices) to edge servers. The access network ensures seamless communication between devices and edge resources [273]. Mobile devices, such as smartphones and tablets, as well as Internet of Things (IoT) devices, generate and consume data. MEC architecture places edge servers in proximity to these devices, enabling faster data processing and real-time interactions. The backhaul network connects edge servers to centralized cloud data centers. It facilitates data synchronization, backup, and offloading between the edge and the core cloud infrastructure. The backhaul network ensures that data is appropriately managed and stored.

2) *Architecture*: Through combining edge servers, access network, mobile and IoT devices and backhaul network, the MEC can provide services to users. Taking content caching as an example. Firstly, when a user on a mobile device requests content, such as a Web page or a video, the request is sent through the access network. Then, the access network routes the content request to the nearest edge server based on factors such as proximity, load, and caching policies. Thirdly, The edge server evaluates whether the requested content is already cached. If the content is present in the cache and is still valid, the edge server can directly serve the cached content to the mobile device, reducing the need to fetch it from a distant data center. This reduces latency and accelerates content delivery. Finally, the edge server periodically updates and manages the cache to ensure that the content remains fresh and relevant. It can evict outdated content or content with lower demand to make space for new content. Last but not least, backhaul connection is optional, to be specific, periodically or under specific conditions, the edge server may synchronize the cached content with the central cloud data center over the backhaul connection. This ensures that the cached content remains consistent across the MEC environment.

3) *Edge Algorithms*: Edge caching involves storing frequently accessed content or data closer to end-users at the edge of the network to reduce latency and improve content delivery. The common edge caching algorithms include least

recently used (LRU), least frequently used (LFU), random replacement, first-in-first-out (FIFO), least recently used with frequency count (LFU-2), greedy dual-size frequency (GDSF), greedy dual-size frequency with aging (GDSF-A), weighted LRU (W-LRU) [274], [275]. LRU is a simple algorithm that removes the least recently accessed content from the cache when it becomes full. It assumes that recently accessed content is more likely to be accessed again soon. LFU removes the least frequently accessed content from the cache. It assumes that less visited content is less likely to be accessed in the future. FIFO removes the oldest content that has been added to the cache. It's a straightforward algorithm but may not consider content popularity. LFU-2 combines elements of LRU and LFU. It tracks both recency and frequency of user's access, evicting content that is both infrequently accessed and was accessed a long time ago. GDSF considers both size and frequency of content access when making eviction decisions. It aims to optimize cache hit rates based on both factors. W-LRU assigns weights to each content based on factors like popularity or importance. When evicting content, it considers both recency and the assigned weight.

4) *Assess Latency and Response Time*: Taking content caching as an example, edge servers can cache frequently requested content, such as videos, images, or Web pages. When a device requests cached content, it can quickly retrieve it from nearby edge servers, thereby reducing the need for long-distance data transmission from the remote cloud server [276], [277].

5) *SDN/NFV*: SDN makes the network programmable by separating the control plane and data plane. The SDN control plane is represented by software called the SDN controller, which is responsible for determining the processing of underlying data traffic. The data plane, as a network device, is responsible for forwarding data according to a set of rules. The SDN controller allows the creation and management of these rules through a programming interface in the northbound interface. The separation of data plane and control plane makes network policy execution simple, flexible, and easy to configure and develop [278]. Virtualization is a foundational technology in MEC, which can enable the efficient utilization of resources, dynamic scaling, and the seamless deployment of services at the edge of the network [279], [280]. It forms the basis for achieving the flexibility and low-latency capabilities of MEC. To be specific, NFV abstracts storage resources as virtual entities that can be allocated to different services or applications at the network edge, and this abstraction can better utilize hardware resources for dynamic allocation.

6) *Edge AI*: Integrating AI into edge caching can further enhance the ability of MEC. Edge based artificial intelligence (AI) and machine learning (ML) algorithms and frameworks are crucial for directly implementing intelligent caching decisions at network edges [281], [282]. These algorithms and frameworks allow devices and edge servers to process data locally, thereby reducing the need to transfer data to centralized data centers. Common edge based AI/ML algorithms and frameworks include reinforcement learning, LSTM (Long Short-Term Memory) for temporal patterns, collaborative filtering, content popularity prediction and so on.

C. Literature Review

In CETCN, many scholars and researchers have always been concerned about caching. With the growth of mobile data traffic, caching content at the network edge is a promising way to release traffic congestion. Many studies have shown that 80% of the traffic consumption in the network comes from users' repeated acquisition of popular content, and popular content often accounts for only a small part of all contents. Therefore, caching popular content in advance can effectively relieve the tremendous traffic pressure on the backhaul link. At the same time, due to the limited storage resources of edge devices, how to implement distributed collaborative caching in CETCN is of great concern.

In recent years, reinforcement learning has shown its great potential to learn content popularity in complex environments. By accurately expressing content popularity, better caching decisions can be made. Therefore, using reinforcement learning to make content caching decisions is of increasing interest to researchers compared to traditional methods [283], [284], [285]. Likewise, at the end of this section, we summarize the literature on DRL-based cloud-edge-terminal collaborative caching in Table VII.

In order to alleviate the impact of the increasing in cache types on cache performance, the authors in [286] propose a collaborative caching mechanism that combines remote radio heads and baseband units. This mechanism is based on the high flexibility and scalability of SDN. Then, due to the difficulty of considering all environmental factors in traditional rule-based and heuristic methods, a Q-learning based method has been proposed for designing caching mechanisms. Finally, the author proposes an unit cache capacity hit efficiency to evaluate cache performance. Extensive experimental results indicate that the Q-learning based caching strategy can improve the system performance. In [287], the authors propose a novel storage architecture. The architecture consists of edge servers and edge users, and the edge user is composed of data producer and data consumer. The edge server includes a master container and several data containers. The master container manages the data containers and assigns storage tasks to data containers. To storage data at the edge server efficiently, the master container first identifies the data's popularity, then utilize the Q-learning algorithm to determine which place to store the data. Secondly, the master node determines which data should be updated through a scalable recommendation algorithm. Finally, the maser container will clean its redundant storage content when the data containers reach its capacity. Extensive experimental results demonstrate that the proposed intelligent storage architecture is more efficient than the comparison schemes. In [288], the author considers the joint allocation of cache, computing, and communication resources. Unlike the above work, this work not only considers cloud edge collaboration, but also considers the impact of traffic distribution on the popularity of cached content. The author uses queuing theory to model the content distribution process as a delay minimization problem, and then uses Q-learning to make the best caching decision to improve network content delivery and reduce latency.

TABLE VII
SUMMARY OF CLOUD-NETWORK-EDGE COLLABORATIVE CACHING BASED ON AI

reference	Proposed solution	Cooperation type	Objective	Evaluation method	Model
[286]	Q-Learning	Edge-cloud	Reduce the traffic and transmission delay	Simulation analysis	MDP
[287]	Q-Learning	Terminal-edge-cloud	Maximize the capacity of edge servers	Simulation analysis	MDP
[288]	Q-Learning	Edge-cloud	Minimize the delay	Simulation analysis	MDP
[289]	DQN	Terminal-edge-cloud	Maximize the hit ratio	Simulation analysis	MDP
[290]	DQN	Edge-edge	Minimize the average download delay	Numerical analysis	MDP
[291]	DQN	Edge-edge	Minimize the system delay	Simulation Analysis	MDP
[292]	DDQN	Terminal-edge-cloud	Maximize the long-term profit	Simulation analysis	MDP
[293]	DDQN	Edge-cloud	Maximize the long-term average content fetching delay	Numerical analysis	MDP
[294]	DDQN	Edge-cloud	Improve the self-adaptive ability	Simulation analysis	MDP
[295]	AC	Edge-cloud	Maximize the hit ratio	Simulation analysis	MDP
[296]	AC	Terminal-edge-cloud	Minimize the redundant content and transmission delay	Simulation analysis	MDP
[297]	DDPG	Edge-cloud	Maximize the transmission delay	Simulation analysis	MDP
[298]	DDPG	Edge-cloud	Maximize the saved cost	Simulation analysis	MDP
[299]	DDPG	Edge-cloud	Minimize the access delay and traffic cost	Real-world analysis	MDP
[300]	DQN	Terminal-edge-cloud	Minimize the access delay and traffic cost	Simulation analysis	POMDP
[301]	MAAC	Terminal-edge-cloud	Minimize the access delay and traffic cost	Simulation analysis	POMDP
[302]	MADDPG	Terminal-edge-cloud	Minimize the execution latency	Simulation analysis	Dec-POMDP

In order to meet the user's demand for multimedia service access, In [289], the authors introduce D2D communication technology to the edge caching problem. Specifically, the authors propose a D2D-assisted collaborative caching framework in the heterogeneous network where flexible cooperation can be carried out between user devices, edge servers and cloud servers. Secondly, through modeling the node selection and content placement problem as an MDP, the authors propose a deep Q-learning method to solve the cooperative caching problem. Finally, extensive simulation results demonstrate the effectiveness of the method at reducing access latency of content and improving hit rate. In order to reduce the transmission delay and traffic pressure of the backbone network, the author proposes a fog node collaborative caching strategy based on DQN in [290]. Firstly, a topic model is used to predict user preferences to obtain the expected content popularity. Secondly, considering the coupling of variables in the optimization problem, the predicted content popularity results are combined with original content popularity to obtain the optimal content caching strategy. Finally, numerical results indicate that the proposed caching scheme can reduce the average delay. The author in [291] proposes a low latency edge caching method to reduce user access latency and effectively cache content in edge networks. Firstly, the author models the cache problem between base stations as a delay minimization problem and transforms it into a maximum cache reward problem, through utilizing the greedy algorithm of origin dual interior points to obtain the strategy for the cache problem. Secondly, in order to improve service quality and balance communication and migration costs, a joint communication

and task migration cost model is established and a task transfer scheme based on reinforcement learning is proposed. Finally, a large number of comparisons are conducted to verify that the proposed caching method can reduce latency, and at the same time, the proposed migration algorithm can reduce communication overhead.

Ren et al. [292] consider the allocation of computation and cache resources between multi-level edge nodes. Users can offload computing tasks to edge servers and optimize the task offloading and resource allocation strategies of edge servers to meet their needs for low latency services. The complexity of this problem is that when computing, caching and communication resource allocation are considered at the same time, the complexity of edge computing systems will become very high. In order to solve the complex resource allocation problem of edge servers, the author proposes a DDQN algorithm, which can achieve fast decision-making while maximize long-term benefits. The experimental results show that the algorithm can effectively allocate resources in edge networks. In [293], the authors propose a content caching and request routing scheme at the network edge to meet the growing demand for emerging services and propose a DDQN algorithm for cooperative content caching to minimize the content fetching delay. The difference from the above work is that the proposed method does not require any prior knowledge of the popularity distribution of the content. Simulation results demonstrate that the proposed algorithm is superior than other baseline caching algorithms. Centralized optimization methods based on learning will consume excessively network resources during training and data transmission. To address

this challenge, the authors in [294] propose a new federated caching framework based on DRL. This framework takes the first round of training parameters of the base station as the initial input parameters for local training, enabling the base station to collaborate in learning shared prediction models. Then, the terminal devices upload the nearly optimal local parameters to the base station to participate in the next round of global training. By introducing federated learning, data is decoupled from the training process to achieve rapid decision-making. The simulation results show that the proposed collaborative caching framework can reduce average latency, alleviate traffic pressure on backbone networks, and improve cache hit rates.

In [295], the authors consider the cooperative content replacement problem in a realistic scenario where the content popularity is unknown to the edge servers, in order to obtain the optimal caching replacement strategy in this scenario, they design a content replacement scheme based on multi-agent AC algorithm where the reward of the agent is the saved delay, extensive numerical results indicate that the proposed cooperative content replacement algorithm can improve the cache hit ratio compared with existing schemes. Mobile edge caching can reduce access latency and redundant traffic transmission through storing content at network edges. Unlike the above work, the authors in [296] consider the complex and ever-changing edge network environment, such as a large number of dynamic and diverse request patterns, and they also consider the high similarity of requests between adjacent edge nodes. In this context, they propose an intelligent edge caching framework that can provide a diverse and distributed caching environment to reduce access latency and traffic costs. At the same time, through utilizing MADRL, each node can learn the optimal caching strategy. Finally, the real-driven evaluation demonstrates that the proposed scheme outperforms other state-of-the-art caching solutions.

In [297], the authors propose an edge caching scheme based on DRL. First, the edge server determines its cache operation according to its cache state, and then the centralized server evaluates the edge server's cache decision to guide the edge server's subsequent decisions. On this basis, considering a scenario where the size of actual and reasonable data items is inconsistent, the authors propose a new multi-agent role-criticism caching algorithm. Extensive experimental results indicate that the algorithm can promote collaboration between edge servers and improve the system hit ratio. Aiming to tackle the issue of redundancy of caching content in different regions and retransmission delays on the communication links caused by the mobility of users, Xu et al. [298] propose a cooperative caching scheme based on DRL, which can realize the collaborative caching between edge servers and cloud server. Specifically, first of all, the predictive model is adopted to describe the users' content preference and movement trajectory. Secondly, the DRL algorithm is used to reduce the rate of redundant content and transmission delay. The author in [299] proposes a method that utilizes the cache capacity of base stations to minimize the average latency of user request content and the energy consumption of the

network. Specifically, the delay of user requests for content is mainly composed of the time spent on wireless and fiber optic links. Energy consumption mainly comes from the forwarding process of content between base stations and servers. Due to the unpredictability of network state and the huge network dimension, the author adopts the DDPG algorithm to learn the optimal caching strategy through interaction with the environment. The simulation results show that the proposed caching framework can improve the cache hit rate and reduce the energy consumption of the network.

In order to satisfy the users' requirements for latency-sensitive and computation-intensive services in IoT-enabled MEC system, the authors in [300] propose a cache assisted task offloading framework that enables joint optimization of communication, caching, and computing resources for the IoT-enabled MEC system. Specifically, first of all, the application services are first pre-stored on edge servers, and thus reduce the latency cost of obtaining and initializing services. Secondly, due to the fact that each base station intelligently observes partial information of the environment and in order to minimize the latency of computation offloading and resource allocation, the author models the multi-agent decision-making problem as a POMDP. Then, they propose a graph DRL algorithm to solve the problem. This algorithm utilizes a graph neural network to obtain the relationships between nodes and inputs these relationships into a deep Q network for resource allocation decisions. Finally, the simulation results indicate that the algorithm can effectively reduce latency by optimizing resource allocation. In [301], the author models the collaborative caching problem as a POMDP to jointly optimize the cost for users to obtain content from local servers, nearby servers, and remote servers. Firstly, the author designs a multi-agent actor-critic framework, in which agents reduce data interaction between agents by sharing actions. Secondly, considering the variation of content popularity over time, the author proposes a popularity estimation method based on variational recurrent neural networks, which can learn content popularity without understanding user preferences. At the same time, due to the mobility of the terminal devices, the author utilizes LSTM to simulate the user's access status. The simulation results indicate that the proposed scheme can improve the benefits of BSs. The author in [302] proposes a collaborative task offloading, service caching, and transmission power allocation problem to reduce terminal computing latency and energy consumption. Unlike the above work, the author considers that obtaining all information about the network in a decentralized MEC network is impractical. Therefore, the author models the resource allocation problem as a Dec-POMDP problem, where each edge base station can cooperate to obtain the optimal solution in situations where only partial information about the network can be obtained for a typical edge server. In addition, in order to better learn cooperative information, the author introduces graph neural networks into the multi-agent reinforcement learning framework, allowing agents to selectively obtain information from other agents. Extensive simulation results demonstrate the proposed scheme outperforms other solutions at average reward, cache hit ratio, and offloading cost.

D. Lesson Learned and Future Directions

In this survey, we investigate the issue of cooperative caching in the CETCN. Taking advantage of edge computing, caching popular content to edge nodes can provide users with low-latency services. However, due to the limited capacity of edge nodes, designing a reasonable caching strategy has become the focus of academia. The traditional method of solving integer programming is computationally intensive and difficult to obtain the optimal solution, but luckily, with the further development of artificial intelligence, DRL has become a promising method for solving the problem of cooperative caching. However, there are still existing a few challenging issues that need to be solved in cooperative content caching domain. We summarize several key points as follows.

- **Towards privacy preserving intelligent caching:** Similarly, when dealing with the edge caching problem, we often assume that the entire edge network knows the content requests of all users, and uses the request information to optimize collaborative caching. However, due to users' privacy requirements, users often do not want to disclose their content preferences. Therefore, how to design a reasonable cooperative caching strategy when only knowing the content requested by the edge node itself will also become one of the future research directions.
- **Data Placement and Replication:** Determining which data should be cached at which locations and how to replicate data efficiently is a complex task. Making the right decisions about data placement and replication is crucial to ensure that frequently accessed data is available at the right location without causing excessive data duplication.
- **Dynamic Workload Variation:** The workload on the network can vary significantly over time, and cooperative caching systems need to adapt to these variations. Balancing the cache content and placement in response to changing user demands and network conditions is a non-trivial task.
- **Heterogeneous Devices:** Edge devices in a cooperative caching network can be heterogeneous in terms of their capabilities, storage capacities, and communication bandwidth. Designing caching algorithms that can efficiently utilize the varying resources of these devices is a challenge.
- **Cache Consistency and Synchronization:** Maintaining cache consistency across distributed caches is important to avoid serving stale or inconsistent data to users. Implementing efficient cache synchronization protocols that minimize overhead and guarantee data consistency is a challenge.
- **Caching from an economic vision:** Since content caching involves the interests of multiple parties, equipment providers need to provide hardware facilities that can be utilized to cache content, content providers need to cache some content in hardware system according to the popularity of these content, and operators need to provide users with necessary connection services. Users get the content they need by paying a certain fee. Therefore, how

to reasonably divide the benefits from the users will also be one of the future possible research directions.

VII. MOBILITY MANAGEMENT FOR AI-ENABLED CLOUD-EDGE-TERMINAL COLLABORATIVE NETWORK

The requirements of emerging applications for high throughput and low latency make mobility management more and more challenging. In the CETCN, the intensive deployment of edge base stations enables users to access multiple base stations at the same time. Although it increases the complexity of connection, the flexibility of user mobility is also increased through multi-cell connection. Therefore, in this section, we first discuss the existing mobility management standards in Section VII-A. Then, we briefly describe an example of mobility management problem in subsection in Section VII-B. Then, Section VII-C reviews the efforts of researchers that have been put in this domain. Finally, in Section VII-D, we summarize the mobility management issues in the CETCN and point out the possible research directions in the future.

A. Existing Mobility Management Scheme in 3GPP standards

In 5G, the mobility management scheme defined by the 3rd Generation Partnership Project (3GPP) standards includes several procedures and functions that enable seamless connectivity for user devices as they move within the network. The mobility management scheme in 5G builds upon the concepts and enhancements introduced in previous generations while also incorporating new features to improve performance, efficiency, and flexibility. Some key elements of the 5G mobility management scheme are as follows [303], [304], [305]:

- **Intra-RAT (Radio Access Technology) Handover:** Inter-RAT handover refers to the process of transferring an ongoing data session from one cell to another cell within the same type of radio network. In other words, it is a handover that occurs between two cells that are part of the same radio technology, such as one 5G NR to another 5G NR cell. Intra-RAT handover schemes include hard handover, soft handover and softer handover.
- **Inter-RAT Handover:** Inter-RAT handover schemes include inter-frequency handover and 4G to 5G handover and vice versa. 5G networks often coexist with previous-generation networks like LTE or even older technologies. Inter-RAT handover procedures allow user devices to smoothly transition between 5G and these legacy networks, ensuring continuous connectivity and service availability.
- **Idle Mode Mobility:** Idle-Mode mobility schemes include cell reselection and tracking area update. In idle mode, when a user device is not actively using the network, mobility management procedures are in place to manage tracking area updating to ensure the device can be reached efficiently when needed.
- **Xn-based Handover:** Xn-based Handover is used within the 5G New Radio (NR) architecture and involves direct communication between two gNodeBs (5G base station) using Xn interface.

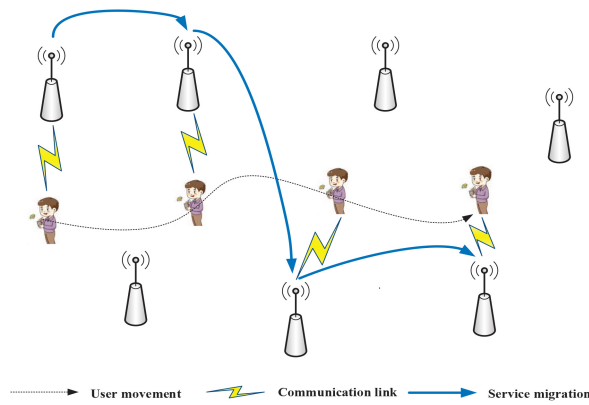


Fig. 9. A toy example of mobility management.

- **NG-based Handover:** Xn-based handover enables seamless transmission for a User Equipment (UE) between different gNodeBs.
- **Dual Connectivity:** Devices can maintain simultaneous connections with two different cells (usually one for the control plane and one for the user plane) to ensure smooth handover and better reliability.
- **Network Slicing-based Mobility Management:** With the introduction of network slicing in 5G, multiple virtual networks with different characteristics can coexist over the same physical infrastructure. Mobility management schemes must ensure that devices can smoothly move between different network slices while maintaining service continuity.
- **Hierarchical Mobility Management:** In this scheme, different layers of hierarchy in the network handle different aspects of mobility. For example, local mobility could be handled by a micro-controller closer to the edge, while broader events are managed by a central entity.
- **IP-based Mobility Management:** Mobile IP and its variants like Proxy Mobile IP can be used for IP session continuity during the handover.

In addition, the mentioned mobility management schemes can be adapted or combined to accommodate specific use cases or network conditions.

B. Challenges of Mobility Management

Mobility is an inherent attribute of many emerging applications, when a user is watching a video on a mobile device, such as a phone, and is moving around as shown in Fig. 9, several mobility management challenges can arise due to the changing network conditions and the requirements of maintaining a seamless user experience. We list specific technical problems and challenges of mobility management problems as follows [306], [307], [308]:

- **Handover Disruption:** As the user moves, the device might need to switch between different cells or access points. If the handover process is not smooth and quick enough, it can lead to disruptions in the video stream, causing buffering or even disconnecting the session.
- **Handover Optimization:** Seamless handovers between different access points and network layers are essential

to maintain a stable connection. However, in collaborative networks, determining the optimal handover decision considering factors like signal strength, available resources, and user mobility can be complex.

- **Resource Management:** Efficiently managing and allocating resources across terminal, edge, and cloud nodes is crucial for performance optimization. This includes managing computing resources, storage capacity, and network bandwidth, and ensuring proper load balancing.
- **Latency and Quality of Service (QoS):** Mobility can introduce latency variations and fluctuations in network quality. Video streaming requires consistent bandwidth and low latency to provide a good quality experience. If the network cannot maintain adequate QoS during mobility, the video quality may suffer.
- **Energy Efficiency:** Terminal devices often have limited battery capacity. Effective mobility management should aim to minimize energy consumption during handovers and data offloading to extend the battery life of mobile devices.
- **Packet Loss and Jitter:** Movement between cells can lead to packet loss and jitter, which can further degrade the video streaming experience. Lost packets or inconsistent packet arrival times can result in pixelation, freezing, or audio-video sync issues.
- **Load Balancing:** When many users are simultaneously streaming videos or performing data-intensive tasks, the network needs to balance the load across different cells or sectors. Inefficient load balancing can lead to congestion and reduced performance.
- **Signal Strength and Coverage:** Movement can affect the signal strength and coverage. In areas with weaker signals, the video stream might degrade or pause entirely. The handover process needs to ensure that the new cell provides adequate signal strength for video streaming.
- **Interference and Heterogeneous Networks:** In urban areas or places with overlapping coverage from different technologies (e.g., 5G, Wi-Fi, LTE), interference and handover decisions become more complex. Managing handovers between different types of networks can impact video streaming.
- **Frequent Handovers:** Rapid movement or switching between cells can lead to frequent handovers, which can put additional load on the network and impact the device's battery life.
- **Context-Aware Handover:** The network needs to make intelligent decisions about when and where to perform handovers based on the user's context, application requirements, and available resources. This procedure requires sophisticated mobility management algorithms.

Mobility management issue has been extensively investigated in traditional cellular networks [309], [310], [311]. In these previous works, user mobility is modeled with connection probabilities or link reliability using information such as the user's moving speed and directions. However, this simple information is not sufficient for CETCN system due to its complex, dynamic and heterogeneous environment. Addressing these challenges requires a holistic approach,

TABLE VIII
SUMMARY OF CLOUD-NETWORK-EDGE COLLABORATIVE MOBILITY MANAGEMENT BASED ON AI

Reference	Proposed solution	Cooperation type	Objective	Evaluation method	Model
[314]	Q-Learning	Edge-cloud	Mobility management	Simulation analysis	MDP
[315]	Q-Learning	Edge-cloud	Optimize the handover	Simulation analysis	MDP
[316]	Q-Learning	Terminal-edge	Minimize the number of handoff	Simulation analysis	MDP
[317]	Q-Learning	Terminal-edge	Maximize the delay	Simulation analysis	MDP
[318]	DQN	Terminal-edge	Minimize the average delay	Simulation Analysis	MDP
[319]	DQN	Terminal-edge-cloud	Maximize the long-term profit	Numerical analysis	MDP
[320]	DQN	Terminal-edge	Handover optimization	Numerical analysis	MDP
[321]	DDQN	Terminal-edge-cloud	Minimize the delay and energy consumption	Simulation analysis	MDP
[322]	DDQN	Edge-cloud	Maximize the handover frequency	Simulation analysis	MDP
[323]	DDQN	Terminal-edge-cloud	Minimize the utility	Simulation and numerical analysis	MDP
[324]	AC	Edge-cloud	Maximize the utility	Simulation analysis	MDP
[325]	DDPG	Edge-cloud	Minimize the migration cost and service delay	Simulation analysis	MDP
[326]	DQN	Terminal-Edge	Maximie the information receive rate	Simulation analysis	POMDP

combining advanced intelligent decision-making algorithms, and effective coordination among different network layers in terminal-edge-cloud collaborative networks. Therefore, in the following, we do an investigation about mobility management using DRL to address the mentioned challenges.

C. Literature Review

Mobility management can provide users with a ubiquitous and seamless connection. Due to the dense deployment of small base stations, traditional switch control methods will lead to various issues, such as frequent handover and failed handovers. Therefore, if the mobile terminal decides to offload tasks to the CETCN, it is very important to ensure the continuity of the business.

In general, the mobility assumption based on the Markov chain model is often oversimplified in reality [312], [313]. In fact, the mobility of terminal devices in the real world is more complicated. Therefore, it is difficult to describe the mobility of terminal equipment with a standard mathematical model.

DRL is considered to be the most promising approach to solve the mobility management problem at present, since DRL can extract and identify hidden information from complex states, and perform optimal operations in a single framework. Therefore, in this survey, we mainly summarize the issues of mobility management and its DRL solutions in CETCN system. A summary is presented in Table VIII.

The authors in [314] consider the issues of switching and business migration in large-scale deployment of mobile application scenarios. Firstly, they propose a mobile user trajectory prediction method that combines Q-learning and short-term memory networks, enabling the training process of the model to be automated. Then, in order to reduce the cost of predicting user trajectories with similar movement patterns, a group user trajectory predictor is developed. Finally, in order to verify the superiority of the proposed mobility management scheme, they propose a service migration scheme based on trajectory prediction. The experimental results show that the predictor

can not only improve QoE, but also effectively reduce return flow. Ultra-dense network is a promising solution to satisfy the growing demand of users' requirements, and expand the coverage of the network, but the interference, additional unnecessary handover and signaling overhead makes the quality of services worse. Hence, in [315], the authors propose a Q-learning based handover optimization scheme to minimize the effect of frequent and unnecessary handovers. Extensive experiments show that the proposed algorithm can minimize the probability of radio link failure and maximize throughput compared to other methods. For cellular-connected unmanned aerial systems, a new switching mechanism is proposed in [316], this novel switching mechanism ensures robust wireless connectivity and mobility support for drone terminals. Dynamic optimization is performed through reinforcement learning, and experimental results demonstrate that the dynamic optimization method can significantly reduce the number of switching times and ensure effective connections. Different from [315], the authors in [317] investigate the mobility management issue in UDNs through combining task offloading process. Due to the multiple options available for providing access and computing services to users in UDNs, mobility management becomes more difficult. Therefore, the authors propose an online optimization scheme based on Q-learning. Firstly, the optimal switching decisions can be made by predicting upcoming information. Secondly, through using the switching cost of offloading tasks as a penalty, the offloading delay can be minimized. Finally, a large number of experimental results indicate that the proposed switching scheme can effectively reduce the average latency and switching rate of user computing tasks.

In [318], the authors propose a novel dynamic programming-based mobility management scheme to minimize the average delay under the energy consumption constraint, then a cooperative transmission mechanism is adopted to reduce the transmission delay. Furthermore, they propose a mobility management scheme based on DQN algorithm. Finally, the simulation results indicate that the proposed mobility management scheme can obtain optimal

delay performance and reduce the frequency of handover. In dense wireless local area networks (WLAN), traditional switching mechanisms can cause significant delays during the switching process. To address the issues of when to trigger switching and which access point to connect to in WLAN, the authors in [319] first design an SDN based WLAN framework. Secondly, a handover management scheme based on DQN is proposed, which can learn the actual user behavior and network state based on the time-varying WLAN environment. In addition, the authors also introduce CNN and RNN to extract fine-grained discriminative features during switching. Finally, the simulation results indicate that the proposed switching scheme can effectively improve data rate. In [320], the authors propose a handover management scheme for millimeter wave communication under the heterogeneous network. In order to solve the significant fluctuations of the signals, they propose a DQN-based handover scheme which takes the users' connectivity and throughput into account, the simulation results reveal that the proposed algorithm can reduce the handover rate up to 70% and can achieve higher throughput compared to other solutions.

In IoT scenarios, due to the limited size of sensor nodes and the difficulty of charging or replacing batteries of sensors, multi-hop data transmission often has high latency and imbalance business load. The authors in [321] propose a UAV assisted IoT architecture that utilizes UAVs to achieve low latency, seamless coverage during sensor data collection process. Secondly, a decision layer, management layer, and aggregation layer are introduced into the architecture to handle machine learning, UAV management, and data aggregation issues, respectively. Considering that different sensor nodes have different sensitivity to delay, the author proposes a real-time path planning scheme based on DDQN. Finally, a grid based method is adopted to reduce the complexity of the scheme. The experimental results show that this scheme can effectively improve network performance. The authors in [322] consider the network switching problem caused by dense base station deployment in millimeter wave scenarios. The increase in the number of switches has led to a decrease in QoS and QoE, as well as higher signaling overhead. Therefore, in this article, the authors propose a DDQN algorithm to reduce switching frequency in millimeter wave networks. In addition, this article adopts offline learning algorithm to reduce the computational costs caused by online learning. The simulation results show that the proposed scheme can effectively improve the QoS of users. End-to-end network slicing has become a key technology in 5G networks. It can enable traditional networks to support diversified businesses in vertical industries. Therefore, the author in [323] proposes a heterogeneous cellular network system that supports network slicing function. In this system, dynamically switching network slices is adopted to improve system performance. Specifically, the author first models the end-to-end network switching problem as an MDP based on the delay requirements of different tasks to maximize user service profits and reduce switching costs. Then, a network slicing switching algorithm based on DDQN is proposed to train the intelligent agent. Finally, the simulation

results indicate that the algorithm has better performance compared to the baseline scheme.

Different from the above work, the authors in [324] consider a blockchain based mobility management framework in the ultra dense edge computing scenario. Under this framework, not only the continuity of mobile users' computing business can be guaranteed, but also the security of offloaded tasks can be guaranteed. Specifically, the first step is to utilize distributed ledgers to reduce duplicate verification issues between edge servers. In addition, blockchain storage of communication keys ensures communication security between base stations. Secondly, the wireless handover and service migration between base stations are modeled as multi-objective optimization problems, and then the Lyapunov optimization method is used to transform this problem into a long-term delay minimization problem. Finally, a DRL algorithm based on the actor critic framework is used to learn the optimal wireless switching strategy. The experimental results show that the proposed scheme can ensure the quality of service for users while reducing the number of switching between base stations. In [325], in order to tackle the problem of dynamic handover between edge servers in mobile edge computing, the authors propose a handover optimization in MEC by DRL and they construct an adaptive weight DDPG algorithm to optimize the migration cost and service latency. The authors in [326] consider a trajectory prediction problem for UAVs in a dynamic environment, such as moving nodes, exiting existing nodes, and adding emergency nodes. Specifically, a two-level DRL framework is proposed. In the first level, the framework models a dynamically changing environment. In the second layer, a DQN algorithm is used to perform trajectory planning based on the environmental features obtained from the first layer. Furthermore, due to the fact that in practical scenarios, UAVs may not know all the information about the environment, the author models the trajectory planning problem as a POMDP problem and uses DRL based algorithms to solve the optimal trajectory prediction problem. Finally, the simulation results indicate that the algorithm has a higher success rate in data collection.

D. Lesson Learned and Future Directions

The fifth-generation mobile communication technology has spawned many emerging application scenarios that expect high throughput and low latency. Many researchers have extensively investigated the mobility management issue in the CETCN, but there are still many challenges in optimizing mobility management issue using DRL. This section briefly discusses the existing challenges and possible future research directions.

- **Real-time Decision-making:** Mobility events require quick decision-making to ensure seamless connectivity. DRL algorithms must make decisions in real time to optimize handovers and resource allocation, which can be challenging due to the computational demands of DRL training.
- **Network Heterogeneity:** Cloud-edge-terminal collaborative networks consist of heterogeneous resources with

varying capabilities and latency. Coordinating between different types of resources to ensure seamless mobility requires sophisticated algorithms.

- **Resource Allocation:** DRL algorithms need to decide how to allocate resources (e.g., computing, storage, bandwidth) between the cloud and edge to optimize mobility management.
- **Interference and Signal Quality:** DRL algorithms need to account for interference and signal quality variations when making mobility decisions. Poor handover decisions could lead to degraded service quality.
- **Edge Computing Constraints:** Edge resources might have constraints in terms of processing power, memory, and storage. DRL algorithms should consider these limitations when making mobility-related decisions.

VIII. APPLICATIONS FOR AI-ENABLED CLOUD-EDGE-TERMINAL COLLABORATIVE NETWORK

With the further development of intelligent transportation (ITS), industrial Internet of Things (IIoT), smart health and digital agriculture, the computing capacity of various terminal devices has also been further enhanced. Therefore, according to the different requirements of users on performance indicators, these tasks can be processed on the terminal, edge or cloud. In this section, we make a comprehensive investigation on task offloading, resource allocation, caching and mobility management in intelligent transportation systems, IIoT, smart health and digital agriculture in Sections VIII-A–VIII-D. We also point out several challenges and possible future directions in Section VIII-E. Summaries of this section are shown in Tables IX, X, XII, XIII, XI, XIV and XVI.

A. Intelligent Transportation Systems

Advanced communication technologies, IoT sensor networks, and modern decision theory make ITS increasingly efficient under CETCN. Hence, in this subsection, we summarize the tasks scheduling, resource allocation, and discuss the challenges and future directions.

In Fig. 10, we depict a typical vehicle Ad-hoc network. In this network, vehicles are equipped with various processing units that can provide computation, caching, and communication capacities for users. Road Side Units (RSUs) can provide more powerful capacities compared with vehicles and cloud servers are considered to provide the most powerful computing, caching, and communication capabilities. In the following, we summarize related topics under the ITS system.

1) **Task Offloading and Resource Allocation:** In the vehicle network, due to the lack of continuous power grid connection for roadside units, their charging mainly relies on human intervention or energy collection technology. Therefore, in order to ensure the continuous operation of the network, it is crucial to make reasonable use of roadside unit energy before the next charging cycle arrives. Therefore, the author in [327] proposes a Q-learning algorithm, with the aim of conducting reasonable electricity scheduling within a cycle of roadside unit electricity consumption, in order to maximize the satisfaction of service requests. Extensive experimental results

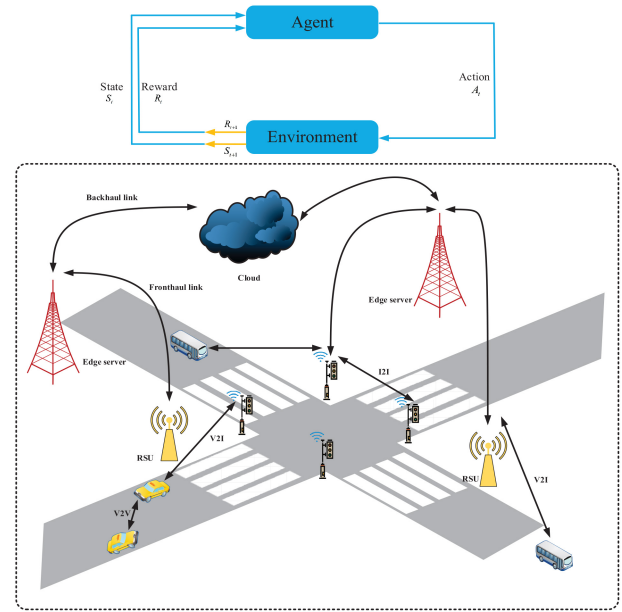


Fig. 10. An introduction of intelligent transportation systems.

indicate that the Q-learning based algorithm can increase the amount of served vehicles. Yan et al. [328] propose an access technology that aggregates multiple wireless access technologies to maximize the long-term throughput of the network while meeting different QoS requirements. However, in order to cope with the complex and dynamic network environment, they formulate the access problem as a Markov model, and then solve the problem through using multi-agent reinforcement learning technology. In addition, the authors also introduce Nash equilibrium to obtain feasible solutions while reducing the search space of multi-agent systems. Finally, the effectiveness of the proposed algorithm has been verified through a large number of simulation experiments. The authors in [329] propose an intelligent task offloading scheme based on DRL to address the high dynamism of vehicle networks and resource constraints of vehicles. Specifically, firstly, they establish an optimization problem for task offloading and resource allocation, taking into account energy consumption and latency. Secondly, due to the complexity of the established problem, the authors divide it into two subproblems. For the task offloading subproblem, a bilateral matching scheme is proposed to maximize utility. The authors propose a DQN based algorithm to solve resource allocation subproblem, and introduce dropout regularization to solve the overestimation problem of DQN. Finally, the simulation results show that the improved DQN algorithm outperforms the baseline scheme under different parameter settings. The authors in [330] propose an intelligent offloading framework for 5G vehicle networks under the coexistence of multiple access technologies. Firstly, considering the performance needs of users for various applications, a task offloading problem with cost minimization is proposed. Secondly, due to the complexity of the problem, they decompose it into two sub problems, i.e., task offloading between vehicles and roadside units, and task offloading between vehicles and infrastructure. Then, they

TABLE IX
AI-GUIDED CLOUD-EDGE-TERMINAL COLLABORATIVE RESOURCE ALLOCATION AND TASK OFFLOADING IN ITS

reference	problems	Model/ Algorithm	State	Action	Reward
[327]	-task offloading	MDP/ Q-learning	-vehicles residing within communication range	-the weight of vehicle	the number of downloaded bits
[328]	-task offloading -resource allocation	MDP/ Q-learning	-service rate -the number of sub-flows -the load of one channel	-the channel and the number of sub-flows	long-term network throughput
[329]	-task offloading	MDP/ DQN	-channel gain -computation capability	-the RSU or BS to process the offloaded task	the comprehensive QoE of vehicles
[330]	-task offloading	MDP/ DQN	-interference degree of all users	-cellular channels for data transmission	offloading cost
[331]	-task offloading	MDP/ DQN	-data size for computational task -size for the computational result -the number of CPU cycles	-binary computational offloading decision	the system cost
[332]	-resource allocation	MDP/ DDPG	-the situation of vehicles	-decide how many requested contents are executed at different MEC	QoS performance
[333]	-task offloading -resource allocation	MDP/ DDPG	-task size -buffer queues -the ratio between delay and queue length	-allocated power -transmitting power	total cost of energy consumption and delay
[334]	-task offloading -resource allocation	MDP/ DDPG	-computing data amount -average vehicle speed	-the index of received RSU -deliver RSU	the overall service time
[335]	-resource allocation	MDP/ DDPG	-the number of vehicles -the coordinates of each MEC	-computing and caching resource allocation	satisfy the QoS requirement
[336]	-task offloading	POMDP/ AC	-task information -longitude and latitude -maximum computation capacities -maximum uplink transmission power -the queuing delay of the current task -the channel state information	-transmission power control policy -transmission channel selection policy -local computing resource allocation policy	Minimize the system cost
[337]	-resource allocation	POMDP/ PPO	-required task memory -required task computing power -total computing memory -the amount of tasks	-accessible candidate computing satellites	Minimize the service delay
[338]	-task offloading	POMDP/ AC	-local sell/buy price -demanding amount of resources -current computing frequency -current local load current basic task utility vehicle delay sensitive parameter	-participating willingness -individual intention	the weighted external cost
[339]	-task offloading -resource allocation	POMDP/ DDPG	-the location and task queue -the location of vehicle -channel state	-the number of vehicles -computing power allocation	minimize the task execution delay

propose a bidirectional matching algorithm to solve the first subproblem, and propose a distributed DQN algorithm to solve the second subproblem. Compared with the centralized DRL algorithm, the proposed algorithm can significantly reduce communication overhead. Finally, the effectiveness of the proposed solution in task offloading is demonstrated through a real-world car rental scenario. Khayyat et al. [331] propose a computational offloading algorithm for multi-level vehicle edge networks based on DRL. Firstly, in order to reduce the energy consumption of vehicles and ensure the effective utilization of resources among multiple vehicles, the author

models the task offloading and resource allocation problem as a binary problem. Secondly, due to the problem is NP-hard, they propose an intelligent algorithm based on DQN to find the approximate optimal solution of the problem. Finally, simulation results demonstrate that the proposed architecture can reduce the energy consumption compared to the baseline solutions.

The introduction of edge computing and content caching in ITS can cope with time-varying network states and various vehicle applications. Therefore, in [332], the authors propose an intention based traffic control system, which can

TABLE X
AI-GUIDED CLOUD-EDGE-TERMINAL COLLABORATIVE CACHING IN ITS

reference	problems	Model/ Algorithm	State	Action	Reward
[340]	-caching	MDP/ DQN	-computation capability -caching cost	-connection control	the cost of task offloading
[341]	-caching	MDP/ DQN	-contest request -content request state -content popularity	-content replacement	hit ratio
[342]	-caching	MDP/ DQN	-the placement of content -the participation level of delivery of requested content	-migrating -caching -dropping the content	magration cost
[343]	-caching	MDP/ DDPG	-V2V communication rate -content latency -energy consumption -vehicle's driving direction -required caching resource -caching capacity	-match caching pairs	system utility
[344]	-caching	MDP/ DDPG	-requesting content type -content delivery deadline -remaining size of requesting content -content popularity	-content caching decision -allocated bandwidth	total cost of content caching
[345]	-caching	MDP/ DDPG	-resource state -vehicle state -task type	-caching decision	network overhead

dynamically coordinate computing tasks and content caching, and maximize the profits of network operators. Specifically, this profit function fully considers the revenue of mobile operators and the QoE level of users. Secondly, a mobility aware task offloading and resource allocation scheme based on DDPG algorithm is proposed for the mobile vehicles. Finally, extensive experimental results verify the scheme can maximize the profits of MNOs and ensure the users' QoS under the condition of real traffic data. In order to solve the problem of energy consumption and delay caused by wireless channel changes over time under the Internet of Vehicles (IoVs), the authors in [333] propose a task offloading framework assisted by edge computing to improve the efficiency of task offloading. Firstly, they design a task offloading model for heterogeneous vehicle networks, which considers multiple random tasks and the variation of wireless channels over time. Secondly, in order to achieve a balance between energy consumption and processing delay, and avoid the curse of dimensionality caused by excessive action space, an adaptive computing offloading method based on DDPG algorithm is proposed. Finally, the simulation results show that the proposed method outperforms DDQN and greedy algorithms. In order to reduce the service delay of the vehicle network and improve the reliability of the service, the authors in [334] design a MEC collaborative computing framework. Firstly, a task scheduling algorithm is proposed to determine the execution order of tasks on edge servers. Secondly, they formulate the task offloading problem as an MDP and propose a DDPG algorithm to determine the task offloading strategy for vehicles to reduce the cost of collaborative computing. Finally, simulation results show that the proposed DDPG-based collaborative computing scheme can better response to the change of the environment

compared to the baseline solutions. Unlike the above work, in [335], the authors propose a joint allocation problem of communication, computing, and storage resources in MEC based vehicle networks. Firstly, for different IoVs applications, the authors consider two typical MEC scenarios of placing MEC servers on macro stations and edge nodes, and construct corresponding multidimensional resource optimization frameworks. Secondly, due to the high computational complexity and solution time of the problem, they develop a DDPG based algorithm to solve the optimization problem. Through offline training, this algorithm can quickly obtain the optimal allocation strategy to meet the QoS requirements of different applications. The experimental results show that the algorithm can achieve high QoS satisfaction.

In order to solve the problem of long task processing time caused by a large number of offloading tasks, the authors in [336] propose a distributed task offloading method based on transformer and policy decoupling. Firstly, the task offloading problem in the vehicle network is modelled as a mixed cooperation competition multi-agent behavior coordination problem, with the aim of selecting the optimal offloading node. Secondly, an improved MADDPG algorithm is proposed to solve the optimization problem. Specifically, a long sequence prediction network is introduced in this algorithm to predict the future queuing delay of the network, the actor network is redesigned using a transformer based time feature extraction network (TTFEN) and a policy decoupling network. Due to the use of TTFEN, this algorithm can adapt to any size of input. Finally, training on real datasets shows that the proposed algorithm can not only quickly adapt to new vehicle network environments, but can also effectively reduce costs. The authors in [337] consider a novel remote intelligent transportation system, which is mainly aimed at various applications of

TABLE XI
AI-GUIDED CLOUD-EDGE-TERMINAL COLLABORATIVE MOBILITY MANAGEMENT IN ITS

reference	problems	Model/ Algorithm	State	Action	Reward
[346]	-Mobility management	MDP/ DDQN	- the locations of different types of vehicle nodes -available computation resources of each edge node -new generated service requests -the association between vehicle nodes and edge nodes	-the index of the edge node that receives the service request	the negative number of the total service time
[347]	-Mobility management	MDP/ DDQN	-UEs location and speed	-the connected BS	the throughput of served services
[348]	-Mobility management	MDP/ DDPG	-the locations of all VUEs -the locations of all VAPs -HUC clustering status	-the HUC clustering policy	the trade-off of utility function
[349]	-Mobility management	MDP/ A3C	-the last k channel observations -the threshold violation indicator -the experienced rate of vehicles	-the optimal vehicle-RSU association	the trade-off of average rate per vehicle
[350]	-Mobility management	POMDP/ A3C	-the transmission rate of wireless network -the required CPU cycles of computation tasks -the size of transmission data	-the migrated service	the negative sum of delay
[351]	-Mobility management	POMDP/ SAC	-s the computing task contained in each vehicle -the mobility model of the vehicle	-task migration decision	the sum of estimated migration cost computed cost and energy consumption

aircraft and ships. Firstly, due to the insufficient ground access network to provide flexible coverage for these applications, the author proposes using low orbit satellites to provide computing offloading services for aircraft and ships. Secondly, in order to optimize the resource allocation of different satellite servers, they propose an optimization problem of joint delay and price. Due to the fact that each satellite intelligently obtains local observation data, and at the same time, the decision of a single satellite can affect global optimality, the author describes the problem as a POMDP with global excitation. To solve this problem, they design a MADRL algorithm with attention mechanism to obtain the optimal offloading decision. Finally, a large number of simulation results have proven the effectiveness of the proposed algorithm. Unlike the above work, the authors in [338] propose a vehicle to vehicle (V2V) distributed dynamic task offloading framework, taking the personal service and offloading intentions of private vehicles into account, to improve the utilization of edge resources. Firstly, in order to collaborate in a more effective manner among vehicles, a stable cooperative relationship between vehicles is established using alliance games and pricing mechanisms. Secondly, in order to achieve stable and efficient offloading services and fully tap into the potential of local computing capability, they describe the transaction process as a POMDP. At the same time, a multi-agent based actor critic algorithm is designed to obtain the optimal offloading strategy. Finally, the experimental results show that the proposed algorithm outperforms the baseline scheme

in dynamic environments. In [339], the authors investigate the dynamic distributed computing offloading problem in heterogeneous road networks, where multiple heterogeneous computing entities simultaneously provide computing services to vehicles. Firstly, considering the autonomous offloading mode of vehicles, they model task offloading and resource allocation problem into a distributed Stackelberg game, where MEC servers responsible for allocating resources and vehicles responsible for offloading tasks. Secondly, due to the information obtained by each vehicle is often incomplete, the authors design an optimal strategy search scheme based on partially observable Markov processes to maximize long-term returns. On this basis, an MADDPG algorithm is proposed to learn the optimal task offloading and resource allocation scheme. Finally, numerical results indicate that the proposed algorithm is superior than other baseline solutions in terms of response time.

2) *Caching*: The author in [340] proposes a resource allocation strategy that combines communication, computation, and caching to improve the cost efficiency of vehicle networks. Firstly, the author proposes a mobile aware encoding caching and computing offloading scheme, where computing tasks can only be offloaded to roadside units when the vehicle can obtain content from them. Otherwise, the content required by the vehicle and the services needed to be offloaded must be completed by the base station. Then, considering the limited storage capacity and computing power of roadside units, as well as the end-to-end delay of services required

TABLE XII
AI-GUIDED CLOUD-EDGE-TERMINAL COLLABORATIVE TASK OFFLOADING AND RESOURCE ALLOCATION IN IIoT

reference	problems	Model/ Algorithm	State	Action	Reward
[357]	-task offloading	MDP/ DQN	-the F-AP selection state -channel gain -data volume -CPU cycles	-F-AP selection	total system energy consumption
[358]	-task offloading -resource allocation	MDP/ DQN	-current work mode	-work mode decision	energy consumption
[359]	-task offloading	MDP/ DQN	-the number of task type	-offloading tasks to base station or not	task completion rate
[360]	-resource allocation	MDP/ DDPG	-wireless channel gain -computing amount -data size -remaining battery energy	-transmission power of devices -computing resources allocated to MEC server	long-term reward of the system
[361]	-resource allocation	MDP/ A3C	-connection relationship -transmission power	-allocated transmission power	long-term system performance
[148]	-task offloading	POMDP/ DQN	-queue state -channel state information	-channel contention -task offloading decision	task delay and energy consumption
[362]	-task offloading -resource allocation	POMDP/ DDQN	-channel gain -queue length	-task offloading -sub-channel assignment	task delay
[363]	-task offloading -resource allocation	POMDP/ DQN	-each users residual data	-bandwidth allocation -power allocation	energy consumption
[364]	-resource allocation	Dec-POMDP/ PPO	-transmission power -transmission rate	-channel assignment	energy efficiency
[365]	-resource allocation	Dec-POMDP/ DDPG	-neighbor node set -link status	-node selection	maximize network lifetime
[366]	-task offloading	Dec-POMDP/ DDQN	- task generation state -data queue state -communication channel state -previous resource allocation state	-resource allocation decision	the expected utility

by vehicles, a resource allocation problem of minimizing cost for joint caching and offloading is formulated. At the same time, in order to reduce the complexity of the problem caused by the large action space, they develop a dual time scale optimization algorithm to configure the parameters of cache and computing resources. Specifically, large time scales use particle swarm optimization method to allocate cache, while small time scales use DQN algorithm for computing resource optimization. Finally, numerical results show that the proposed algorithm can achieve significant performance gain compared with other baseline solutions. The authors in [341] propose a collaborative caching algorithm for vehicle networks based on federated reinforcement learning to address the content replacement issue, and thus adapt the high dynamics of vehicle networks and rapid changes in content popularity. Firstly, each vehicle collects data separately and trains using its own computing unit. Then, each vehicle uploads the trained parameters to the roadside unit and the roadside unit aggregates the collected parameters. After aggregation, all roadside units upload the parameters to the base station for global aggregation. Finally, the base station sends globally aggregated parameters to each vehicle for training. In this manner, the computational and communication costs of vehicles can be greatly reduced. In addition, in order to meet the cache capacity constraint, the author proposes a DQN based

algorithm to maximize the cache hit rate. Simulation results show that the proposed algorithm can improve cache hit rate while accelerating convergence speed. In [342], the author considers a vehicle node assisted content delivery network and proposes a content migration strategy, the proposed strategy can help users obtain the required content from the edge network easily. However, in order to fully utilize the cache resources of edge networks and determine which edge server is used to offload content, they propose a DQN based algorithm to minimize costs and ensure the reasonable utilization of edge resources. Secondly, this article not only considers the popularity of content, but also considers the impact of content priority on cost. At the same time, in order to adapt to the dynamic nature of the vehicle network, the author also adds a LSTM network to the DQN network to automatically learn and update its parameters. Finally, simulation results indicate that the DQN-based approach can reduce 70% content access delay compared to the traditional methods.

The high-speed movement of vehicles and the dynamic changes in wireless channels pose significant challenges to the design of caching strategies, and for many sensitive information, vehicles may not be willing to cache to untrusted roadside units. To address the above issues, the author in [343] proposes a distributed content caching framework based on blockchain authorization. Blockchain can establish a secure

TABLE XIII
AI-GUIDED CLOUD-EDGE-TERMINAL COLLABORATIVE SERVICE PLACEMENT IN IIoT

reference	problems	Model/ Algorithm	State	Action	Reward
[367]	-service placement	MDP/ DQN	-current resource status -current deployment status	-mapping scheme	weighted sum of the bandwidth consumption
[368]	-service placement	MDP/ DQN	-service placement decisions -corresponding computational resources	-desired service placement decisions on one edge server	delay-latency rate
[369]	-service placement	MDP/ DDQN	-communication state -computation state	-service placement decision	redundant cost
[370]	-service placement	MDP/ A3C	-channel power gain -idle VNF instance capacity -idle link capacity -SFC mapping result	-choose which server to active	long-term per-package latency
[371]	-service placement	MDP/ DDPG	-remaining memory size -remaining bandwidth of link	-service placement	the combination of delay and bandwidth consumption
[372]	-service placement	POMDP/ PPO	-request information -caching status -available bandwidth -CPU cycles of each MEC	-service placement	average service delay

TABLE XIV
AI-GUIDED CLOUD-EDGE-TERMINAL COLLABORATIVE MOBILITY MANAGEMENT IN IIoT

reference	problems	Model/ Algorithm	State	Action	Reward
[373]	-Mobility management	MDP/ DQN	-the number of UEs -the average throughput the number of edge users	-the bias values of the directional beam	the throughput for all UEs and the handover cost for all UEs
[374]	-Mobility management	MDP/ DQN	-channel state realization -available energy -computation capability	-task offload decision server selection handover decision	total computation delay cost
[375]	-Mobility management	MDP/ DDPG	-the number of UEs -handover margin	-handover trigger points	the total latency for all users
[376]	-Mobility management	MDP/ DDPG	-user velocity -possible user data rate -average service time transmission distances with the same SINR user direction in the n -th sample	-SINR ratios	minimize packet delivery latency for all users
[377]	-Mobility management	MDP/ PPO	-the size of offloading task data -the number of CPU cycles the maximum delay time to complete the task the distance between the user device to each MEC server -the CPU frequency of each MEC server the number of CPU cycles required for uncompleted tasks	-the optimal handover scheme	the requirement of offloading efficiency
[378]	-Mobility management	MDP/ DQN	-the distance between IIoT node and UEs -path loss	-channel selection	the probability of selecting the best channel
[379]	-Mobility management	Dec-POMDP/ DQN	-the set of CSIs -the set of the RSUs locations	-select the associated RSU transmit power level	the long-term per-user trade-off

caching environment, and when a vehicle need to cache content, the base station will maintain a licensed blockchain. Then, an optimal content caching scheme for mobile awareness is designed based on DDPG algorithm. Finally, utility

proof is used to select the validators of the block. The numerical results show that the DRL-enabled caching strategy performs better than the baseline strategies. The authors in [344] propose a collaborative caching scheme between

TABLE XV
AI-GUIDED CLOUD-EDGE-TERMINAL TASK OFFLOADING AND RESOURCE ALLOCATION IN SMART HEALTH

reference	problems	Model/ Algorithm	State	Action	Reward
[382]	-resource allocation	MDP/ DQN	-task size -computing capacity of edge nodes	-the resource allocation decision	energy consumption
[383]	-task offloading -resource allocation	MDP/ DQN	-the number of mobile nodes -the number of fog nodes	-the task offloading decision -the resource allocation decision	processing time
[384]	-task offloading	MDP/ DQN	-maximum residual delay	-the number of frames -the number of offloading task	total delay
[385]	-task offloading -resource allocation	MDP/ DDQN	-the state of nodes -the number of data packet	-threshold -wake-up probability	energy consumption and average AoI
[386]	-task offloading -resource allocation	MDP/ AC	-the channel state quality -the task offloading profile	-the task offloading decision -the resource allocation decision	energy consumption
[387]	-task offloading	MDP/ AC	-queue information	-the set of nodes that can be chosen to offloaded	long term throughput
[388]	-task offloading -resource allocation	MDP/ AC	-the number of remaining challels -computing resources -data volume -delay sensitivity of the service	-the number of resources -the number of channels	the offloading advantages
[389]	-task offloading -resource allocation	MDP/ AC	-the remaining tasks -the available computing resources -the available bandwidth -the maximum gas limit	-task offloading decisions -resource allocation decisions -transmission rate of user -the amount of gas consumed by user	the sum cost of delay and energy
[390]	-resource allocation	MDP/ MADDPG	-maximum tolerable latency -channel strength -available resource block	-resource allocation decisions -transmission power	long-term reward

macro base stations, roadside units and intelligent vehicles to jointly optimize the content placement of vehicle edge computing networks. They describe the optimization problem as an MDP with dual time scales. Firstly, compared to the mobility of vehicles and changes in the network, the frequency of content popularity changes is relatively low. Therefore, on a large time scale, content placement strategies are determined based on the popularity of content, the trajectory of vehicles, and the availability of resources. Secondly, on a small time scale, a bandwidth allocation scheme is designed to meet the latency of content acquisition, with the aim of minimizing the cost of accessing content. As this problem is an MINLP problem, they propose a DDPG algorithm to obtain suboptimal solutions with lower computational complexity. Finally, extensive experimental results indicate the proposed scheme can reduce the system cost and content delivery time compared to the other caching schemes. In [345], the authors propose a hierarchical terminal-edge collaborative vehicle network caching framework to minimize the network overhead. Then, a DDPG based near-optimal solution is proposed to solve the problem. Finally, the performance analysis indicates that the framework outperforms other baseline solutions at reducing the network overhead.

3) *Mobility Management*: Mobile edge computing (MEC)-based heterogeneous vehicular networks (HetVNs) are attractive technologies to support low-latency applications in vehicles. However, in practical systems, joint vehicle-to-edge (V2E) association and resource allocation is an urgent problem, since precise prior knowledge of the dynamic wireless environment is difficult to obtain. Therefore, in [346], the author first uses random geometry to simulate real

scenarios. Specifically, the intersection is first set up as two vertical streets, and then the distribution of vehicles on the streets is modeled as different independent poisson point processes. Secondly, considering the delay for different types of vehicles to receive services, they propose a DDQN algorithm to solve the connection problem between vehicles and edge networks. On the one hand, this algorithm considers the transmission delay, computational delay, and switching cost. On the other hand, it can minimize service time while meeting computational resource constraints. Finally, the simulation results indicate that the proposed algorithm has a shorter service time than the baseline scheme. Unlike the above work, the authors in [347] consider a practical deployment scenario to validate the algorithm for handover management. Firstly, based on configurable input parameters in cellular networks, a DDQN algorithm is proposed for handover management, and the performance of the proposed algorithm is evaluated using the network simulator ns3 and LTE module. At the same time, a real network is set up in the city center of Glasgow, and vehicle trajectories generated by a route movement model and Google Maps are configured for algorithm performance evaluation. The simulation results and real network validation show that the proposed algorithm outperforms the baseline scheme in terms of packet loss rate and service quality. A user centered cluster(UC) enhances the connectivity of mobile users by grouping an appropriate number of access points. However, for high-speed vehicles, UC clustering may lead to excessively frequent switching, thereby reducing throughput and increasing energy consumption. In order to solve the problem caused by excessive switching, the author in [348] proposes a new cluster migration strategy, which can perform horizontal and

TABLE XVI
AI-GUIDED CLOUD-EDGE-TERMINAL TASK OFFLOADING AND RESOURCE ALLOCATION IN DIGITAL AGRICULTURE

reference	problems	Model/ Algorithm	State	Action	Reward
[393]	-resource allocation -task offloading	MDP/ DQN	-the type of task -energy consumption of each task -transmission latency	-selected processing device	the sum of energy consumption
[394]	-task offloading -resource allocation	MDP/ DQN	-task offloading decisions -QoS requirements	-optimal sequence	computation cost
[395]	-task offloading -resource allocation	MDP/ DDQN	-the CPU power of mobile devices wireless network status	-the task offloading decisions	energy consumption and calculation delay

vertical switching between V2V and V2I. Specifically, in order to achieve a balance between connectivity and switching times, the author formulates an optimal switching problem under the minimum data rate and association constraints. Secondly, a DDPG algorithm is proposed to determine the migration scheme for UC clusters. Finally, the simulation results indicate that the proposed scheme can achieve higher data rates than the baseline scheme. Khan et al. [349] investigate the correlation problem between vehicles and cells in millimeter wave communication scenarios, with the aim of maximizing the average speed of vehicles while ensuring relatively low vehicle overhead. To address the above issues, the author first formulates an optimization problem that maximizes the average speed of all vehicles while minimizing the probability of the average speed falling below a certain threshold. Secondly, an actor critic based reinforcement learning algorithm is proposed to obtain the approximate optimal solution. In the proposed algorithm, each roadside unit makes independent decisions based on its own observations and utilizes offline training schemes to reduce computational load at the same time. Finally, the simulation results show that the proposed scheme can reduce the interruption probability while meeting the constraint of average rate.

Making service migration decisions based on complete system information to ensure user service quality not only incurs extremely high latency but also lacks scalability. Therefore, in [350], the author models the service migration problem as a POMDP, which does not require all information about the system to make effective migration decisions. At the same time, a DRL based offline actor critic algorithm is used to make migration decisions and improve the stability of training. Secondly, the author introduces the LSTM network to effectively extract hidden information from historical data. Simulation results show that the proposed method outperforms other baseline schemes. Mobile vehicles cannot obtain continuous services by accessing a fixed MEC server. Therefore, the authors in [351] propose a service migration problem in a V2X scenario in this paper. By modeling the service migration problem as a Markov process, and comprehensively considering computational costs, migration costs, and energy consumption of vehicles during movement. Second, an actor-critic based deep reinforcement learning algorithm is proposed to solve the Markov process. Finally, simulation results show that the proposed algorithm is more robust.

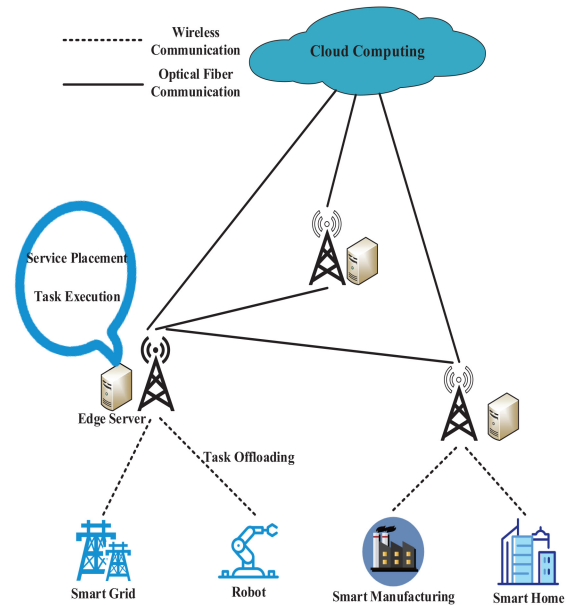


Fig. 11. A toy description of IIoT.

B. Industrial IoT

Normally, there are a huge number of devices with limited computing and storage resources in an IIoT scenario, and these devices generate a large amount of data that need to be processed. For example, as shown in Fig. 11, typical IIoT scenarios include smart grid, smart manufacturing and so on, the processing devices in these scenarios have limited computing power, and tens of millions of data needs to be processed will put a huge burden on these systems.

The complex and real time decision-making process, dynamic environment of IIoT and heterogeneous capacity of devices are the main challenges of IIoT system. For example, the authors in [352] propose a joint task offloading and resource allocation problem. Through considering the instability caused by multihop transmission, a non-convex optimization problem is established to minimize task processing delay, and convex optimization methods are used to solve the problem. However, constructing convex optimization problems requires additional computational and communication costs, which can become a burden in resource constrained IIoT devices. In addition, convex optimization methods require a

long time to solve large-scale task offloading and resource allocation problems with many devices and tasks [353]. The author in [354] proposes a computational offloading strategy based on potential game to minimize the computational time and energy consumption of each task. This method solves the problem of focusing on single optimization objective in convex optimization [355], and has good scalability with the increase of IIoT devices. However, this method requires the assumption that all players participating in the game are rational [356], and in real-world IIoT scenarios, due to equipment failures and other issues, this assumption is not always reasonable. But fortunately, DRL can effectively overcome the problems brought by the traditional methods mentioned above. Therefore, DRL can become a promising method to handle task offloading, resource allocation, service placement and mobility management issues in IIoT environment.

1) *Task Offloading and Resource Allocation*: Traditional computing offloading methods are not suitable for IIoT scenarios due to their high computational latency. In [357], the author proposes a task offloading method based on DRL, which minimizes long-term offloading energy consumption in scenarios with multiple IIoT nodes and access points. Specifically, through creating DRL agent for each IIoT node, the action space can be reduced. Secondly, after selecting access points, each access point executes a low complexity greedy algorithm to determine which tasks should be offloaded to the cloud for processing under computational power constraints. Through offline training in the cloud and online decision-making processes, access points can quickly adjust their selection of IIoT devices. Finally, the simulation results show that the proposed scheme is superior to the exhaustive method and genetic algorithm. The authors in [358] consider a machine to machine assisted NOMA based computing offloading scheme, where IIoT devices can offload tasks to devices with idle computing resources through machine to machine communication. Specifically, they propose a DRL based sleep scheduling mechanism, which optimizes communication and computing resources through sleep mechanisms while satisfying task delay constraint and energy consumption constraint. Secondly, an iterative optimization algorithm is designed to solve the resource allocation problem. Finally, the numerical results indicate that the scheme can efficiently reduce the system's energy consumption and satisfy the computation requirements of users at the same time. Due to the fact that in the field of IIoT, computing tasks generated by various devices often have different service requirements. The distribution of these tasks has a basic internal pattern and varies depending on regional, seasonal, and temporal factors. Therefore, in [359], the authors consider developing specific task offloading schemes for specific tasks to reduce task execution delays. In order to learn these generation patterns and improve task completion rate, the authors develop a DRL based task offloading algorithm. At the same time, a multi-layer computing architecture supporting SDN is introduced to efficiently allocate task features based on the obtained factors. Finally, the proposed algorithm can improve the task completion rate and achieve near-optimal solution in IIoT scenarios. Due to the dynamic and continuous nature of task completion in the IIoT scenarios, in [360], the

authors investigate the dynamic resource management problem of joint power control and computing resource allocation. Firstly, total latency generated by all tasks is optimized in the process of dynamically generating tasks. Secondly, the authors transform the long-term delay minimization problem to an MDP. Considering the dynamic characteristics of task generation, a DDPG algorithm is proposed to obtain the optimal solution. Finally, through extensive simulations, the proposed method can reduce the long-term average delay of the tasks compared to other baseline solutions. Unlike the above work, in [361], the authors propose an intelligent driven green resource allocation mechanism for IIoT in heterogeneous networks from the perspective of efficiency and energy conservation. Firstly, the authors propose an end-to-end self-organizing resource allocation framework for IIoT. Secondly, under the premise of meeting the quality of service, with the goal of maximizing energy efficiency, a green resource allocation model is constructed with the number of devices and base station resources as variables. Simultaneously, a DRL algorithm based on actor critic framework is used to solve the optimization problem. Simulation results indicate that the proposed method can achieve better performance at quality of service compared with other traditional methods.

The authors in [148] study the task offloading problem of IoT systems under rapidly time-varying wireless channel conditions. Unlike assumptions that consider IoT devices can obtain channel state information, due to hardware limitations, short sensor sensing time, and network connectivity issues, it is difficult for IoT devices to have a perfect understanding of dynamic channel conditions. Therefore, the authors propose a partially observable IoT task offloading scheme in this article, which enables IoT devices to make optimal offloading decisions under imperfect channel conditions. Secondly, the authors formulate the optimization problem as a POMDP that minimizes the energy consumption of IoT devices while meeting task processing delays. In order to find the optimal strategy, an offline algorithm based on deep recursive Q-network is proposed. Finally, the simulation results indicate that the proposed algorithm can effectively reduce the energy consumption of IoT devices while ensuring computational latency. Unlike the previous work, the authors in [362] not only consider the characteristics of wireless channels changing over time, but also consider the challenges of unknown dynamic IIoT environments such as the diversity of task requests, limited resources, and dynamic load changes on task offloading decisions and NOMA user pairing. Therefore, in order to address the aforementioned challenges, the authors first consider task offloading problem and NOMA subchannel allocation problem in a dynamic environment. This problem describes the effects of task transmission delay, task processing delay, and dynamic characteristics of the environment on user satisfaction. Secondly, due to the signaling overhead required for each edge server to obtain all information about IIoT devices is large, the authors consider a distributed DRL algorithm where each IIoT device interacts with the environment as an agent and makes the best decision. Due to this method transforming the original problem into a POMDP problem, they use recursive neural networks to predict the real-time

load of all subchannels and servers, and the agent use the predicted results to make decision. Finally, the simulation results indicate that the proposed scheme is superior than baseline schemes in terms of task satisfaction ratio. Since the IoT systems assisted by low orbit satellites can overcome terrain obstacles, they are suitable for computing offloading in scenarios such as sparsely populated rural areas and disaster sites. However, sharing communication and power resources with users and satellites is a challenging task. Therefore, the author in [363] studies the task offloading problem in the LEO-IoT scenario, with the aim of making the fast offloading decisions and improving energy efficiency. Specifically, firstly, the author models the collaborative offloading problem as an optimization problem with resource constraints. Secondly, the author describes the problem as a POMDP problem and proposes an algorithm for multi-agent information broadcasting and judgment. In addition, the resource allocation decision is obtain through the collaboration among agents. Finally, extensive numerical results indicate that the proposed algorithm achieves a better performance in transmission latency and energy consumption than several baseline algorithms. The authors in [364] propose a wireless resource allocation scheme in a controllable IoT scenario to achieve strict requirements for service quality in the uRLLC scenario. Firstly, they consider a cellular heterogeneous network and develop a dynamic mixed nonlinear integer programming resource allocation problem to maximize long-term average energy consumption while meeting reliability, latency, and transmission rate constraints. Secondly, a distributed MADRL resource allocation algorithm is proposed, while using Dec-POMDP to model some observable characteristics of agents in large-scale networks, where each agent does not require all information about the environment. Finally, large amount of results show that the proposed approach can achieve a better performance in maximizing energy efficiency than other traditional methods. The authors in [365] investigate the distributed resource allocation problem of energy-saving data forwarding in resource constrained IIoT scenarios. Firstly, the authors formulate the problem as a Dec-POMDP based decision problem and propose an intelligent resource allocation algorithm based on a dual attention mechanism. Due to the introduction of a dual attention mechanism, it overcomes the difficulties of feature extraction in partially observable environments and non-stationary network environments. In addition, the multi-scale convolutional attention mechanism is used for local state feature extraction, while the multi-head attention mechanism is used to extract global state information. Numerical results show the proposed scheme outperforms other baseline solutions in network lifetime, transmission reliability, and network stability. Tang et al. [366] study the distributed task scheduling problem of IoT in serverless edge computing. Due to heterogeneous edge servers are rational individuals, the edge servers will maximize their own utility by observing local data considering the security issues. Secondly, since there is no complete information about the environment, the authors model the task scheduling problem as a stochastic game, in which each serverless edge computing node schedules the tasks and allocates resources in a non-cooperative manner

according to the available computing resources and task queue status, in order to maximize its own utility while reducing delay and energy consumption. Then, in order to solve the POSG problem, the authors propose an intelligent algorithm based on DDQN to obtain the approximate optimal solution of the problem. Finally, simulation experiments show that the proposed algorithm can achieve effective task scheduling decisions with lower complexity.

The further development of the IIoT also requires edge servers to have caching capabilities. Unlike the content caching described in the previous chapters, caching in the IIoT scenario pays more attention to service placement, to provide vertical industries with the services they need nearby. Therefore, this section summarizes the DRL-based service placement problem in the IIoT scenarios.

2) *Service Placement*: In [367], in order to support the reliable operation in IIoT, the authors propose an NFV-enabled cloud-edge collaborative architecture to provide flexible services for IIoT in the form of the service function chain (SFC). However, it is difficult to provide IIoT services with high qualities due to the limited caching capacity of edge servers. Hence, in order to handle the problem, a DQN-based method is presented to solve the SFC placement problem and hence provide reliable services to IIoT applications. Simulation results indicate that the proposed method can better balance the resource consumption and perform well in providing delay-sensitive IIoT services compared to other baseline solutions. Hazra et al. in [368] design an intelligent service deployment strategy to simultaneously handle dynamic requests and edge resources allocation in IIoT networks. Firstly, the authors describe the problem as the sum of weighted energy delays under QoS constraints, and formulate the service deployment problem as an MINLP. Then, a heuristic task execution strategy is designed and a DRL algorithm is used to intelligently process the generated IIoT requests at the network edge. Finally, the simulation results indicate that the proposed algorithm can achieve effective service deployment strategies. The dynamic arrival of task requests and limited edge network resources in the IIoT make service scheduling problems become a challenging issue. Therefore, the authors in [369] introduces a service function chain (SFC) orchestration framework based on NFV technology to overcome the expensive cost of applying and expanding traditional networks. Specifically, first of all, considering timeliness and resource availability, the authors describe the problem as an utility maximization problem, and divide it into SFC request selection and SFC dynamic scheduling problems. Secondly, for the first problem, a dynamic pricing mechanism and matching algorithm are proposed to select the current SFC request that should be executed. For the second problem, the DDQN algorithm is used to approximate the optimal cost. Finally, the numerical results show that the proposes method is more efficient than other baseline solutions. On the basis of the [369], the authors in [370] consider the deployment of SFC for wireless sensing scenario. Firstly, the authors describe a formulaic expression of wireless sensing deployment problem to minimize the service latency, the expression can not only control the traffic passing through wired links, but also control

the traffic passing through wireless links. Then, the allocation of wireless resources in the network and resources on IIoT servers are modeled as an MDP, and a DRL algorithm based on the actor critical framework is proposed to find the approximate optimal strategy. Secondly, in order to solve the high-dimensional problem of the action space, an optimization sub-algorithm based on norm penalty is introduced in the calculation of the reward function, thereby reducing the size of the action space and improving the convergence speed. Finally, numerical results indicate that the proposed method superior to other solutions. In [371], the authors provide a method for virtual network function (VNF) placement with routing issues. Firstly, the VNF placement and routing problem is formulated to minimize service latency and resource consumption costs, and then decompose the problem into task placement subproblem and routing subproblem. Then a new MADRL framework is proposed to solve these two sub problems, which consists of a VNF placement module and a routing module. At the same time, the DDPG algorithm is introduced to find the optimal solution for these two modules. Finally, extensive experiments show that the proposed method superior than others in terms of service cost and delay, and can offer higher flexibility for personalized services. The authors in [372] propose a dual time scale adaptive request scheduling and collaborative service caching joint optimization problem. Firstly, considering the limited computing and caching resources on MEC, the authors model the problem as a problem of maximizing the service latency reduction rate. Meanwhile, due to the fact that the time scale of content popularity changes is often greater than the time scale of request scheduling, they separate the service caching problem from the joint optimization problem. Then, in order to adapt to the unavailability of information, they formulate the service caching problem and request scheduling problem as a POMDP, and design a DRL algorithm to solve the approximate optimal solution. Furthermore, the Lagrangian multiplier method is used to allocate computing resources, thereby further minimizing the average service delay. Finally, the simulation results indicate that the proposed algorithm can serve more requests while reduce service latency.

3) *Mobility Management*: Since emerging mobile IoT services require extremely high transmission bandwidth and extremely low transmission latency, the authors in [373] investigate the problem of user access control in mmWave-enabled edge computing systems. They consider the impact of system throughput and switching efficiency on the optimal extension range at the same time, and proposes a DQN algorithm to optimize the optimal extension range. The authors in [374] consider a multi-user wireless mobile edge computing system scenario, where users can transmit tasks to edge base stations for processing through wireless channels. The user's task offloading decision can be adjusted according to the time-varying network environment. Since there are a large number of users and edge servers in the actual network, the author proposes a server selection, task offloading and switching strategy based on deep reinforcement learning. The simulation results show that the proposed algorithm can achieve better performance than traditional methods. In addition, the ultra-dense network is also considered as a key enabling technology

in the fifth generation mobile communication. Ultra dense networks not only bring high data transmission but also problems of high interference and complex switching. In order to achieve seamless connection of base stations and provide high-quality services for users, in [375], the author proposes an efficient handover algorithm based on DDPG. Specifically, this algorithm accurately determines the switching trigger point by dynamically adjusting key switching parameters such as switching margin. The simulation results show that the proposed algorithm improves switching performance while ensuring high throughput and low latency at different speed of vehicles. Due to the susceptibility of millimeter wave links to user and terrain influences, the author proposes a switching scheme based on jumping Markov linear systems and DRL in [376] to achieve connectivity between devices and base stations and avoid unnecessary switching at the same time. In order to reduce training samples, the author introduces meta learning technology, which is a technology that makes new decisions by automatically reusing data from past or related tasks. Finally, the simulation results indicate that the proposed scheme can select more reliable links and reduce the number of handovers. The articles mentioned above mostly focus on improving user service quality through switching optimization. However, the author proposes an optimal switching scheme that ensures security while improving service quality in [377]. Specifically, firstly, a blockchain based mobility management scheme is proposed, which not only achieves user mobility management but also reduces the number of switches. Secondly, in order to improve the security of task offloading process, the author establishes a security model that includes confidentiality, integrity, and availability standards. Then, based on this standard, a differential evolution algorithm is designed that can ensure security and reduce costs. Finally, an intelligent algorithm based on DRL is proposed to solve the adaptive switching scheme in MEC environment, which can improve QoS while reducing switching risk.

The scarcity of wireless spectrum resources leads to fierce competition among users. Therefore, in [378], the authors propose an efficient dynamic spectrum access technique to ensure a fair allocation of available network resources. Then, aim to provide users with the most suitable spectrum access, a DRL based approach is proposed to evaluate channel occupancy and determine channel quality. The results show that the proposed method can significantly reduce communication latency and improve system throughput compared to traditional methods. Unlike the previous references on reducing the number of handovers to improve service quality, the authors in [379] propose an active edge correlation technique that can improve wireless connectivity by increasing the frequency and energy consumption of handovers. Due to the dependence of this method on a large amount of private information sharing, this paper investigates the joint edge association problem in environmental uncertainty and privacy protection. Firstly, due to the uncertainty of the environment, the author models connectivity, switching, and long-term energy consumption as a Dec-POMDP problem. Secondly, they propose a multi-agent federated reinforcement learning algorithm that shares encrypted local Q values. Specifically, when some vehicles

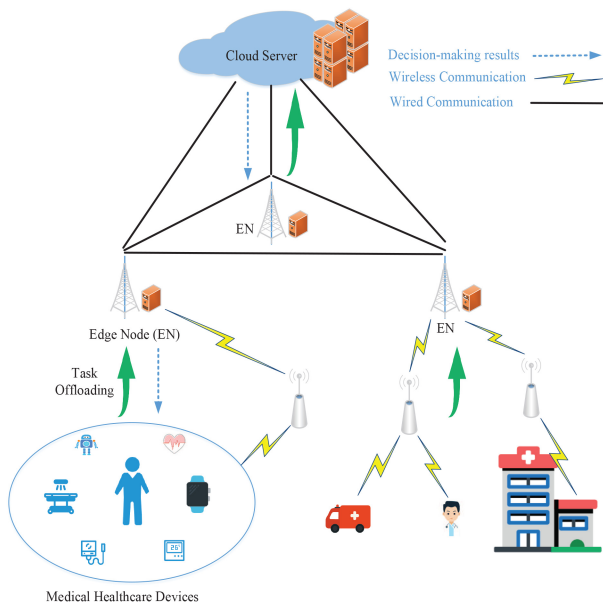


Fig. 12. CETCN Framework for Smart Health.

cannot learn independently, distributed decisions can be made based on the results of federated training. Simulation results show that the proposed solution improves user privacy compared with other baseline solutions.

C. Smart Health

Smart health has become one of the important directions in today's social development as shown in Fig. 12. The growth of medical data and the heterogeneity of data have become the main challenges of smart medicine, especially in processing tasks and storing data, which makes it necessary to use edge computing and AI to make smart health more efficient. To be specific, DRL can optimize scheduling and coordination of healthcare operations, such as appointment scheduling, patient flow in hospitals. At the same time, in a smart health ecosystem, DRL can be used to dynamically allocate medical resources, such as hospital beds, surgical facilities, and medical staff, to ensure efficient and effective patient care. Next, we will introduce how to use DRL technology to solve the challenges of task offloading and resource allocation issue in smart healthcare. More relevant issues and challenges can be found in [380], [381] for interested readers.

The detection of early symptoms in heart disease patients has always been a challenge for healthcare providers, and at the same time, traditional machine learning models are no longer able to meet the development of medical big data. Therefore, the author in [382] proposes a computing paradigm based on cloud fog collaboration to provide low latency and energy-saving computing for healthcare systems. Simultaneously, the DQN algorithm is used to construct a heart disease prediction model. Finally, validation on a heart disease dataset shows that the proposed algorithm can improve prediction accuracy, reduce energy consumption and latency. The authors in [383] develop a nested word remote procedure call based on reinforcement learning and federated

learning, with the aim of designing a consumer centric semi robot efficient training and testing system to execute the entire application with minimal latency in the medical IoT. Compared with traditional machine learning algorithms, this algorithm can effectively reduce the local processing time of tasks. The authors in [384] propose a computing framework for ultra high reliability and extremely low latency healthcare applications in 6G networks. Firstly, they combine the wireless body area network (WBAN) and edge computing to enable IoMT to sense, process and transmit large amounts of data. Secondly, due to the excessive control frames generated during transmission process in WBAN, it leads to high latency and high energy consumption. Therefore, they propose a frame aggregation and task offloading method based on DQN. The algorithm divides different services based on QoS, and then the algorithm dynamically selects the frame aggregation length and the number of offloading tasks according to the requirements of latency and energy consumption. The simulation results show that the proposed algorithm can effectively reduce latency and energy consumption, and improve the throughput of WBAN. In healthcare systems, the freshness of data is crucial for monitoring the vital signs of patients. Therefore, the author in [385] considers a resource management solution in an intelligent healthcare system. Specifically, edge nodes process data from wearable medical devices and then send the processed data to the medical center. The goal of edge nodes is to find the optimal resource management strategy that maximizes information freshness. Firstly, the author models the problem as a Markov process, and then designs an optimal strategy for minimizing system energy consumption and information freshness using DDQN algorithm. At the same time, the author also proposes a sleep mechanism to improve the energy efficiency of each wearable device. The simulation results show that the proposed scheme effectively balances energy consumption and information freshness.

The wireless virtual reality medical treatment (WVMT) system that combines virtual reality technology with the medical IoT is a promising application in the future medical industry. However, due to the high requirements of medical services, improving the efficiency of the WVMT system is an urgent problem need to be solved. Therefore, in [386], the author first proposes a blockchain supported task offloading problem. Then, the problem is described as a Markov decision problem with block consistency, content correlation, and channel fluctuation constraints. Finally, a quantum inspired actor critic algorithm is used to adaptively allocate resources according to the requirements of content transmission. The simulation results show that this algorithm can improve convergence speed and reduce energy consumption. The Health IoT has become a promising electronic health approach by uploading a large number of computationally intensive services to edge servers. Considering that current computing offloading technologies are unable to meet the growing health needs of people in ultra reliable and low latency communication environment, the authors in [387] propose a multi-agent based task offloading and resource allocation scheme that can maximize throughput while minimizing terminal power consumption. Firstly, the authors use Lyapunov

optimization method to decouple the problem into task offloading subproblem and resource allocation subproblem. Secondly, the discrete multi-agent SAC algorithm and queue squeezing perception algorithm are used to solve two sub problems, respectively. Finally, a large number of simulation results indicate that the proposed algorithm outperforms traditional DRL algorithms in terms of data arrival rate and latency. Unlike the above work, the author in [388] considers the real-time monitoring problem in the medical IoT. Due to the additional energy consumption will be generated by offloading services from medical monitoring units to edge servers, a service offloading problem with A3C algorithm is proposed to optimize energy perception. This algorithm can dynamically adjust delay energy correction based on business class and system requirements. Finally, the effectiveness of the algorithm is verified through actual data. The author in [389] proposes a healthcare data offloading scheme based on blockchain technology, which can achieve safe and efficient data offloading. Specifically, the first step is to introduce a smart contract mechanism for security and integrity supervision of user data. Secondly, an optimization problem integrating offloading decisions, resource allocation, and blockchain data audit costs is proposed. Finally, a DRL algorithm is used to obtain the optimal solution of the problem. The simulation results show that the proposed scheme can ensure the security of user data while reducing costs. The author in [390] considers UAV assisted medical IoT systems in remote network coverage or edge server overload/emergency situations. Firstly, in order to ensure the security of user data and minimize system costs while considering latency and energy consumption, the author proposes a multi-agent federated reinforcement learning framework. In this framework, medical IoT devices train their own models and send parameters to UAVs to aggregate the global model. Secondly, the author proposes a resource optimization problem for joint task offloading and information age, and transforms this problem into an MDP. Finally, a multi-agent reinforcement learning algorithm based on MADDPG is developed to solve the optimization problem. Last but not least, the proposed algorithm combines federated learning models to obtain the optimal solution while ensuring user data security.

D. Digital Agriculture

Smart agriculture has become a key solution for the sustainable development of agriculture and solving the global food crisis. Machine learning and deep learning are crucial for the development of smart agriculture. Machine learning algorithms can be used for crop selection and management, and deep learning can be used for crop yield prediction. In these applications, a large amount of data is sent through terminal devices to edge servers and cloud servers for processing as shown in Fig. 13. In order to better utilize the resources of edge devices, reasonable task offloading and resource allocation schemes should be developed. Therefore, this section will conduct a survey on the task offloading and resource allocation issues designed in smart agriculture. Interested readers can

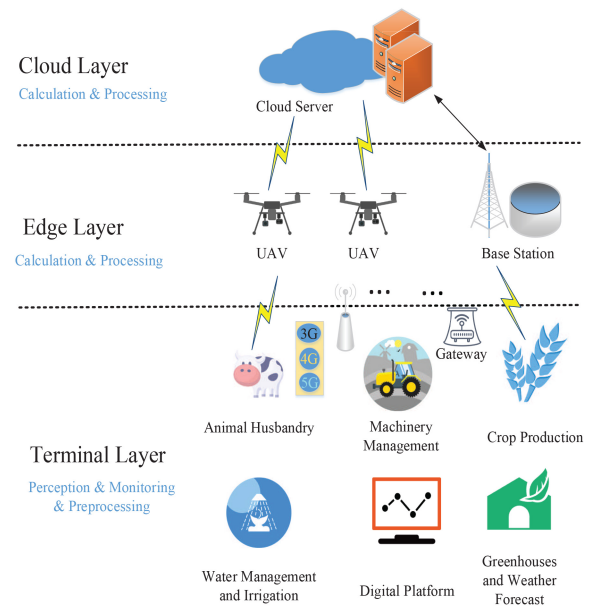


Fig. 13. CETCN Framework for Digital Agriculture.

refer to [391], [392] on other issues and challenges in smart agriculture.

The deep learning architecture with IoT, UAVs, and edge clouds has made smart agriculture become an important component of the agricultural revolution. In [393], the author proposes a two-stage end-to-end DRL smart agriculture system. Specifically, in the first stage, a DQN model based on ant colony algorithm is proposed to complete task offloading issues such as fire detection, pest detection, crop growth, and irrigation scheduling. This model offloads tasks to UAVs, edge clouds, or core clouds based on energy consumption, latency, and computing power. In the second stage, the task of predicting and monitoring various agricultural activities is carried out through using DQN algorithm. A large number of simulation experiments show that the proposed scheme can improve the accuracy and precision of smart agricultural systems. Due to advanced carrier aggregation (CA) technology and MEC technology, 5G and 6G networks can be used to support resource and computationally intensive services of smart agriculture. CA technology achieves higher peak rates by aggregating multiple frequency bands, while MEC reduces latency and costs by offloading tasks. Therefore, in [394], the author studies the use of CA and MEC technologies in 5G and 6G networks to solve the task offloading problem of computationally intensive services in smart agriculture. Firstly, the author proposes a reinforcement learning based sequential decision algorithm to solve the offloading problem, with the aim of achieving a balance between maximum transmission rate and minimum transmission power. In addition, the author also maximizes network capacity and improves network coverage by activating/disabling composite carriers. The simulation results show that the proposed algorithm can achieve effective spectrum resource management and improve data rate while reducing energy consumption and latency. By

introducing deep learning into smart agriculture, it can be well used for identifying crop growth processes. However, using deep learning to detect crop growth on mobile devices requires high energy consumption, thereby reducing device runtime. At the same time, implementing deep learning algorithms on remote servers may lead to higher latency and lower quality of service. Therefore, the authors in [395] propose a low complexity and low computational delay wheat growth period detection model, and then a task migration algorithm based on DQN algorithm is proposed to minimize the weighted sum of energy consumption and delay in task migration. The experimental results show that the proposed algorithm can reduce energy consumption and improve efficiency. The author in [396] proposes a smart agricultural IoT system based on DRL to increase food production and reduce the consumption of freshwater resources. Firstly, the system consists of data acquisition layer, edge computing layer, data transmission layer and cloud computing layer. Secondly, by integrating DRL algorithms into the cloud computing layer, real-time decision-making is provided for the production of smart agriculture, such as determining the required irrigation water amount for crop growth environment. Finally, the author also discusses the potential challenges and development directions of DRL in smart agriculture.

E. Lesson Learned and Future Directions

This section reviews related applications of cloud-edge intelligent collaboration, such as intelligent transportation systems (ITS), Industrial Internet of Things (IIoT), smart health and digital agriculture. The CETCN can provide flexible and dynamic resource configurations for these emerging applications. Not only that, the introduction of DRL makes the CETCN more intelligent. Specifically, delay-sensitive services need to provide extremely low latency for users, while some traffic-intensive applications require extremely high bandwidth. Although DRL-assisted CETCN in related applications is better than other traditional methods, many problems still need to be solved. Next, we will list several possible research directions for the CETCN in future applications.

- **More emerging applications:** For example, in bearing fault detection, such computationally intensive tasks may require low response latency, low energy consumption, and high processing precision. In addition to traditional performance indicators such as delay and energy consumption, various emerging applications in the future may derive various performance indicators according to different tasks. Flexible indicator configuration and higher-dimensional performance optimization problems will be a potential research direction.
- **Security problem:** Due to the increasing number of various terminal devices, more and more data is generated, collected, and processed at the terminal. Therefore, data security is becoming one of the critical obstacles which affects the further development of the CETCN. As a distributed and secure ledger, blockchain technology has been used by more and more scholars to overcome existing security problems in recent years. With the

pseudonym mechanism of the blockchain, edge nodes can protect privacy well when transmitting data to other nodes other than local nodes. At the same time, since edge computing and blockchain are both distributed solutions, they can be well integrated without causing additional scalability problems.

IX. SECURITY ISSUES FOR AI-ENABLED CLOUD-EDGE-TERMINAL COLLABORATIVE NETWORK

With the continuous improvement of model accuracy, the leakage of personal privacy has become increasingly serious. How to ensure that the training process of artificial intelligence models can comply with data protection regulations and protect personal privacy has become an urgent problem that needs to be solved. Differential privacy is a technique used to protect the privacy of individuals' sensitive information while allowing aggregated statistical analysis on the data. It ensures that individual data points cannot be re-identified, even when combined with other external datasets [397]. The application of differential privacy to end-edge-cloud collaboration network can play a crucial role in safeguarding users' sensitive information while enabling efficient and secure data processing across the entire network. Therefore, in Section IX-A, we give a simple example of differential privacy. Then, we make a overview of applying differential privacy to end-edge-cloud collaboration to protect users sensitive information in IX-B. Finally, we give the lesson learned and future directions in Section IX-C.

A. A Toy Example

In the context of end-edge-cloud collaboration network, the collaboration involves multiple entities, such as end-user devices, edge servers, and cloud data centers. These entities exchange data and perform computations collaboratively to provide various services and applications. Differential privacy can be applied at different stages of data processing within this collaboration to protect users' privacy:

- **Data Collection at the End-User Devices:** When sensitive data is collected from end-user devices, differential privacy techniques can be used to add noise to the data before it is transmitted to the edge or cloud. By adding controlled noise, individual data points are obfuscated, ensuring that the original data cannot be inferred from the collected data.
- **Aggregation at the Edge Servers:** In an end-edge-cloud collaboration, edge servers often aggregate data from multiple end-user devices to perform local computations. Differential privacy can be applied to the aggregation process, ensuring that no individual user's data can be distinguished in the aggregated results.
- **Secure Computation in the Cloud:** Differential privacy can also be applied to computations performed in the cloud data centers. When performing joint computations involving data from multiple sources, differential privacy guarantees that the sensitive information of individual users remains protected.

B. Literature Review

In the following, we will do a brief review on differential privacy in intelligent CETCN.

1) *Differential Privacy in RL*: Local differential privacy (LDP), as a powerful privacy protection tool, breaks the constraints of trusted third parties and allows users to encrypt data locally, providing more powerful privacy protection [398]. Reinforcement learning agents collect data and learn the strategies through interaction with the environment. However, the data provided by the environment may involve personal information, such as the private layout of the interior. For example, in [399], the author proposes environmental dynamic search based on genetic algorithm and candidate inference based on shadow strategy to study whether trained DRL strategies will leak privacy information. They have confirmed that the proposed algorithm can accurately infer floor plans.

The author in [400] proposes a LDP algorithm for reinforcement learning, which achieves good robustness in distributed private environments. Specifically, in the proposed A3C algorithm, randomness is introduced through the Laplace method and random projection method. At the same time, randomness is also introduced in the distribution gradient that satisfies LDP to prevent information leakage. The author in [401] studies the problem of random multi-armed bandit (MAB) problem, with the aim of observing the maximum sum of rewards after random destruction. At the same time, they provide an upper bound for the expected regret of any bandit algorithm that incorporates random noise, and provide a lower bound for regret through frequency algorithm and Bayesian algorithm. The experimental results show the effectiveness of the proposed algorithm in recommendation systems that protect user privacy using random perturbations. The author introduces a unified privacy definition for multi-arm bandit algorithms in [402] and derives the lower bound of errors for algorithms that meet these definitions. The proof results indicate that the regret of the intelligent agent increases by a multiplication factor that depends on the level of privacy. The author in [403] proposes an online DRL algorithm based on differential privacy to address the security issues that may arise during DRL training. Specifically, firstly, Gaussian noise is added to the neural network based on privacy budget, and secondly, an automatic encoder is trained to protect the data in the original environment. At the same time, they also demonstrate the budget formula for online deep learning based on differential privacy. Finally, the experiment shows that the proposed algorithm has good decision-making ability while protecting data privacy.

2) *Differential Privacy in CETCN Task Offloading and Resource Allocation*: Due to the limited computing power of various sensors in the IIoT scenario, it is necessary to offload tasks on the production line to the edge server. However, uploading data to the edge server poses a privacy breach issue. Therefore, the author [404] proposes a DRL algorithm based on LDP to provide personalized privacy protection. Due to the ability of LDP to generate different levels of noise, it can meet various privacy requirements for task offloading. At the same time, the author adds priority experience replay into the algorithm to reduce the impact of noise on the QoS

of offloading tasks. Finally, the simulation results show that the algorithm can achieve good privacy protection ability and cost efficiency. The authors in [405] propose a novel online privacy protection offloading mechanism to address the privacy leakage issue in honest but curious MEC server computing offloading, and thus generating real-time and efficient offloading strategies for users. Firstly, they design a DRL based offloading model that allows users to determine satisfactory disturbance offloading ratios based on time-varying channel states, in order to balance user privacy and computational costs. Specifically, in order to protect the real data being offloaded and prevent untrusted MEC servers from leaking user privacy, a range constrained Laplace transform is proposed to blur the original offloading ratio of each user. Finally, a large number of experiments have shown the effectiveness of the proposed algorithm in protecting user privacy. In [406], the author not only considers latency and energy consumption, but also considers a privacy protection offloading scheme that minimizes packet loss rate. Specifically, the author describes the dynamic computing offloading problem as an MDP, and then develops an offloading algorithm based on differential privacy DQN to solve the MDP problem. By injecting noise into the generated offloading decisions, the pattern privacy problem is solved, which prevents attackers from inferring queue information and user usage patterns of edge servers through eavesdropping on offloading decisions. In addition, the generation of noise is achieved by modifying the DQN algorithm through a Gaussian process. Finally, the simulation results show that the algorithm outperforms the greedy algorithm and DQN algorithm in terms of performance. Due to the fact that attackers with user history data in ITS can easily infer the real-time location of users, [407] proposes a privacy oriented task offloading scheme in ITS to resist attackers' attacks. Firstly, the author defines local computing models, communication models, and privacy loss models, where the privacy loss model is defined as the probability of successful attack by an attacker. Then, the optimal offloading problem is formulated as an MDP and a PPO based DRL algorithm is proposed to obtain the optimal solution. Finally, the simulation results indicate that this method can handle large-scale task offloading problems and achieve good offloading strategies while reducing privacy loss, energy consumption, and latency. The authors in [408] propose an offloading scheme for privacy protection in satellite ground collaborative computing. Specifically, this article uses different offloading targets, such as completion time, energy consumption, and reliability, to balance mode privacy and location privacy. Then, the authors introduce a privacy protection algorithm based on DRL to solve the optimal offloading strategy. The experimental results show that the proposed algorithm outperforms the baseline scheme in terms of completion time, energy consumption, and privacy protection level. In order to reduce offloading preference inference attacks in DRL, the authors in [409] propose a privacy aware joint task offloading and power optimization scheme to minimize system costs. By incorporating the designed differential privacy mechanism into the process of action selection and policy update in DRL, privacy protection is achieved. Theoretical analysis shows that this

method can enable each vehicle to make offloading decisions without compromising privacy and achieve Nash equilibrium. Finally, the effectiveness of this method in protecting privacy and improving cost-effectiveness is verified through a large number of experiments.

3) *Differential Privacy in CETCN Caching*: The authors in [410] propose a distributed caching and routing strategy with privacy protection mechanism. Firstly, they propose a distributed Gaussian Seidel iteration method to achieve global optimization of caching and routing problem. Secondly, they design a differential privacy mechanism to protect routing information in the base station and analyze the performance of the proposed algorithm while ensuring performance. Finally, simulation of real-world data demonstrates that the proposed algorithm can significantly reduce costs. In order to cache content without compromising user privacy. The authors in [411] propose a content caching scheme based on LDP. Firstly, content providers collect LDP content preferences of specific users and provide economic incentives to access points that provide content to maximize the benefits of content providers. Secondly, by using an optimized local hash method to add noise to the preferred content, and using the reference distribution constructed from noisy content preference data to predict the popularity of the content. Finally, they provide a data-driven cache benefit scheme and verify its effectiveness through simulation. Due to insufficient state collection between edge servers and mobile users, as well as the presence of malicious attacks, existing edge caching schemes not only struggle to improve video experience quality, but also easily leak personal privacy. To address this issue, the author in [412] proposes a video caching framework based on privacy preserving Q-learning. Firstly, the author uses LDP schemes to protect users' location and preferences. Then, a data aggregation model is proposed to balance aggregation accuracy and privacy. At the same time, the popularity of videos in a certain region is obtained by calculating the service model, and then edge caching with low latency and high hit rate is achieved by setting a caching objective function. Finally, a large number of simulation results indicate that the proposed scheme can protect user privacy and improve the quality of user video experience.

4) *Differential Privacy in CETCN Mobility Management*: Vehicle trajectory data is crucial for location-based services, however, containing private information about families and individuals can raise concerns about data security. By utilizing differential privacy, we can solve this problem by generating synthetic trajectories from raw sensitive data while protecting personal privacy. The existing synthetic trajectories often contain aggregation level information, which loses personal level movement patterns. Therefore, in [413], a new private trajectory solution is proposed, which integrates the public geographic structure of the target area into the private movement trajectory to capture more accurate movement patterns while preserving aggregation level movement behavior. A large number of simulation results indicate that the proposed scheme can improve data utility by 37%. Users' mobile data may contain trade secrets of mobile providers (MP), and mobile data can also provide insights for municipal authorities

(MA) on how to operate, regulate, and improve transportation networks. Due to data security considerations, MP may not be willing to provide datasets to MA. In [414], in order to provide MA with insights on mobile data, a differential privacy based interaction protocol between MP and MA is proposed, and MP does not need to disclose trade secrets. Specifically, the first step is to generate encryption commitments through Merkel commitments and aggregated traffic metrics, while the second step checks the correctness of the received data through the use of commitments and zero knowledge proofs. The author also provides a private version of the protocol suitable for large query mechanisms, which can be easily extended to scenarios with multiple MPs through secure multi-party computation. In order to optimize task allocation in mobile crowdsourcing without exposing user location information, the authors in [415] propose a geographically obfuscated location privacy protection task allocation framework. Specifically, they enable participants to obfuscate their reported locations without the need for any actual third-party trusted entities, while ensuring differential and distorted privacy. In addition, under the constraints of differential privacy and distorted privacy, a MINLP problem is constructed to minimize the expected travel distance of the operator. Finally, a large number of simulation experiments have demonstrated that the proposed scheme can shorten the average transmission distance under the same level of privacy protection. With the development of smart cities, better location-based services are provided by analyzing people's trajectory data. However, the release of raw data poses a serious security threat to individuals. Privacy protection frameworks based on differential privacy often assume that individuals have the same preferences for privacy. The authors in [416] propose a personalized differential privacy publishing mechanism for trajectory data based on individual privacy requirements. It utilizes Hilbert curves to extract the distribution characteristics of each trajectory data, and proposes personalized privacy protect algorithms for trajectories with different privacy preferences. Finally, experiments on real-world trajectory datasets confirm that the proposed mechanism can better balance data privacy and data utility.

C. Lesson Learned and Future Directions

Differential privacy has emerged as a promising technique for privacy protection in various data analysis and sharing scenarios. As the field of data privacy and security continues to evolve, several future directions and advancements are expected in the area of differential privacy:

- **Improved Privacy Guarantees**: Future research will focus on refining and strengthening the privacy guarantees provided by differential privacy. Efforts will be made to reduce the amount of noise introduced in the data while still maintaining a high level of privacy protection.
- **Dynamic Privacy Budget Management**: Managing the privacy budget is critical in differential privacy. Future developments may explore dynamic mechanisms that allocate privacy budget based on the specific data and analysis requirements to achieve better utility while preserving privacy.

- **Privacy for Complex Data Types:** Current differential privacy techniques are well-suited for numerical data, but extending these methods to handle complex data types like text, images, and graphs remains a challenge. Future research will focus on developing privacy-preserving mechanisms for these data types.
- **Privacy in Machine Learning:** Integrating differential privacy with machine learning algorithms is an active area of research. Future directions will explore ways to protect individual data points during the training process while still achieving accurate and useful models.
- **Privacy in Real-Time and Streaming Data:** As data processing becomes more real-time and streaming-oriented, future research will focus on developing differential privacy techniques for streaming data that can handle continuous data updating and adapt to changing data distributions.
- **Federated Differential Privacy:** Federated learning and privacy are closely related concepts. Future developments may explore combining federated learning with differential privacy to achieve privacy-preserving collaborative machine learning across multiple entities.

X. STANDARDIZATION EFFORTS OF EDGE INTELLIGENCE

Standardization refers to the formulation, publication, and implementation of standards for repetitive things and concepts in social practices such as economy, technology, science, and management, in order to achieve the best order and social benefits. The important significance of standardization work lies in improving the practicality of products, processes, and services, preventing trade barriers, and promoting technological cooperation. Moving from 5G to 6G networks, data-driven machine learning and artificial intelligence have become increasingly important for real-time applications, including but not limited to personal intelligence devices and smart factories. At the same time, how to effectively run AI applications at the edge of the network has become a research hotspot. There is currently no unified definition of edge intelligence in the industrial and academic communities. In 2019, the International Electrotechnical Commission defined edge intelligence as the ability to use machine learning algorithms for data collection, storage, analysis, and aggregation at the edges [417]. Therefore, in this subsection, we will review relevant standardization works and technical challenges of the edge intelligence.

The standardization process of edge intelligence is being promoted by ITU-T. ITU-T is responsible for studying technical, operating and tariff questions and issuing recommendations on them with a view to standardizing telecommunications on a worldwide basis [418]. Due to the data loss and network delay caused by the network bottleneck between the terminal and the cloud device, ITU-T suggests to provide enhanced networking capabilities for emerging services and applications through artificial intelligence technology and edge computing technology [419], [420]. For the architecture development of intelligent edge computing, the recommendation [421] specifies the signaling

architecture, protocol interface, protocol process and message format of intelligent edge computing based on microservices. Furthermore, edge cloud platform services can assist in smart agriculture and achieve edge cloud assisted smart agriculture systems. Reference [422] defines the data management interface of intelligent agricultural services based on intelligent edge computing. Recently, the *Third Generation Partner Program* (3GPP) has shown increasing interest in incorporating artificial intelligence into its 5G system, and has identified and reported functional support for the 5G system in recent technical specification documents [423].

A. Standardization Efforts of IEEE

The IEEE has been actively involved in the development of standards related to edge intelligence and edge computing. Standard [424] defines the OpenFog reference architecture, which is a framework aimed at supporting fog computing, including aspects related to edge intelligence. Specifically, in this standard, four aspects related to edge intelligence are defined. Firstly, it defines the ability to perform data analysis and apply machine learning algorithms at the edge of the network. Secondly, the machine learning model can be deployed on edge devices. The third is the ability to perform real-time data processing. Finally, there can be inheritance between edge intelligence and cloud resources to meet complex analysis and data aggregation. Apart from real-time analysis, [425] highlights the possibility of executing complex computational tasks on mobile devices, rather than relying solely on remote cloud servers or data centers. While this standard does not directly address edge intelligence, the concepts related to task offloading are consistent with edge computing. To be specific, edge intelligence may involve dynamically decide whether to execute a task locally on the mobile device or offload it to a nearby edge server or cloud resource based on factors like device resources, network conditions, and user preferences. In [426], IEEE defines the essential concepts and terminology for the IoT. Within the context of IoT, edge intelligence is an integral aspect where data is processed and analyzed at the edge of the network, which is closer to the data source. The standard may acknowledge the importance of edge intelligence in IoT scenarios, where devices and sensors generate vast amounts of data that need to be analyzed, interpreted, and acted upon in real-time at the network edge. Although standard [427] does not focus on edge intelligence directly, time-sensitive network aims to provide deterministic and low-latency communication in Ethernet-based networks, which is critical for supporting time-critical applications such as those used in edge computing and edge intelligence scenarios.

B. Standardization Efforts of ITU

The International Telecommunication Union (ITU) has been actively involved in various standardization efforts related to edge computing and edge intelligence. Although the ITU does not have specific standards solely dedicated to edge intelligence, we summarize the standards of edge computing which involve edge intelligence.

In [428], it points out the requirements for the deployment of edge computing which including the capability of edge intelligence. This standard highlights the importance of edge intelligence in supporting real-time data processing and decision-making at the network edge, enabling timely responses to time-critical applications. In addition, this standard also addresses the need for inter-operability between different edge computing components, including edge intelligence modules, to facilitate seamless integration and data exchange. Standard [429] addresses the essential architectural aspects of edge computing, including highlighting the distribution of computation, storage, and data processing capabilities should be closer to the edge of the network. This standard discusses the orchestration of services and applications within the edge computing architecture, where edge intelligence can dynamically allocate resources and manage services based on changing conditions.

C. Technical Challenges and Requirements

In this subsection, we will discuss several technical challenges and requirements of edge intelligence.

- *Limited Computing Resources*: Edge devices often have limited processing power, memory, and energy resources. Executing sophisticated machine learning algorithms and artificial intelligence tasks on resource-constrained edge devices can be challenging.
- *Data Privacy and Security*: Edge devices generate and process sensitive data, raising concerns about data privacy and security. Safeguarding data at the edge and ensuring secure communication between edge devices and cloud resources are essential challenges.
- *Latency and Real-time Processing*: Edge intelligence applications often require real-time processing and quick decision-making. Ensuring low latency and timely response is a significant challenge.
- *Scalability and Management*: Deploying and managing a large number of edge devices in diverse environments pose challenges related to scalability, orchestration, and configuration management.
- *Cost and Energy Efficiency*: Edge intelligence solutions must consider the cost and energy implications of performing computation and analytics at the edge versus offloading tasks to the cloud.

D. Edge Intelligence for 6G

It is expected that edge intelligence will play a significant role in the development of 6G networks. As with previous generations of wireless communication technologies, 6G is expected to support an even higher level of data throughput, lower latency, and increased connectivity. To achieve these objectives, edge computing and intelligence are likely to be essential components of the 6G ecosystem. Some potential aspects of edge intelligence could be relevant to 6G.

- *Ultra-Low Latency*: Edge intelligence can enable real-time data processing and decision-making at the network edge, contributing to the achievement of ultra-low latency requirements in 6G networks.

- *Enhanced User Experience*: Edge intelligence can personalize services and applications for individual users based on real-time data analysis, leading to enhanced user experiences in 6G networks.
- *Distributed Machine Learning*: Edge intelligence can enable collaborative machine learning and distributed training at the edge, improving the efficiency and scalability of machine learning models in 6G environments.

XI. CHALLENGES AND FUTURE DIRECTIONS

In this section, we discuss challenges and future directions of applying AI to the CETCN.

A. DRL Training Problem

Most research assumes DRL agents are trained in a centralized manner. However, in the CETCN, the DRL agent can be implemented in a cloud server and this centralized training in a cloud server may lead to the following problems: 1) The training data needs to be transmitted to a central server, which makes users more concerned about the leakage of private data. At the same time, transmitting massive data also adds burdens to the communication network and consumes huge amount of energy in general. 2) The limited computing power of the terminals leads to the failure of the training process. Therefore, how to appropriately train the DRL agents need further research.

Federated learning is a promising method to overcome the resource limitation of terminals and release the privacy concerns. First, a well-designed computational offloading framework can enable the training process to be performed in resource-constrained terminal devices [430]. Secondly, federated learning can also be integrated into the distributed training process, which can obtain knowledge from a large amount of data and ensure user privacy at the same time [431].

B. The Integration of Emerging Communication Technologies and DRL

The rapid growth of a variety of emerging applications is driving the development of next-generation wireless technologies, including massive multiple-input multiple-output (Massive MIMO), millimeter-wave communications and reflected intelligent surface (RIS) [432]. However, achieving resource optimization in such a time-varying and unpredictable environment is a very challenging task [433]. DRL is seen as a viable tool to solve dynamic resource optimization. However, with the deployment of ultra-dense networks, the action space will become larger and larger, resulting in the curse of dimensionality. Therefore, it is necessary to find new ways to solve the dimensionality explosion problems, such as multi-agent reinforcement learning (MARL) and game theory [434].

C. The Synchronization Issue of Multi-Agent

The centralized training of MARL puts forward higher requirements for the synchronization of states between agents. In [435], an ideal collection framework has been proposed to

synchronize the state information of all the agents. However, the CETCN is distributed and decentralized in a wide range, and it is difficult for the network vendors to obtain such the collection the global information simultaneously to learn the policy. Furthermore, the unreliability of wireless communication links, such as V2V links and IIoT links, is also an main obstacle to realize the synchronization among agents. Because of these factors, the agents have to learn from past state information and thus inevitably lead to a non-optimal policy.

D. Mixed Reward Design and Multi-Objective Optimization

The design of the return function is crucial to the learning process of the agent. A well-designed return function can greatly increase the efficiency of agent learning. However, the CETCN involves multiple network entities with inconsistent objectives. For example, users want to get the ultimate service experience, while content providers want to get the highest revenue from users. Designing effective reward functions for different entities in the network is still an open issue. The reviewed literature typically define a global reward for all the network vendors or a local reward for an individual agent. Both schemes have their own advantages and disadvantages. The global return can encourage the cooperation between network entities, but it will also generate lazy network entities. At the same time, designing the local return function will make each agent in the network pay all their efforts to obtain the maximum benefits for themselves, but it will also cause the competition of agents for the limited resources in the network. Due to these reasons, the mixed reward has been proposed to improve the learning performance of agents [436]. Furthermore, it is common to design a conflict reward for the CETCN, for example, maximizing the revenue for the content providers while minimizing the service delay of the users.

E. Multi-Agent State Information Extraction

In the MARL framework, each agent makes decisions by using the past state information. However, due to the limited processing capacity of various edge devices, the state information obtained by the agent is often not preprocessed, and resulting in the low learning efficiency of multi-agent reinforcement learning. There exist several methods to extract the state information, such as GNN [437] and attention mechanism [438]. However, these state information extraction technologies usually need to collect information from other agents frequently, thus increasing the burden of communication links. Therefore, it is still an urgent problem to design effective state information extraction technology for the CETCN.

F. Large Scale Heterogeneous Agents Cooperation

The definition of mean-field Nash equilibrium discussed above corresponds to non-cooperative game, in which each agent aims to maximize its own utility which trying to join in the behavior of other selfish agents. However, in the CETCN, edge nodes sometimes tries to cooperative to minimize the total transmission delay of all end users which in line with

cooperative game. At the same time, because the mean field control assumes that players are isomorphic, that is, only by studying a representative player can learn the optimal strategy. However, this assumption is impractical in the cloud edge collaboration system. For example, there are multiple edge computing devices in the edge system, the computing capacity and cache capacity of these edge computing devices are often very different. If the calculation delay per unit time is taken as the return function, the return function of each edge device will be different, and the whole system will no longer be homogeneous. Therefore, reinforcement learning methods in large-scale heterogeneous multi-agent environment need to be further studied.

G. Quantum Computing-Assisted CETCN (QCCETCN)

Quantum computing is a revolutionary technology. It uses quantum mechanism to solve the problems that traditional computers cannot solve, such as traditional encryption and decryption problems. It has greatly influence on existing technology fields, such as artificial intelligence (AI) [439], security [440], and finance [441]. By deploying quantum computers at the edge of the network, users' tasks will be processed faster, so that users can obtain the ultimate service experience from quantum computer providers.

Although quantum computing is in its infancy, the QCCETCN still has the following advantages compared with traditional CETCN. On the one hand, quantum computer can perform complex calculations while transmitting and processing data. For example, the Google Sycamore nuclear machine can complete a task within 200 seconds, while a traditional computer takes 10000 years to complete the task [442]. On the other hand, DRL algorithms need a heavy learning phase, which require huge computing capacities [443], [444]. Therefore, the computing power for processing massive amount of data is hard to reach for traditional CETCN. But fortunately, quantum machine learning (QML) utilizes the principles of quantum mechanics to develop new algorithms that can solve complex problems in a very fast manner [445]. In this sense, quantum computing is suitable for the CETCN to accomplish computationally heavy and real-time applications.

H. The Convergence Proof of Applying DRL in CETCN

Deep reinforcement learning obtains the optimal policy through iterative interaction with the complex and dynamic environment, but this optimization algorithm that obtains the optimal solution through interactive learning leads to difficulties in proving the convergence of the algorithm. In the existing literature that use DRL to solve the optimization problem in CETCN, few of them mention the proof of convergence [446], and most of them illustrate the convergence of the algorithm through simulation. It is generally not convincing to illustrate the convergence of the algorithm through simulation. Therefore, when using the DRL algorithm to solve the CETCN problem, a more rigorous mathematical proof of convergence needs to be further investigated.

I. Lightweight Privacy Protection Schemes

Massive emerging applications in CETCN that need to exchange sensitive data. For example, household electricity consumption reports contain massive private information. This privacy information needs to be accessed by different professions and organizations, including electrical distribution personnel, and employees of the national grid. In this case, easy access to data is required in order to make real-time decisions. At the same time, data access should be protected to prevent unauthorized modification. However, existing methods to protect data privacy, such as blockchain and asymmetric key encryption, are computationally intensive and greatly reduce the ability to interact in real-time when identifying access requests and modifications to authorized data [447], [448], [449]. Therefore, it is necessary to develop lightweight data protection methods that can support real-time interaction while ensuring secure data management.

XII. CONCLUSION

As an emerging network paradigm, the CETCN exploits advanced communication technology to enable the system to collect, exchange, analyze and process data at the terminal layer, edge layer and cloud layer, thus solving the problem of the high latency and high energy consumption in the traditional networks. At the same time, AI also provides the CETCN with high-value and real-time decision-making ability in an unprecedented way. In this paper, we provide a review of the literature on AI-enabled CETCN, especially in collaborative task offloading, resource allocation, caching, mobility management, and their applications in emerging scenarios. Firstly, we review the related concepts in the AI-enabled CETCN and provide a comprehensive tutorial for task offloading, resource allocation, collaborative caching and mobility management and give two use cases in AI-enabled collaborative cloud-edge-terminal collaborative network. Secondly, we point out a few research challenges and future directions in each part of this tutorial. Finally, we highlight the research challenges of reinforcement learning and point out several possible development directions in the convergence of emerging technologies and the CETCN.

REFERENCES

- [1] Y. Wang, S. Yang, X. Ren, P. Zhao, C. Zhao, and X. Yang, "IndustEdge: A time-sensitive networking enabled edge-cloud collaborative intelligent platform for smart industry," *IEEE Trans. Ind. Informat.*, vol. 18, no. 4, pp. 2386–2398, Apr. 2022.
- [2] X. Zhou, W. Liang, J. She, Z. Yan, and K. I.-K. Wang, "Two-layer federated learning with heterogeneous model aggregation for 6G supported Internet of Vehicles," *IEEE Trans. Veh. Technol.*, vol. 70, no. 6, pp. 5308–5317, Jun. 2021.
- [3] T. Yu, X. Wang, and J. Hu, "A fast hierarchical physical topology update scheme for edge-cloud collaborative IoT systems," *IEEE/ACM Trans. Netw.*, vol. 29, no. 5, pp. 2254–2266, Oct. 2021.
- [4] Z. He, T. Zhang, and R. B. Lee, "Attacking and protecting data privacy in edge-cloud collaborative inference systems," *IEEE Internet Things J.*, vol. 8, no. 12, pp. 9706–9716, Jun. 2021.
- [5] D. Yang, E. Cui, H. Wang, and H. Zhang, "EH-edge—An energy harvesting-driven edge IoT platform for online failure prediction of rail transit vehicles: A case study of a cloud, edge, and end device collaborative computing paradigm," *IEEE Veh. Technol. Mag.*, vol. 16, no. 2, pp. 95–103, Jun. 2021.
- [6] J. Yuan, H. Xiao, Z. Shen, T. Zhang, and J. Jin, "ELECT: Energy-efficient intelligent edge-cloud collaboration for remote IoT services," *Future Gen. Comp. Syst.*, vol. 147, pp. 179–194, Oct. 2023.
- [7] H. Liao et al., "Cloud-edge-device collaborative reliable and communication-efficient digital twin for low-carbon electrical equipment management," *IEEE Trans. Ind. Informat.*, vol. 19, no. 2, pp. 1715–1724, Feb. 2023.
- [8] W. Saad, M. Bennis, and M. Chen, "A vision of 6G wireless systems: Applications, trends, technologies, and open research problems," *IEEE Netw.*, vol. 34, no. 3, pp. 134–142, May/Jun. 2020.
- [9] Y. Wei and M. B. Blake, "Service-oriented computing and cloud computing: Challenges and opportunities," *IEEE Internet Comput.*, vol. 14, no. 6, pp. 72–75, Nov./Dec. 2010.
- [10] H. Takabi, J. B. D. Joshi, and G.-J. Ahn, "Security and privacy challenges in cloud computing environments," *IEEE Security Privacy*, vol. 8, no. 6, pp. 24–31, Nov./Dec. 2010.
- [11] A. C. Baktir, A. Ozgovde, and C. Ersoy, "How can edge computing benefit from software-defined networking: A survey, use cases, and future directions," *IEEE Commun. Surveys Tuts*, vol. 19, no. 4, pp. 2359–2391, 4th Quart., 2017.
- [12] X. Lin, Y. Wang, Q. Xie, and M. Pedram, "Task scheduling with dynamic voltage and frequency scaling for energy minimization in the mobile cloud computing environment," *IEEE Trans. Services Comput.*, vol. 8, no. 2, pp. 175–186, Mar. 2015.
- [13] H. Zhang, Q. Zhang, and X. Du, "Toward vehicle-assisted cloud computing for smartphones," *IEEE Trans. Veh. Technol.*, vol. 64, no. 12, pp. 5610–5618, Dec. 2015.
- [14] J. Zheng, Y. Cai, Y. Wu, and X. Shen, "Dynamic computation offloading for mobile cloud computing: A stochastic game-theoretic approach," *IEEE Trans. Mobile Comput.*, vol. 18, no. 4, pp. 771–786, Apr. 2019.
- [15] X. He and W. Dou, "Offloading deadline-aware task in edge computing," in *Proc. IEEE 13th Int. Conf. Cloud Comput. (CLOUD)*, Beijing, China, Oct. 2020, pp. 28–30.
- [16] Z. Wu and D. Yan, "Deep reinforcement learning-based computation offloading for 5G vehicle-aware multi-access edge computing network," *China Commun.*, vol. 18, no. 11, pp. 26–41, Nov. 2021.
- [17] X. Duan, F. Xu, and Y. Sun, "Research on offloading strategy in edge computing of Internet of Things," in *Proc. Int. Conf. Comput. Netw. Electron. Autom. (ICCNEA)*, Xi'an, China, Sep. 2020, pp. 206–210.
- [18] L. Lin, X. Liao, H. Jin, and P. Li, "Computation offloading toward edge computing," *Proc. IEEE*, vol. 107, no. 8, pp. 1584–1607, Aug. 2019.
- [19] C. Kai, H. Zhou, Y. Yi, and W. Huang, "Collaborative cloud-edge-end task offloading in mobile-edge computing networks with limited communication capability," *IEEE Trans. Cogn. Commun. Netw.*, vol. 7, no. 2, pp. 624–634, Jun. 2021.
- [20] G. Zhou, L. Zhao, Y. Wang, G. Zheng, and L. Hanzo, "Energy efficiency and delay optimization for edge caching aided video streaming," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 14116–14121, Nov. 2020.
- [21] J. Ren, G. Yu, Y. He, and G. Y. Li, "Collaborative cloud and edge computing for latency minimization," *IEEE Trans. Veh. Technol.*, vol. 68, no. 5, pp. 5031–5044, May 2019.
- [22] Q. Jiang et al., "Game theory-based task offloading and resource allocation for vehicular networks in edge-cloud computing," in *Proc. IEEE Int. Conf. Web Services (ICWS)*, Chicago, IL, USA, Nov. 2021, pp. 341–346.
- [23] J. Zhao, Q. Li, Y. Gong, and K. Zhang, "Computation offloading and resource allocation for cloud assisted mobile edge computing in vehicular networks," *IEEE Trans. Veh. Technol.*, vol. 68, no. 8, pp. 7944–7956, Aug. 2019.
- [24] J. Lee, D. Kim, and D. Niyato, "Market analysis of distributed learning resource management for Internet of Things: A game-theoretic approach," *IEEE Internet Things J.*, vol. 7, no. 9, pp. 8430–8439, Sep. 2020.
- [25] D. Wu, R. Bao, Z. Li, H. Wang, H. Zhang, and R. Wang, "Edge-cloud collaboration enabled video service enhancement: A hybrid human-artificial intelligence scheme," *IEEE Trans. Multimedia*, vol. 23, pp. 2208–2221, Mar. 2021.
- [26] Y. Dai, D. Xu, S. Maharjan, G. Qiao, and Y. Zhang, "Artificial intelligence empowered edge computing and caching for Internet of Vehicles," *IEEE Wireless Commun.*, vol. 26, no. 3, pp. 12–18, Jun. 2019.
- [27] G. Manogaran, S. Mumtaz, C. X. Mavromoustakis, E. Pallis, and G. Mastroakis, "Artificial intelligence and blockchain-assisted offloading approach for data availability maximization in edge nodes," *IEEE Trans. Veh. Technol.*, vol. 70, no. 3, pp. 2404–2412, Mar. 2021.

- [28] S. Borst, V. Gupta, and A. Walid, "Distributed caching algorithms for content distribution networks," in *Proc. IEEE INFOCOM*, San Diego, CA, USA, Mar. 2010, pp. 1–9.
- [29] X. Wang, C. Wang, X. Li, V. C. M. Leung, and T. Taleb, "Federated deep reinforcement learning for Internet of Things with decentralized cooperative edge caching," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 9441–9455, Oct. 2020.
- [30] M. E. Crovella and A. Bestavros, "Self-similarity in World Wide Web traffic: Evidence and possible causes," *IEEE/ACM Trans. Netw.*, vol. 5, no. 6, pp. 835–846, Dec. 1997.
- [31] M. Y. Cha, K. Haewoon, R. Pablo, A. Yong-Yeol, and M. Sue, "I tube, you tube, everybody tubes: Analyzing the worlds largest user generated content video system," in *Proc. ACM SIGCOMM Conf. Internet Meas. (SIGCOMM)*, New York, NY, USA, Oct. 2007, pp. 1–14.
- [32] D. F. Towsley, "Mobility models for wireless networks: Challenges, pitfalls, and successes," in *Proc. 22nd Workshop Princ. Adv. Distrib. Simulat. (PADS)*, Rome, Italy, Jun. 2008, p. 3.
- [33] C. Meurisch et al., "Temporal coverage analysis of router-based cloudlets using human mobility patterns," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Singapore, Dec. 2017, pp. 1–6.
- [34] I. Farris, T. Taleb, M. Bagaa, and H. Flick, "Optimizing service replication for mobile delay-sensitive applications in 5G edge network," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Paris, France, May 2017, pp. 1–6.
- [35] T. Q. Dinh, J. Tang, Q. D. La, and T. Q. S. Quek, "Offloading in mobile edge computing: Task allocation and computational frequency scaling," *IEEE Trans. Commun.*, vol. 65, no. 8, pp. 3571–3584, Aug. 2017.
- [36] B. P. Rimal, D. P. Van, and M. Maier, "Mobile-edge computing versus centralized cloud computing over a converged FiWi access network," *IEEE Trans. Netw. Service Manag.*, vol. 14, no. 3, pp. 498–513, Sep. 2017.
- [37] A. N. Al-Shuwaili and O. Simeone, "Energy-efficient resource allocation for mobile edge computing-based augmented reality applications," *IEEE Wireless Commun. Lett.*, vol. 6, no. 3, pp. 398–401, Jun. 2017.
- [38] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, "A survey on mobile edge computing: The communication perspective," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 4, pp. 2322–2358, 4th Quart., 2017.
- [39] P. Mach and Z. Becvar, "Mobile edge computing: A survey on architecture and computation offloading," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 3, pp. 1628–1656, 3rd Quart., 2017.
- [40] J. Ni, K. Zhang, X. Lin, and X. Shen, "Securing fog computing for Internet of Things applications: Challenges and solutions," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 1, pp. 601–628, 1st Quart., 2018.
- [41] J. Moura and D. Hutchison, "Game theory for multi-access edge computing: Survey, use cases, and future trends," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 1, pp. 260–288, 1st Quart., 2019.
- [42] H. Wang et al., "Architectural design alternatives based on cloud/edge/fog computing for connected vehicles," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 4, pp. 2349–2377, 4th Quart., 2020.
- [43] X. Wang, Y. Han, V. C. M. Leung, D. Niyato, X. Yan, and X. Chen, "Convergence of edge computing and deep learning: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 2, pp. 869–904, 2nd Quart., 2020.
- [44] T. Qiu, J. Chi, X. Zhou, Z. Ning, M. Atiquzzaman, and D. O. Wu, "Edge computing in Industrial Internet of Things: Architecture, advances and challenges," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 4, pp. 2462–2488, 4th Quart., 2020.
- [45] T. K. Rodrigues, K. Suto, H. Nishiyama, J. Liu, and N. Kato, "Machine learning meets computation and communication control in evolving edge and cloud: Challenges and future perspective," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 1, pp. 38–67, 1st Quart., 2020.
- [46] L. Lei, Y. Tan, K. Zheng, S. Liu, K. Zhang, and X. Shen, "Deep reinforcement learning for autonomous Internet of Things: Model, applications and challenges," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 3, pp. 1722–1760, 3rd Quart., 2020.
- [47] D. E. Sarmiento, A. Lebre, L. Nussbaum, and A. Chari, "Decentralized SDN control plane for a distributed cloud-edge infrastructure: A survey," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 256–281, 1st Quart., 2021.
- [48] Y. Siriwardhana, P. Porambage, M. Liyanage, and M. Ylianttila, "A survey on mobile augmented reality with 5G mobile edge computing: Architectures, applications, and technical aspects," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 2, pp. 1160–1192, 2nd Quart., 2021.
- [49] Q. Luo, S. Hu, C. Li, G. Li, and W. Shi, "Resource scheduling in edge computing: A survey," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 4, pp. 2131–2165, 4th Quart., 2021.
- [50] W. Chen, X. Qiu, T. Cai, H.-N. Dai, Z. Zheng, and Y. Zhang, "Deep reinforcement learning for Internet of Things: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 3, pp. 1659–1692, 3rd Quart., 2021.
- [51] T. Li et al., "Applications of multi-agent reinforcement learning in future Internet: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 24, no. 2, pp. 1240–1279, 2nd Quart., 2022.
- [52] R. Jain and S. Paul, "Network virtualization and software defined networking for cloud computing: A survey," *IEEE Commun. Mag.*, vol. 51, no. 11, pp. 24–31, Nov. 2013.
- [53] R. Yu, G. Xue, V. T. Kilar, and X. Zhang, "Network function virtualization in the multi-tenant cloud," *IEEE Netw.*, vol. 29, no. 3, pp. 42–47, Jun. 2015.
- [54] B. Han, V. Gopalakrishnan, L. Ji, and S. Lee, "Network function virtualization: Challenges and opportunities for innovations," *IEEE Commun. Mag.*, vol. 53, no. 2, pp. 90–97, Feb. 2015.
- [55] A. Gumaste, T. Das, K. Khandwala, and I. Monga, "Network hardware virtualization for application provisioning in core networks," *IEEE Commun. Mag.*, vol. 55, no. 2, pp. 152–159, Feb. 2017.
- [56] Z. Liu et al., "Cloud-based video-on-demand services for smart TV," in *Proc. 7th Int. Conf. Informat. Sci. Technol. (ICIST)*, May 2017, pp. 81–84.
- [57] X. Wu, M. C. Wang, W. Zhang, and Y. Guo, "Cloud program with a pricing strategy for IaaS in cloud computing," in *Proc. IEEE 26th Int. Parallel Distrib. Process. Symp. Workshops Ph.D. Forum (IPDPSW)*, Shanghai, China, Aug. 2012, pp. 2316–2319.
- [58] J. Seol, S. Jin, and S. Maeng, "Secure storage service for IaaS cloud users," in *Proc. 13th IEEE/ACM Int. Symp. Cluster Cloud Grid Comput. (CCGRID)*, Delft, The Netherlands, Jun. 2013, pp. 190–191.
- [59] T. Wang, X. Chang, and B. Liu, "Performability analysis for IaaS cloud data center," in *Proc. 17th Int. Conf. Parallel Distrib. Comput. Appl. Technol. (PDCAT)*, Guangzhou, China, Jun. 2016, pp. 91–94.
- [60] C. Pahl, "Containerization and the PaaS Cloud," *IEEE Cloud Comput.*, vol. 2, no. 3, pp. 24–31, May 2015.
- [61] C. Mouradian et al., "An IoT platform-as-a-service for NFV-based hybrid cloud/fog systems," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 6102–6115, Jul. 2020.
- [62] Y. Verginadis et al., "Context-aware policy enforcement for PaaS-enabled access control," *IEEE Trans. Cloud Comput.*, vol. 10, no. 1, pp. 276–291, Jan.–Mar. 2022.
- [63] L. Wu, S. K. Garg, S. Versteeg, and R. Buyya, "SLA-based resource provisioning for hosted software-as-a-service applications in cloud computing environments," *IEEE Trans. Services Comput.*, vol. 7, no. 3, pp. 465–485, Jul.–Sep. 2014.
- [64] S. Sharif, P. Watson, J. Taheri, S. Nepal, and A. Y. Zomaya, "Privacy-aware scheduling SaaS in high performance computing environments," *IEEE Trans. Parallel Distrib. Syst.*, vol. 28, no. 4, pp. 1176–1188, Apr. 2017.
- [65] S. Aleem, F. Ahmed, R. Batool, and A. Khattak, "Empirical investigation of key factors for SaaS architecture," *IEEE Trans. Cloud Comput.*, vol. 9, no. 3, pp. 1037–1049, Jul.–Sep. 2021.
- [66] D. C. Marinescu, *Cloud Computing: Theory and Practice*. Cambridge, MA, USA: Morgan Kaufmann, 2022.
- [67] D. Puthal, B. P. S. Sahoo, S. Mishra, and S. Swain, "Cloud computing features, issues, and challenges: A big picture," in *Proc. Int. Conf. Comput. Intell. Netw.*, Odisha, India, Jan. 2015, pp. 116–123.
- [68] S. Singh, I. Chana, and R. Buyya, "STAR: SLA-aware autonomic management of cloud resources," *IEEE Trans. Cloud Comput.*, vol. 8, no. 4, pp. 1040–1053, Oct.–Dec. 2020.
- [69] J. Mateo-Forns, F. Solsona-Tehs, J. Vilaplana-Mayoral, I. Teixid-Torrelles, and J. Rius-Torrent, "CART, a decision SLA model for SaaS providers to keep QoS regarding availability and performance," *IEEE Access*, vol. 7, pp. 38195–38204, 2019.
- [70] D. Kimovski, R. Math, J. Hammer, N. Mehran, H. Hellwagner, and R. Prodan, "Cloud, fog, or edge: Where to compute?" *IEEE Internet Comput.*, vol. 25, no. 4, pp. 30–36, Jul./Aug. 2021.
- [71] S. K. Grewal and N. Mangla, "Deadline and cost optimization based task scheduling (DCOTS) in cloud computing environment," in *Proc. 4th Int. Conf. Intell. Eng. Manag. (ICIEM)*, London, U.K., Jul. 2023, pp. 1–6.
- [72] N. Dimitri, "Pricing cloud IaaS computing services," *J. Cloud Comput.*, vol. 9, no. 1, p. 14, Mar. 2020.
- [73] P. Kumar and R. Kumar, "Issues and challenges of load balancing techniques in cloud computing: A survey," *ACM Comput. Surveys*, vol. 51, no. 6 pp. 1–35, Feb. 2019.

- [74] H. Tabrizchi and M. K. Rafsanjani, "A survey on security challenges in cloud computing: Issues, threats, and solutions," *J. Supercomput.*, vol. 76, no. 12, pp. 9493–9532, Feb. 2020.
- [75] E. Jonas et al., "Cloud programming simplified: A Berkeley view on serverless computing," Feb. 2019, *arXiv:1902.03383*.
- [76] H. Shafiei, A. Khonsari, and P. Mousavi, "Serverless computing: A survey of opportunities, challenges, and applications," *ACM Comput. Surveys*, vol. 54, no. 11, pp. 1–32, Nov. 2022.
- [77] Y. Li, Y. Lin, Y. Wang, K. Ye, and C. Xu, "Serverless computing: State-of-the-art, challenges and opportunities," *IEEE Trans. Services Comput.*, vol. 16, no. 2, pp. 1522–1539, Mar./Apr. 2023.
- [78] A. Alnoman, S. K. Sharma, W. Ejaz, and A. Anpalagan, "Emerging edge computing technologies for distributed IoT systems," *IEEE Netw.*, vol. 33, no. 6, pp. 140–147, Nov./Dec. 2019.
- [79] G. H. S. Carvalho, I. Woungang, A. Anpalagan, M. Jaseemuddin, and E. Hossain, "Intercloud and HetNet for mobile cloud computing in 5G systems: Design issues, challenges, and optimization," *IEEE Netw.*, vol. 31, no. 3, pp. 80–89, May/Jun. 2017.
- [80] M. Aazam, S. Zeadally, and K. A. Harras, "Fog computing architecture, evaluation, and future research directions," *IEEE Commun. Mag.*, vol. 56, no. 5, pp. 46–52, May 2018.
- [81] E. Ahmed et al., "Bringing computation closer toward the user network: Is edge computing the solution?" *IEEE Commun. Mag.*, vol. 55, no. 11, pp. 138–144, Nov. 2017.
- [82] E. K. Markakis et al., "Efficient next generation emergency communications over multi-access edge computing," *IEEE Commun. Mag.*, vol. 55, no. 11, pp. 92–97, Nov. 2017.
- [83] P. Zhang, C. Wang, C. Jiang, and A. Benslimane, "UAV-assisted multi-access edge computing: Technologies and challenges," *IEEE Internet Things Mag.*, vol. 4, no. 4, pp. 12–17, Dec. 2021.
- [84] "Mobile-edge computing introductory technical white paper, white paper, mobile-edge computing industry initiative," ETSI, Sophia Antipolis, France, White Paper, Sep. 2014. [Online]. Available: <https://portal.etsi.org/portals/0/tpages/mec/docs/mobile-edgecomputing-introductory-technical-white-paperv1>
- [85] P. Corcoran and S. K. Datta, "Mobile-edge computing and the Internet of Things for consumers: Extending cloud computing and services to the edge of the network," *IEEE Consum. Electron. Mag.*, vol. 5, no. 4, pp. 73–74, Oct. 2016.
- [86] Y. Yu, "Mobile edge computing towards 5G: Vision, recent progress, and open challenges," *China Commun.*, vol. 13, no. 2, pp. 89–99, Feb. 2016.
- [87] T. X. Tran, A. Hajisami, P. Pandey, and D. Pompili, "Collaborative mobile edge computing in 5G networks: New paradigms, scenarios, and challenges," *IEEE Commun. Mag.*, vol. 55, no. 4, pp. 54–61, Apr. 2017.
- [88] D. Sabella, N. Nikaiein, A. Huang, J. Xhembulla, G. Malnati, and S. Scarpina, "A hierarchical MEC architecture: Experimenting the RAVEN use-case," in *Proc. IEEE 87th Veh. Technol. Conf. (VTC-Spr)*, Porto, Portugal, Jul. 2018, pp. 1–5.
- [89] E. Schiller, N. Nikaiein, E. Kalogiton, M. Gasparyan, and T. Braun, "CDS-MEC: NFV/SDN-based application management for MEC in 5G systems," *Comput. Netw.*, vol. 135, pp. 96–107, Apr. 2018.
- [90] Y. Jararweh, A. Doulat, O. AlQudah, E. Ahmed, M. Al-Ayyoub, and E. Benkhelifa, "The future of mobile cloud computing: Integrating cloudlets and mobile edge computing," in *Proc. 23rd Int. Conf. Telecommun. (ICT)*, Thessaloniki, Greece, Jun. 2016, pp. 1–5.
- [91] L. Li, Y. Li, and R. Hou, "A novel mobile edge computing-based architecture for future cellular vehicular networks," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, San Francisco, CA, USA, May 2017, pp. 1–6.
- [92] N. Abbas, Y. Zhang, A. Taherkordi, and T. Skeie, "Mobile edge computing: A survey," *IEEE Internet Things J.*, vol. 5, no. 1, pp. 450–465, Feb. 2018.
- [93] S. Hur, Y.-J. Cho, J. Lee, N.-G. Kang, J. Park, and H. Benn, "Synchronous channel sounder using horn antenna and indoor measurements on 28 GHz," in *Proc. IEEE Int. Black Sea Conf. Commun. Netw. (BlackSeaCom)*, Odessa, Ukraine, Jul. 2014, pp. 83–87.
- [94] J. N. Murdock, E. Ben-Dor, Y. Qiao, J. I. Tamir, and T. S. Rappaport, "A 38 GHz cellular outage study for an urban outdoor campus environment," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Paris, France, Jun. 2012, pp. 3085–3090.
- [95] L. M. Contreras and C. J. Bernardos, "Overview of architectural alternatives for the integration of ETSI MEC environments from different administrative domains," *Electronics*, vol. 9, no. 9, p. 1392, Aug. 2020.
- [96] P. Porambage, J. Okwuibe, M. Liyanage, M. Ylianttila, and T. Taleb, "Survey on multi-access edge computing for Internet of Things realization," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 2961–2991, 4th Quart., 2018.
- [97] W. Duan, J. Gu, M. Wen, G. Zhang, Y. Ji, and S. Mumtaz, "Emerging technologies for 5G-IoV networks: Applications, trends and opportunities," *IEEE Netw.*, vol. 34, no. 5, pp. 283–289, Sep./Oct. 2020.
- [98] J. Ahn, J. Lee, D. Niyato, and H.-S. Park, "Novel QoS-guaranteed orchestration scheme for energy-efficient mobile augmented reality applications in multi-access edge computing," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 13631–13645, Nov. 2020.
- [99] D. Sabella et al., "Mobile-edge computing architecture: The role of MEC in the Internet of Things," *IEEE Consum. Electron. Mag.*, vol. 5, no. 4, pp. 84–91, Oct. 2016.
- [100] M. P. Anastasopoulos, A. Tzanakaki, and D. Simeonidou, "Scalable service chaining in MEC-assisted 5G networks," *J. Lightw. Technol. J.*, vol. 37, no. 16, pp. 4115–4124, Jun. 15, 2019.
- [101] J. Lee and H. Kim, "Security and privacy challenges in the Internet of Things [security and privacy matters]," *IEEE Consum. Electron. Mag.*, vol. 6, no. 3, pp. 134–136, Jun. 2017.
- [102] M. Satyanarayanan, P. Bahl, R. Caceres, and N. Davies, "The case for VM-based cloudlets in mobile computing," *IEEE Pervasive Comput.*, vol. 8, no. 4, pp. 14–23, Oct.–Dec. 2009.
- [103] *Understanding 5G: Perspectives on Future Technological Advancements in Mobile*. London, U.K.: GSMA Intell., Dec. 2014.
- [104] A. Somov and R. Giaffreda, "Powering IoT devices: Technologies and opportunities," *IEEE IoT Newslett.*, Nov. 2015.
- [105] R. Kemp et al., "EyeIdentify: Multimedia cyber foraging from a smartphone," in *Proc. IEEE Int. Symp. Multimedia (ISM)*, San Diego, CA, USA, Dec. 2009, pp. 392–399.
- [106] B. Shi, J. Yang, Z. Huang, and P. Hui, "Offloading guidelines for augmented reality applications on wearable devices," in *Proc. 23rd ACM Int. Symp. Multimedia (MM)*, Brisbane, QLD, Australia, Oct. 2015, pp. 1271–1274.
- [107] H. Suo, Z. Liu, J. Wan, and K. Zhou, "Security and privacy in mobile cloud computing," in *Proc. IEEE Int. Wireless Commun. Mobile Comput. Conf. (IWCMC)*, Cagliari, Italy, Jul. 2013, pp. 655–659.
- [108] "Mobile-edge computing (MEC): Service scenarios." 2021. [Online]. Available: <http://www.etsi.org/deliver/etsigs/MEC-IEG/001099/004/01.01.0160/gsmec-ieg004v010101p.pdf>
- [109] F. Bonomi, R. Milito, J. Zhu, and S. Addepalli, "Fog computing and its role in the Internet of Things," in *Proc. 1st MCC Workshop Mobile Cloud Comput.*, Helsinki, Finland, Aug. 2012, pp. 13–16.
- [110] C. Mouradian, D. Naboulsi, S. Yangui, R. H. Glitho, M. J. Morrow, and P. A. Polakos, "A comprehensive survey on fog computing: State-of-the-art and research challenges," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 1, pp. 416–464, 1st Quart., 2017.
- [111] N. Bessis and C. Dobre, *Big Data and Internet of Things: A Roadmap for Smart Environments*. Cham, Switzerland: Springer, Jun. 2014, pp. 169–186.
- [112] "Openfog architecture overview," Openfog Consortium Archit. Working Group, Tokyo, Japan, White Paper, 2016.
- [113] C. Kai, H. Zhou, Y. Yi, and W. Huang, "Collaborative cloud-edge-end task offloading in mobile-edge computing networks with limited communication capability," *IEEE Trans. Cogn. Commun. Netw.*, vol. 7, no. 2, pp. 624–634, Jun. 2021.
- [114] H. Xiao, J. Huang, Z. Hu, M. Zheng, and K. Li, "Collaborative cloud-edge-end task offloading in MEC-based small cell networks with distributed wireless backhaul," *IEEE Trans. Netw. Service Manag.*, early access, Apr. 11, 2023, doi: [10.1109/TNSM.2023.3266238](https://doi.org/10.1109/TNSM.2023.3266238).
- [115] Z. Hu, C. Fang, Z. Wang, S.-M. Tseng, and M. Dong, "Many-objective optimization based-content popularity prediction for cache-assisted cloud-edge-end collaborative IoT networks," *IEEE Internet Things J.*, early access, Jun. 29, 2023, doi: [10.1109/JIOT.2023.3290793](https://doi.org/10.1109/JIOT.2023.3290793).
- [116] Z. Wang, Z. Zhou, H. Zhang, G. Zhang, H. Ding, and A. Farouk, "AI-based cloud-edge-device collaboration in 6G space-air-ground integrated power IoT," *IEEE Wireless Commun.*, vol. 29, no. 1, pp. 16–23, Feb. 2022.
- [117] D. C. Nguyen et al., "Enabling AI in future wireless networks: A data life cycle perspective," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 553–595, 1st Quart., 2021.
- [118] K. Samdanis, A. N. Abbou, J. Song, and T. Taleb, "AI/ML service enablers & model maintenance for beyond 5G networks," *IEEE Netw.*, early access, Jan. 9, 2023, doi: [10.1109/MNET.129.2200417](https://doi.org/10.1109/MNET.129.2200417).
- [119] S. Sen, E. J. Husom, A. Goknil, S. Tverdal, P. Nguyen, and I. Mancisidor, "Taming data quality in AI-enabled Industrial Internet of Things," *IEEE Softw.*, vol. 39, no. 6, pp. 35–42, Nov./Dec. 2022.

- [120] W. Tong and G. Y. Li, "Nine challenges in artificial intelligence and wireless communications for 6G," *IEEE Wireless Commun.*, vol. 29, no. 4, pp. 140–145, Aug. 2022.
- [121] M. Merluzzi et al., "The hexa-X project vision on artificial intelligence and machine learning-driven communication and computation co-design for 6G," *IEEE Access*, vol. 11, pp. 65620–65648, 2023.
- [122] X. Wang, H. Li, and Q. Wu, "Optimizing adaptive coding and modulation for satellite network with ML-based CSI prediction," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Marrakesh, Morocco, Oct. 2019, pp. 1–6.
- [123] S. K. Punia, M. Kumar, T. Stephan, G. G. Deverajan, and R. Patan, "Performance analysis of machine learning algorithms for big data classification: ML and AI-based algorithms for big data analysis," *Int. J. e-Health Med. Commun.*, vol. 12, no. 4, pp. 60–75, 2021.
- [124] X.-Y. Zhang, C.-L. Liu, and C. Y. Suen, "Towards robust pattern recognition: A review," *Proc. IEEE*, vol. 108, no. 6, pp. 894–922, Jun. 2020.
- [125] J. M. DeAlmeida et al., "Abnormal behavior detection based on traffic pattern categorization in mobile networks," *IEEE Trans. Netw. Services Manag.*, vol. 18, no. 4, pp. 4213–4224, Dec. 2021.
- [126] P. Guruvareddiar and P. Prasad, "Artificial intelligence based region of interest enhanced video compression," in *Proc. Data Comput. Conf. (DCC)*, Jun. 2020, pp. 373–373.
- [127] S. Bundasak and P. Wittayasirikul, "Predictive maintenance using AI for motor health prediction system," in *Proc. Int. Electron Eng. Congr. (iEECON)*, Mar. 2022, pp. 1–9.
- [128] D. Nallaperuma et al., "Online incremental machine learning platform for big data-driven smart traffic management," *IEEE Trans. Intell. Trans. Syst.*, vol. 20, no. 12, pp. 4679–4690, Dec. 2019.
- [129] C. Zhang, H. Zhang, J. Qiao, D. Yuan, and M. Zhang, "Deep transfer learning for intelligent cellular traffic prediction based on cross-domain big data," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 6, pp. 1389–1401, Jun. 2019.
- [130] X. Li, G. Zhu, K. Shen, W. Yu, Y. Gong, and K. Huang, "Joint annotator-and-spectrum allocation in wireless networks for crowd labeling," *IEEE Trans. Wireless Commun.*, vol. 19, no. 9, pp. 6116–6129, Sep. 2020.
- [131] H. Yang, J. Zhao, K.-Y. Lam, Z. Xiong, Q. Wu, and L. Xiao, "Distributed deep reinforcement learning-based spectrum and power allocation for heterogeneous networks," *IEEE Trans. Wireless Commun.*, vol. 21, no. 9, pp. 6935–6948, Sep. 2022.
- [132] Y. Gong, H. Yao, J. Wang, L. Jiang, and F. R. Yu, "Multi-agent driven resource allocation and interference management for deep edge networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 2, pp. 2018–2030, Feb. 2022.
- [133] A. H. Sodhro et al., "Toward ML-based energy-efficient mechanism for 6G enabled industrial network in box systems," *IEEE Trans. Ind. Informat.*, vol. 17, no. 10, pp. 7185–7192, Oct. 2021.
- [134] V.-L. Nguyen, P.-C. Lin, B.-C. Cheng, R.-H. Hwang, and Y.-D. Lin, "Security and privacy for 6G: A survey on prospective technologies and challenges," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 4, pp. 2384–2428, 4th Quart., 2021.
- [135] F. Bonomi, R. Milito, P. Natarajan, and J. Zhu, "Fog computing: A platform for Internet of Things and analytics," in *Big Data Internet Things: A Roadmap for Smart Environment*. Cham, Switzerland: Springer, May 2014, pp. 169–186.
- [136] S. Patil-Karpe, S. H. Brahmananda, and S. Karpe, "Review of resource allocation in fog computing," in *Smart Intelligent Computation Application (SICA)*. Singapore: Springer, Sep. 2019, pp. 327–334.
- [137] C. Szepesvari, "Algorithms for reinforcement learning," *Synth. Lectures Artif. Intell. Mach. Learn.*, vol. 4, no. 1, pp. 1–103, 2010.
- [138] J. Gedeon, F. Brandherm, R. Egert, T. Grube, and M. Mühlhäuser, "What the fog? Edge computing revisited: Promises, applications and future challenges," *IEEE Access*, vol. 7, pp. 152847–152878, 2019.
- [139] X. Dutreilh, S. Kirgizov, O. Melekhova, J. Malenfant, N. Rivierre, and I. Truck, "Using reinforcement learning for autonomic resource allocation in clouds: Towards a fully automated workflow," in *Proc. 7th Int. Conf. Autom. Autom. Syst. (ICAS)*, Valencia, Spain, Jun. 2021, pp. 67–74.
- [140] K. Arulkumaran, M. P. Deisenroth, M. Brundage, and A. A. Bharath, "Deep reinforcement learning: A brief survey," *IEEE Signal Process. Mag.*, vol. 34, no. 6, pp. 26–38, Nov. 2017.
- [141] H. Mao, M. Alizadeh, I. Menache, and S. Kandula, "Resource management with deep reinforcement learning," in *Proc. ACM Workshop Hot Topic Netw. (HotNets)*, Atlanta, GA, USA, Nov. 2016, pp. 50–56.
- [142] L. P. Kaelbling, M. L. Littman, and A. W. Moore, "Reinforcement learning: A survey," *J. Artif. Intell. Res.*, vol. 4, no. 1, pp. 237–285, 1996.
- [143] K. Arulkumaran, M. P. Deisenroth, M. Brundage, and A. A. Bharath, "A brief survey of deep reinforcement learning," 2017, [arXiv:1708.05866](https://arxiv.org/abs/1708.05866).
- [144] V. Francois-Lavet, P. Henderson, R. Islam, M. G. Bellemare, and J. Pineau, "An introduction to deep reinforcement learning," 2018, [arXiv:1811.12560](https://arxiv.org/abs/1811.12560).
- [145] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*. Cambridge, MA, USA: MIT Press, 2018.
- [146] S. Nath, Y. Li, J. Wu, and P. Fan, "Multi-user multi-channel computation offloading and resource allocation for mobile edge computing," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Dublin, Ireland, Jun. 2020, pp. 1–6.
- [147] G. E. Monahan, "State-of-the-art—A survey of partially observable Markov decision processes: Theory, models, and algorithms," *Manag. Sci.*, vol. 28, no. 1, pp. 1–16, 1982.
- [148] R. Xie, Q. Tang, C. Liang, F. R. Yu, and T. Huang, "Dynamic computation offloading in IoT fog systems with imperfect channel-state information: A POMDP approach," *IEEE Internet Things J.*, vol. 8, no. 1, pp. 345–356, Jan. 2021.
- [149] M. L. Littman, "Markov games as a framework for multi-agent reinforcement learning," in *Proc. Mach. Learn.*, New Brunswick, NJ, USA, Jul. 1994, pp. 157–163.
- [150] L. S. Shapley, "Stochastic games," *Proc. Nat. Acad. Sci.*, vol. 39, no. 10, pp. 1095–1100, Jul. 1953.
- [151] T. Basar and G. J. Olsder, *Dynamic Noncooperative Game Theory*. Philadelphia, PA, USA: SIAM, 1998.
- [152] J. Filar and K. Vrieze, *Competitive Markov Decision Processes*. New York, NY, USA: Springer, 1997.
- [153] S. Kar, J. M. Moura, and H. V. Poor, "QD-learning: A collaborative distributed strategy for multi-agent reinforcement learning through consensus," 2012, [arXiv:1205.0047](https://arxiv.org/abs/1205.0047).
- [154] K. Zhang, Z. Yang, H. Liu, T. Zhang, and T. Basar, "Finite-sample analyses for fully decentralized multi-agent reinforcement learning," 2018, [arXiv:1812.02783](https://arxiv.org/abs/1812.02783).
- [155] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch, "Multiagent actor-critic for mixed cooperative-competitive environments," 2017, [arXiv:1706.02275](https://arxiv.org/abs/1706.02275).
- [156] D. S. Bernstein, S. Zilberstein, and N. Immerman, "The complexity of decentralized control of Markov decision processes," *Math. Oper. Res.*, vol. 27, no. 4, pp. 32–37, Nov. 2002.
- [157] F. A. Oliehoek and C. Amato, *A Concise Introduction to Decentralized POMDPs*. Cham, Switzerland: Springer, 2016.
- [158] Z. Cao, P. Zhou, R. Li, S. Huang, and D. Wu, "Multi-agent deep reinforcement learning for joint multi-channel access and task offloading of mobile edge computing in industry 4.0," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 6201–6213, Jul. 2020.
- [159] C. Zhong, M. C. Gursoy, and S. Velipasalar, "Deep multi-agent reinforcement learning based cooperative edge caching in wireless networks," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Shanghai, China, May 2019, pp. 1–6.
- [160] J. N. Tsitsiklis and B. Van Roy, "An analysis of temporal-difference learning with function approximation," *IEEE Trans. Autom. Control*, vol. 42, no. 5, pp. 674–690, May 1997.
- [161] H. Van Hasselt, A. Guez, and D. Silver, "Deep reinforcement learning with double Q-learning," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, vol. 30, Feb. 2016, pp. 222–227.
- [162] V. Konda and J. Tsitsiklis, "Actor-critic algorithms," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 12, Nov. 1999, pp. 1–8.
- [163] R. S. Sutton, D. A. McAllester, S. Singh, and Y. Mansour, "Policy gradient methods for reinforcement learning with function approximation," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, vol. 12, Denver, CO, USA, Nov. 1999, pp. 1057–1063.
- [164] D. Silver et al., "Deterministic policy gradient algorithms," in *Proc. Int. Conf. Mach. Learn. (PMLR)*, vol. 32, Jun. 2014, pp. 387–395.
- [165] J. Foerster, G. Farquhar, T. Afouras, N. Nardelli, and S. Whiteson, "Counterfactual multi-agent policy gradients," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, vol. 32, Feb. 2018, pp. 2974–2982.
- [166] J. M. Lasry and P. L. Lions, "Mean field games," *Jpn. J. Math.*, vol. 2, pp. 229–260, Jan. 2007.
- [167] M. Y. Huang, R. P. Malhame, and P. E. Caines, "Large population stochastic dynamic games: Closed-loop McKean–Vlasov systems and the Nash certainty equivalence principle," *Commun. Inf. Syst.*, vol. 6, no. 3, pp. 221–252, 2006.

- [168] Y. Yang, R. Luo, M. Li, M. Zhou, W. Zhang, and J. Wang, "Mean field multi-agent reinforcement learning," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 80, Jul. 2018, pp. 5571–5580.
- [169] L. Mathieu et al., "Learning mean field games: A survey," 2022, *arXiv:2205.12944v2*.
- [170] R. Elie et al., "Approximate fictitious play for mean field games," 2019, *arXiv:1907.02633*.
- [171] W. Shi and S. Dustdar, "The promise of edge computing," *Computer*, vol. 49, no. 5, pp. 78–81, May 2016.
- [172] S. Chen et al., "Internet of Things based smart grids supported by intelligent edge computing," *IEEE Access*, vol. 7, pp. 74089–74102, 2019.
- [173] P.-O. Ostberg et al., "Reliable capacity provisioning for distributed cloud/edge/fog computing applications," in *Proc. Eur. Conf. Netw. Commun. (EuCNC)*, Oulu, Finland, Jun. 2017, pp. 1–6.
- [174] R. Wang et al., "Cloud-edge collaborative industrial robotic intelligent service platform," in *Proc. IEEE Int. Conf. Joint Cloud Comput. (JCC)*, Oxford, U.K., Sep. 2020, pp. 71–77.
- [175] Z. Zhang, W. Ma, F. Li, R. Tang, J. Wang, and W. Chen, "An attention-enhanced edge-cloud collaborative framework for multi-task application," in *Proc. IEEE Int. Conf. Internet Things Intell. Syst. (IoTIS)*, Jan. 2021, pp. 109–115.
- [176] Y. Zhang, X. Wang, J. He, Y. Xu, F. Zhang, and Y. Luo, "A transfer learning-based high impedance fault detection method under a cloud-edge collaboration framework," *IEEE Access*, vol. 8, pp. 165099–165110, 2020.
- [177] J. Ing et al., "Edge-cloud collaboration architecture for AI transformation of SME manufacturing enterprises," in *Proc. IEEE/ITU Int. Conf. Artif. Intell. Good (AI4G)*, Geneva, Switzerland, Sep. 2020, pp. 170–175.
- [178] L. Qing, "A 5G PaaS collaborative management and control platform technology based on cloud edge collaboration based on particle swarm optimization algorithm," in *Proc. IEEE Asia-Pac. Conf. Image Process Electron. Comput. (IPEC)*, Dalian, China, Apr. 2021, pp. 144–147.
- [179] W. Xiao, Y. Miao, G. Fortino, D. Wu, M. Chen, and K. Hwang, "Collaborative cloud-edge service cognition framework for DNN configuration toward smart IIoT," *IEEE Trans. Ind. Informat.*, vol. 18, no. 10, pp. 7038–7047, Aug. 2021.
- [180] W. Qu, X. Ding, K. Yang, Y. Bao, and W. Chen, "IDEC: Intelligent distributed edge computing system architecture enabling deep learning across heterogeneous IoT devices," in *Proc. IEEE 6th Int. Conf. Comput. Commun. (ICCC)*, Chengdu, China, Feb. 2020, pp. 926–933.
- [181] S. Lu, Y. Yao, and W. Shi, "CLONE: Collaborative learning on the edges," *IEEE Internet Things J.*, vol. 8, no. 13, pp. 10222–10236, Jul. 2021.
- [182] S. Xu, X. Liu, S. Guo, X. Qiu, and L. Meng, "MECC: A mobile edge collaborative caching framework empowered by deep reinforcement learning," *IEEE Netw.*, vol. 35, no. 4, pp. 176–183, Aug. 2021.
- [183] J. Li, C. Gu, Y. Xiang, and F. Li, "Edge-cloud computing systems for smart grid: State-of-the-art, architecture, and applications," *J. Mod. Power Syst. Clean Energy*, vol. 10, no. 4, pp. 805–817, Jul. 2022.
- [184] M. Babar, M. A. Jan, X. He, M. U. Tariq, S. Mastorakis, and R. Alturki, "An optimized IoT-enabled big data analytics architecture for edge-cloud computing," *IEEE Internet Things J.*, vol. 10, no. 5, pp. 3995–4005, Mar. 2023.
- [185] X. Zhang, Y. Wang, S. Lu, L. Liu, L. Xu, and W. Shi, "OpenEI: An open framework for edge intelligence," in *Proc. IEEE 39th Int. Conf. Distrib. Comput. Syst. (ICDCS)*, Dallas, TX, USA, Oct. 2019, pp. 1840–1851.
- [186] H. Flores, P. Nurmi, and P. Hui, "AI on the move: From on-device to on-multi-device," in *Proc. IEEE Int. Conf. Pervasive Comput. Commun. Workshops (PerCom Workshops)*, Kyoto, Japan, Mar. 2019, pp. 310–315.
- [187] K. Nguyen, S. Drew, C. Huang, and J. Zhou, "Collaborative container-based parked vehicle edge computing framework for online task offloading," in *Proc. IEEE 9th Int. Conf. Cloud Netw. (CloudNet)*, Piscataway, NJ, USA, Feb. 2020, pp. 1–6.
- [188] Y. Liu, S. Xie, and Y. Zhang, "Cooperative offloading and resource management for UAV-enabled mobile edge computing in power IoT system," *IEEE Trans. Veh. Technol.*, vol. 69, no. 10, pp. 12229–12239, Oct. 2020.
- [189] C. Hu, Y. Bai, R. Wang, C. Liu, and X. Wang, "CCIED: Cache-aided collaborative intelligence between edge devices," in *Proc. IEEE 22nd Int. Conf. High Perform. Comput. Commun. IEEE 18th Int. Conf. Smart City IEEE 6th Int. Conf. Data Sci. Syst. (HPCC/SmartCity/DSS)*, Dec. 2020, pp. 668–673.
- [190] S. Shahhosseini et al., "Online learning for orchestration of inference in multi-user end-edge-cloud networks," *ACM Trans. Embedded Comput. Syst.*, vol. 21, no. 6, pp. 1–25, 2022.
- [191] B. Yang, Z. Pang, S. Wang, F. Mo, and Y. Gao, "A coupling optimization method of production scheduling and computation offloading for intelligent workshops with cloud-edge-terminal architecture," *J. Manuf. Syst.*, vol. 65, pp. 421–438, Oct. 2022.
- [192] Z. Zhou, X. Chen, E. Li, L. Zeng, K. Luo, and J. Zhang, "Edge intelligence: Paving the last mile of artificial intelligence with edge computing," *Proc. IEEE*, vol. 107, no. 8, pp. 1738–1762, Aug. 2019.
- [193] S. Deng, H. Zhao, W. Fang, J. Yin, S. Dustdar, and A. Y. Zomaya, "Edge intelligence: The confluence of edge computing and artificial intelligence," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 7457–7469, Aug. 2020.
- [194] M. A. Javed and S. Zeadally, "AI-empowered content caching in vehicular edge computing: Opportunities and challenges," *IEEE Netw.*, vol. 35, no. 3, pp. 109–115, Jun. 2021.
- [195] M. Zwolenski and L. Weatherill, "The digital universe: Rich data and the increasing value of the internet of things," *Telecommun. Digit. Econ. J.*, vol. 2, no. 3, pp. 1–9, Sep. 2014.
- [196] Y. Mao, C. You, J. Zhang, K. Huang, and K. B. Letaief, "A survey on mobile edge computing: The communication perspective," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 4, pp. 2322–2358, 4th Quart., 2017.
- [197] T. Burd and R. Broderon, "Processor design for portable systems," *Kluwer J. VLSI Signal Process. Syst.*, vol. 13, no. 2, pp. 203–221, Aug. 1996.
- [198] W. Yuan and K. Nahrstedt, "Energy-efficient CPU scheduling for multimedia applications," *ACM Trans. Comput. Syst.*, vol. 24, no. 3, pp. 292–331, Aug. 2006.
- [199] K. D. Vogeleer, G. Memmi, P. Jouvelot, and F. Coelho, "The energy/frequency convexity rule: Modeling and experimental validation on mobile devices," in *Proc. Int. Conf. Parallel Process. Appl. Math. (PPAM)*, Warsaw, Poland, Sep. 2013, pp. 793–803.
- [200] W. Zhang, Y. Wen, K. Guan, D. Kilper, H. Luo, and D. O. Wu, "Energyoptimal mobile cloud computing under stochastic wireless channel," *IEEE Trans. Wireless Commun.*, vol. 12, no. 9, pp. 4569–4581, Sep. 2013.
- [201] A. Goldsmith, *Wireless Communications*. New York, NY, USA: Cambridge Univ. Press, 2005.
- [202] L. Suo, H. Li, S. Zhang, and J. Li, "Successive interference cancellation and alignment in K-user MIMO interference channels with partial unidirectional strong interference," *China Commun.*, vol. 19, no. 2, pp. 118–130, Feb. 2022.
- [203] N. H. Mahmood, O. A. Lpez, H. Alves, and M. Latva-Aho, "A predictive interference management algorithm for URLLC in beyond 5G networks," *IEEE Commun. Lett.*, vol. 25, no. 3, pp. 995–999, Mar. 2021.
- [204] Y. Gong, H. Yao, J. Wang, L. Jiang, and F. R. Yu, "Multi-agent driven resource allocation and interference management for deep edge networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 2, pp. 2018–2030, Feb. 2022.
- [205] C. Song, H. Zhao, L. Qin, R. Wen, and S. Shao, "Analysis and optimization of transceiver IQ imbalances in artificial noise shielded FH communication," *IEEE Trans. Signal Process.*, vol. 70, pp. 2798–2813, 2022.
- [206] X. Quan, Y. Liu, P. Fan, and Y. Tang, "Full-duplex transceiver design in the presence of phase noise and performance analysis," *IEEE Trans. Veh. Technol.*, vol. 70, no. 1, pp. 558–571, Jan. 2021.
- [207] J. G. Andrews, H. Claussen, M. Dohler, S. Rangan, and M. C. Reed, "Femtocells: Past, present, and future," *IEEE Commun. Mag.*, vol. 30, no. 3, pp. 497–508, Apr. 2012.
- [208] H. Cui et al., "Space-air-ground integrated network (SAGIN) for 6G: Requirements, architecture and challenges," *China Commun.*, vol. 19, no. 2, pp. 90–108, Feb. 2022.
- [209] A. Doshi, S. Yerramalli, L. Ferrari, T. Yoo, and J. G. Andrews, "A deep reinforcement learning framework for contention-based spectrum sharing," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 8, pp. 2526–2540, Aug. 2021.
- [210] P. Wang, B. Di, and L. Song, "Cellular communications over unlicensed mmWave bands with hybrid beamforming," *IEEE Trans. Wireless Commun.*, vol. 21, no. 8, pp. 6064–6078, Aug. 2022.
- [211] Z. J. Grey, S. Mosleh, J. D. Rezac, Y. Ma, J. B. Coder, and A. M. Dienstfrey, "Bi-criteria radio spectrum sharing with subspace-based Pareto tracing," *IEEE Trans. Commun.*, vol. 70, no. 5, pp. 3512–3525, May 2022.

- [212] W. Jiang, B. Han, M. A. Habibi, and H. D. Schotten, "The road towards 6G: A comprehensive survey," *IEEE Open J. Commun. Soc.*, vol. 2, pp. 334–366, 2021.
- [213] H. Gu and H. Wang, "A distributed caching scheme using non-cooperative game for mobile edge networks," *IEEE Access*, vol. 8, pp. 142747–142757, 2020.
- [214] M. E. Crovella and A. Bestavros, "Self-similarity in world wide Web traffic: Evidence and possible causes," *IEEE/ACM Trans. Netw.*, vol. 5, no. 6, pp. 835–846, Dec. 1997.
- [215] M. Y. Cha, K. Haewoon, R. Pablo, A. Yong-Yeol, and M. Sue, "I tube you tube everybody tubes: Analyzing the worlds largest user generated content video system," in *Proc. ACM SIGCOMM Conf. Internet Meas.*, San Diego, CA, USA, Oct. 2007, pp. 1–25.
- [216] T. Q. Dinh, Q. D. La, T. Q. S. Quek, and H. Shin, "Learning for computation offloading in mobile edge computing," *IEEE Trans. Commun.*, vol. 66, no. 12, pp. 6353–6367, Dec. 2018.
- [217] S. Wang, M. Chen, X. Liu, and C. Yin, "Task and resource allocation in mobile edge computing: An improved reinforcement learning approach," in *Proc. IEEE Globecom Workshops (GC Wkshps)*, Waikoloa, HI, USA, Dec. 2019, pp. 1–6.
- [218] Z. Gao, W. Hao, Z. Han, and S. Yang, "Q-learning-based task offloading and resources optimization for a collaborative computing system," *IEEE Access*, vol. 8, pp. 149011–149024, 2020.
- [219] C. Fang et al., "AI-driven energy-efficient content task offloading in cloud-edge-end cooperation networks," *IEEE Open J. Comput. Soc.*, vol. 3, pp. 162–171, 2022.
- [220] D. Huo et al., "A DQN-based workflow task assignment approach in cloud-fog cooperative considering terminal mobility," in *Proc. 6th Int. Conf. Control Eng. Artif. Intell. (CCEAI)*, Sanya, China, Mar. 2022, pp. 78–82.
- [221] X. Zhao, G. Huang, J. Jiang, L. Gao, and M. Li, "Task offloading of cooperative intrusion detection system based on deep Q-network in mobile edge computing," *Exp. Syst. Appl.*, vol. 206, Nov. 2022, Art. no. 117860.
- [222] Q. Luo, C. Li, T. H. Luan, and W. Shi, "Collaborative data scheduling for vehicular edge computing via deep reinforcement learning," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 9637–9650, Oct. 2020.
- [223] J. Li, Z. Yang, X. Wang, Y. Xia, and S. Ni, "Task offloading mechanism based on federated reinforcement learning in mobile edge computing," *Digit. Commun. Netw.*, vol. 9, no. 2, pp. 492–504, Apr. 2023.
- [224] H. Tang, H. Wu, G. Qu, and R. Li, "Double deep Q-network based dynamic framing offloading in vehicular edge computing," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 3, pp. 1297–1310, May/Jun. 2023.
- [225] F. Qi, L. Zhuo, and C. Xin, "Deep reinforcement learning based task scheduling in edge computing networks," in *Proc. IEEE/CIC Int. Conf. Commun. China (ICCC)*, Chongqing, China, Aug. 2020, pp. 835–840.
- [226] Y. Liu, Q. Cui, J. Zhang, Y. Chen, and Y. Hou, "An actor-critic deep reinforcement learning based computation offloading for three-tier mobile computing networks," in *Proc. 11th Int. Conf. Wireless Commun. Signal Process. (WCSP)*, Xi'an, China, Oct. 2019, pp. 1–6.
- [227] Z. Wang, M. Li, L. Zhao, H. Zhou, and N. Wang, "A3C-based computation offloading and service caching in cloud-edge computing networks," in *Proc. IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPs)*, New York, NY, USA, May 2022, pp. 1–2.
- [228] B. Dai, J. Niu, T. Ren, and M. Atiquzzaman, "Towards mobility-aware computation offloading and resource allocation in end-edge-cloud orchestrated computing," *IEEE Internet Things J.*, vol. 9, no. 19, pp. 19450–19462, Oct. 2022.
- [229] L. Chen, G. Gong, K. Jiang, H. Zhou, and R. Chen, "DDPG-based computation offloading and service caching in mobile edge computing," in *Proc. IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPs)*, New York, NY, USA, May 2022, pp. 1–6.
- [230] S. Chen, H. Chen, J. Ruan, and Z. Wang, "Context-aware online offloading strategy with mobility prediction for mobile edge computing," in *Proc. Int. Conf. Comput. Commun. Netw. (ICCCN)*, Athens, Greece, Jul. 2021, pp. 1–9.
- [231] S. Yao et al., "Blockchain-empowered collaborative task offloading for cloud-edge-device computing," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 12, pp. 3485–3500, Dec. 2022.
- [232] Z. Lin, B. Gu, X. Zhang, D. Yi, and Y. Han, "Online task offloading in UDN: A deep reinforcement learning approach with incomplete information," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Austin, TX, USA, Apr. 2022, pp. 1236–1241.
- [233] Y. Zhan, S. Guo, P. Li, and J. Zhang, "A deep reinforcement learning based offloading game in edge computing," *IEEE Trans. Comput.*, vol. 69, no. 6, pp. 883–893, Jun. 2020.
- [234] H. Huang, Q. Ye, and Y. Zhou, "Deadline-aware task offloading with partially-observable deep reinforcement learning for multi-access edge computing," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 6, pp. 3870–3885, Nov./Dec. 2022.
- [235] T. -H. Nguyen and L. Park, "Multi-agent DRL-based task offloading in hierarchical HAP-LAP networks," in *Proc. 13th Int. Conf. Inf. Commun. Technol. Conver. (ICTC)*, Jeju Island, South Korea, Oct. 2022, pp. 817–821.
- [236] X. Wang, Z. Ning, L. Guo, S. Guo, X. Gao, and G. Wang, "Mean-field learning for edge computing in mobile blockchain networks," *IEEE Trans. Mobile Comput.*, vol. 22, no. 10, pp. 5978–5994, Oct. 2023.
- [237] G. Wu, Y. Zhao, Y. Shen, H. Zhang, S. Shen, and S. Yu, "DRL-based resource allocation optimization for computation offloading in mobile edge computing," in *Proc. IEEE Conf. Comput. Commun. Workshops (INFOCOM WKSHPs)*, New York, NY, USA, Dec. 2022, pp. 1–6.
- [238] J. Hou, M. Chen, H. Geng, R. Li, and J. Lu, "GP-NFSP: Decentralized task offloading for mobile edge computing with independent reinforcement learning," *Future Gener. Comput. Syst.*, vol. 141, pp. 205–217, Apr. 2023.
- [239] S. Song, Z. Fang, and J. Jiang, "Fast-DRD: Fast decentralized reinforcement distillation for deadline-aware edge computing," *Inf. Process. Manag.*, vol. 59, no. 2, Mar. 2022, Art. no. 102850.
- [240] H. Liang, L. X. Cai, D. Huang, X. Shen, and D. Peng, "An SMDP-based service model for interdomain resource allocation in mobile cloud networks," *IEEE Trans. Veh. Technol.*, vol. 61, no. 5, pp. 2222–2232, Jun. 2012.
- [241] C. Papagianni, A. Leivadreas, S. Papavassiliou, V. Maglaris, C. Cervello-Pastor, and A. Monje, "On the optimal allocation of virtual resources in cloud computing networks," *IEEE Trans. Comput.*, vol. 62, no. 6, pp. 1060–1071, Jun. 2013.
- [242] Y. Liu, M. J. Lee, and Y. Zheng, "Adaptive multi-resource allocation for cloudlet-based mobile cloud computing system," *IEEE Trans. Mobile Comput.*, vol. 15, no. 10, pp. 2398–2410, Oct. 2016.
- [243] Z. Cao, J. Lin, C. Wan, Y. Song, Y. Zhang, and X. Wang, "Optimal cloud computing resource allocation for demand side management in smart grid," *IEEE Trans. Smart Grid*, vol. 8, no. 4, pp. 1943–1955, Jul. 2017.
- [244] J. Du, L. Zhao, J. Feng, and X. Chu, "Computation offloading and resource allocation in mixed fog/cloud computing systems with min-max fairness guarantee," *IEEE Trans. Commun.*, vol. 66, no. 4, pp. 1594–1608, Apr. 2018.
- [245] K. Wang, K. Yang, and C. S. Magurawalage, "Joint energy minimization and resource allocation in C-RAN with mobile cloud," *IEEE Trans. Cloud Comput.*, vol. 6, no. 3, pp. 760–770, Jun. 2018.
- [246] W. Wei, X. Fan, H. Song, X. Fan, and J. Yang, "Imperfect information dynamic Stackelberg game based resource allocation using hidden Markov for cloud computing," *IEEE Trans. Services Comput.*, vol. 11, no. 1, pp. 78–89, Feb. 2018.
- [247] Q. Zhang, L. Gui, F. Hou, J. Chen, S. Zhu, and F. Tian, "Dynamic task offloading and resource allocation for mobile-edge computing in dense cloud RAN," *IEEE Internet Things J.*, vol. 7, no. 4, pp. 3282–3299, Apr. 2020.
- [248] X. Gao, R. Liu, and A. Kaushik, "Hierarchical multi-agent optimization for resource allocation in cloud computing," *IEEE Trans. Parallel Distrib. Syst.*, vol. 32, no. 3, pp. 692–707, Mar. 2021.
- [249] S. Xu et al., "RJCC: Reinforcement-learning-based joint communications-and-computational resource allocation mechanism for smart city IoT," *IEEE Internet Things J.*, vol. 7, no. 9, pp. 8059–8076, Sep. 2020.
- [250] I. AlQerm and J. Pan, "Enhanced online Q-learning scheme for resource allocation with maximum utility and fairness in edge-IoT networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 7, no. 4, pp. 3074–3086, Aug. 2020.
- [251] Z. Gao, W. Hao, Z. Han, and S. Yang, "Q-learning-based task offloading and resources optimization for a collaborative computing system," *IEEE Access*, vol. 8, pp. 149011–149024, 2020.
- [252] Q. Chen, Z. Kuang, and L. Zhao, "Multiuser computation offloading and resource allocation for cloud-edge heterogeneous network," *IEEE Internet Things J.*, vol. 9, no. 5, pp. 3799–3811, Mar. 2022.
- [253] S. Wang and K. Ning, "Network resource allocation strategy based on UAV cooperative edge computing," *J. Robot.*, vol. 2022, Mar. 2022, Art. no. 8514235.
- [254] C. Fang et al., "Deep reinforcement learning based resource allocation for content distribution in fog radio access networks," *IEEE Internet Things J.*, vol. 9, no. 18, pp. 16874–16883, Jan. 2022.

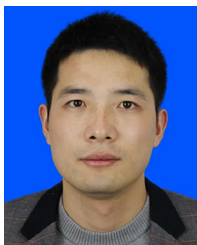
- [255] F. Khoramnejad and M. Erol-Kantarci, "On joint offloading and resource allocation: A double deep Q -network approach," *IEEE Trans. Cogn. Commun. Netw.*, vol. 7, no. 4, pp. 1126–1141, Dec. 2021.
- [256] H. C. Ke, H. Wang, H. W. Zhao, and W. J. Sun, "Deep reinforcement learning-based computation offloading and resource allocation in security-aware mobile edge computing," *Wireless Netw.*, vol. 27, no. 5, pp. 3357–3373, Jul. 2021.
- [257] N. Shan, X. Cui, and Z. Gao, "DRL+ FL: An intelligent resource allocation model based on deep reinforcement learning for mobile edge computing," *Comput. Commun.*, vol. 160, pp. 14–24, Jul. 2020.
- [258] J. Feng, F. R. Yu, Q. Pei, X. Chu, J. Du, and L. Zhu, "Cooperative computation offloading and resource allocation for blockchain-enabled mobile-edge computing: A deep reinforcement learning approach," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 6214–6228, Jul. 2020.
- [259] M. Chen et al., "Intelligent resource allocation management for vehicles network: An A3C learning approach," *Comput. Commun.*, vol. 151, pp. 485–494, Feb. 2020.
- [260] A. Qadeer and M. J. Lee, "DDPG-edge-cloud: A deep-deterministic policy gradient based multi-resource allocation in edge-cloud system," in *Proc. Int. Conf. Artif. Intell. Informat. Commun. (ICAIIIC)*, Feb. 2022, pp. 339–344.
- [261] Y. Gao and Z. Li, "Energy and delay-aware task offloading and resource allocation in mobile edge computing," in *Proc. IEEE 25th Int. Conf. Comput. Supported Cooper. Work Design (CSCWD)*, Hangzhou, China, May 2022, pp. 1335–1340.
- [262] Y. Gong, H. Yao, J. Wang, L. Jiang, and F. R. Yu, "Multi-agent driven resource allocation and interference management for deep edge networks," *IEEE Trans. Veh. Technol.*, vol. 71, no. 2, pp. 2018–2030, Feb. 2022.
- [263] J. Baek and G. Kaddoum, "Heterogeneous task offloading and resource allocations via deep recurrent reinforcement learning in partial observable multifog networks," *IEEE Internet Things J.*, vol. 8, no. 2, pp. 1041–1056, Jul. 2020.
- [264] J. Baek and G. Kaddoum, "Online partial offloading and task scheduling in SDN-fog networks with deep recurrent reinforcement learning," *IEEE Internet Things J.*, vol. 9, no. 13, pp. 11578–11589, Jul. 2022.
- [265] Y. Xiao, Y. Song, and J. Liu, "Towards energy efficient resource allocation: When green mobile edge computing meets multi-agent deep reinforcement learning," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Seoul, South Korea, May 2022, pp. 1–9.
- [266] Q. Cui, X. Zhao, W. Ni, Z. Hu, X. Tao, and P. Zhang, "Multi-agent deep reinforcement learning-based interdependent computing for mobile edge computing-assisted robot teams," *IEEE Trans. Veh. Technol.*, vol. 72, no. 5, pp. 6599–6610, May 2023.
- [267] L. Li et al., "Resource allocation for NOMA-MEC systems in ultra-dense networks: A learning aided mean-field game approach," *IEEE Trans. Wireless Commun.*, vol. 20, no. 3, pp. 1487–1500, Mar. 2021.
- [268] S. Long, Y. Zhang, Q. Deng, T. Pei, J. Ouyang, and Z. Xia, "An efficient task offloading approach based on multi-objective evolutionary algorithm in cloud-edge collaborative environment," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 2, pp. 645–657, Mar. 2023.
- [269] S. Tian, C. Chang, S. Long, S. Oh, Z. Li, and J. Long, "User preference-based hierarchical offloading for collaborative cloud-edge computing," *IEEE Trans. Services Comput.*, vol. 16, no. 1, pp. 684–697, Feb. 2023.
- [270] G. Zhou, L. Zhao, G. Zheng, Z. Xie, S. Song, and K.-C. Chen, "Joint multi-objective optimization for radio access network slicing using multi-agent deep reinforcement learning," *IEEE Trans. Veh. Technol.*, vol. 72, no. 9, pp. 11828–11843, Sep. 2023.
- [271] H. T. Malazi et al., "Dynamic service placement in multi-access edge computing: A systematic literature review," *IEEE Access*, vol. 10, pp. 32639–32688, 2022.
- [272] Z. Zhao, H. Cheng, X. Xu, and Y. Pan, "Graph partition and multiple choice-UCB based algorithms for edge server placement in MEC environment," *IEEE Trans. Mobile Comput.*, early access, Jun. 21, 2023, doi: [10.1109/TMC.2023.3284994](https://doi.org/10.1109/TMC.2023.3284994).
- [273] R. G. Garroppo, M. Volpi, G. Nencioni, and P. V. Wadatar, "Experimental evaluation of handover strategies in 5G-MEC scenario by using AdvantEDGE," in *Proc. IEEE Int. Med. Conf. Commun. Netw. (MediCom)*, Athens, Greece, Sep. 2022, pp. 286–291.
- [274] C. Li, M. Wu, Y. Liu, K. Zhou, J. Zhang, and Y. Sun, "SS-LRU: A smart segmented LRU caching," in *Proc. 59th ACM/IEEE Design Autom. Conf.*, San Francisco, CA, USA, Jul. 2022, pp. 397–402.
- [275] N. K. Panigrahy, P. Nain, G. Neglia, and D. Towsley, "A new upper bound on cache hit probability for non-anticipative caching policies," *ACM Trans. Model. Perfect. Eval. Comput. Syst.*, vol. 7, no. 2, pp. 1–24, 2022.
- [276] J. Yao, T. Han, and N. Ansari, "On mobile edge caching," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2525–2553, 3rd Quart., 2019.
- [277] H. Gu and H. Wang, "A distributed caching scheme using non-cooperative game for mobile edge networks," *IEEE Access*, vol. 8, pp. 142747–142757, 2020.
- [278] D. Kreutz, F. M. V. Ramos, P. E. VerAssimo, C. E. Rothenberg, S. Azodolmolky, and S. Uhlig, "Software-defined networking: A comprehensive survey," *Proc. IEEE*, vol. 103, no. 1, pp. 14–76, Jan. 2015.
- [279] B. Li, F. Hou, G. Yang, H. Zhao, and S. Chen, "Data analysis-oriented stochastic scheduling for cost efficient resource allocation in NFV based MEC network," *IEEE Trans. Veh. Technol.*, vol. 72, no. 5, pp. 6695–6708, May 2023.
- [280] P. Cruz, N. Achir, and A. C. Viana, "On the edge of the deployment: A survey on multi-access edge computing," *ACM Comput. Surveys*, vol. 55, no. 5, pp. 1–34, Dec. 2022.
- [281] J. Taheri, S. Dustdar, A. Zomaya, and S. Deng, "AI/ML on edge," in *Edge Intelligence*. Cham, Switzerland: Springer, 2023, pp. 183–211.
- [282] W. Su, L. Li, F. Liu, M. He, and X. Liang, "AI on the edge: A comprehensive review," *Artif. Intel. Rev.*, vol. 55, no. 8 pp. 6125–6183, Mar. 2022.
- [283] M. Sheraz et al., "Artificial intelligence for wireless caching: Schemes, performance, and challenges," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 1, pp. 631–661, 1st Quart., 2021.
- [284] N. C. Luong et al., "Applications of deep reinforcement learning in communications and networking: A survey," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3133–3174, 4th Quart., 2019.
- [285] B. Feng, H. Zhou, G. Li, Y. Zhang, K. Sood, and S. Yu, "Enabling machine learning with service function chaining for security enhancement at 5G edges," *IEEE Netw.*, vol. 35, no. 5, pp. 196–201, Sep. 2021.
- [286] W. Chien, H. Weng, and C. Lai, "Q-learning based collaborative cache allocation in mobile edge computing," *Future Gener. Comput. Syst.*, vol. 102, pp. 603–610, Jan. 2020.
- [287] F. Qiao, M. Dong, K. Ota, S. Liao, J. Wu, and J. Li, "Making big data intelligent storable at the edge: Storage resource intelligent orchestration," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Waikoloa, HI, USA, Dec. 2019, pp. 1–4.
- [288] C. Fang et al., "Q-learning based delay-aware content delivery in cloud-edge cooperation networks," in *Proc. 7th Int. Conf. Comput. Commun. (ICCC)*, Chengdu, China, Dec. 2021, pp. 1458–1462.
- [289] X. Wang, R. Li, C. Wang, X. Li, T. Taleb, and V. C. M. Leung, "Attention-weighted federated deep reinforcement learning for device-to-device assisted heterogeneous collaborative edge caching," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 1, pp. 154–169, Jan. 2021.
- [290] F. Jiang, J. Wang, and C. Sun, "Deep Q-learning-based cooperative caching strategy for fog radio access networks," in *Proc. 7th Int. Conf. Comput. Commun. (ICCC)*, Chengdu, China, Dec. 2021, pp. 922–927.
- [291] C. Li, C. Qianqian, and Y. Luo, "Low-latency edge cooperation caching based on base station cooperation in SDN based MEC," *Exp. Syst. Appl.*, vol. 191, pp. 116252–116261, Apr. 2022.
- [292] J. Ren, H. Wang, T. Hou, S. Zheng, and C. Tang, "Collaborative edge computing and caching with deep reinforcement learning decision agents," *IEEE Access*, vol. 8, pp. 120604–120612, 2020.
- [293] D. Li et al., "Deep reinforcement learning for cooperative edge caching in future mobile networks," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Marrakesh, Morocco, Apr. 2019, pp. 2346–2354.
- [294] X. Wang, C. Wang, X. Li, V. C. M. Leung, and T. Taleb, "Federated deep reinforcement learning for Internet of Things with decentralized cooperative edge caching," *IEEE Internet Things J.*, vol. 7, no. 10, pp. 9441–9455, Oct. 2020.
- [295] M. Somesula, R. Rout, and D. Somayajulu, "Cooperative cache update using multi-agent recurrent deep reinforcement learning for mobile edge networks," *Comput. Netw.*, vol. 209, May 2022, Art. no. 108876.
- [296] F. Wang, F. Wang, J. Liu, R. Shea, and L. Sun, "Intelligent video caching at network edge: A multi-agent deep reinforcement learning approach," in *Proc. IEEE Conf. Comput. Commun. (INFOCOM)*, Toronto, ON, Canada, Jul. 2020, pp. 2499–2508.
- [297] Y. Zhang et al., "Cooperative edge caching: A multi-agent deep learning based approach," *IEEE Access*, vol. 8, pp. 133212–133224, 2020.
- [298] S. Xu, X. Liu, S. Guo, X. Qiu, and L. Meng, "MECC: A mobile edge collaborative caching framework empowered by deep reinforcement learning," *IEEE Netw.*, vol. 35, no. 4, pp. 176–183, Jul./Aug. 2021.
- [299] Q. Li, Y. Sun, Q. Wang, L. Meng, and Y. Zhang, "A green DDPG reinforcement learning-based framework for content caching," in *Proc. 12th Int. Conf. Commun. Softw. Netw. (ICCSN)*, Chongqing, China, Jun. 2020, pp. 223–227.

- [300] D. Wang, Y. Bai, G. Huang, B. Song, and F. R. Yu, "Cache-aided MEC for IoT: Resource allocation using deep graph reinforcement learning," *IEEE Internet Things J.*, vol. 10, no. 3, pp. 11486–11496, Feb. 2023.
- [301] S. Chen, Z. Yao, X. Jiang, J. Yang, and L. Hanzo, "Multi-agent deep reinforcement learning-based cooperative edge caching for ultra-dense next-generation networks," *IEEE Trans. Commun.*, vol. 69, no. 4, pp. 2441–2456, Apr. 2021.
- [302] Z. Yao, Y. Li, S. Xia, and G. Wu, "Attention cooperative task offloading and service caching in edge computing," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Rio de Janeiro, Brazil, Dec. 2022, pp. 5189–5194.
- [303] *General Packet Radio Service (GPRS); Service Description; Stage 2 (Release 13)*, v12.11.0, 3GPP Standard TS 23.060, Dec. 2015.
- [304] *System Architecture for the 5G System (5GS) (Release 15)*, V16.5.0, 3GPP Standard TS 23.501, Oct. 2020.
- [305] *NR; NR and NG-RAN Overall Description; Stage-2 (Release 16)*, V16.4.0, 3GPP Standard TS 38.300, Jan. 2021.
- [306] M. M. Sajjad, C. J. Bernardos, D. Jayalath, and Y.-C. Tian, "Inter-slice mobility management in 5G: Motivations, standard principles, challenges, and research directions," *IEEE Commun. Stand. Mag.*, vol. 6, no. 1, pp. 93–100, Mar. 2022.
- [307] A. Zaidi, H. Farooq, A. Rizwan, A. Abu-Dayya, and A. Imran, "A framework to address mobility management challenges in emerging networks," *IEEE Wireless Commun.*, vol. 30, no. 4, pp. 90–97, Aug. 2023, doi: [10.1109/MWC.015.2100666](https://doi.org/10.1109/MWC.015.2100666).
- [308] I. Shayea et al., "Key challenges, drivers and solutions for mobility management in 5G networks: A survey," *IEEE Access*, vol. 8, pp. 172534–172552, 2020.
- [309] D. Lopez-Perez, I. Guvenc, and X. Chu, "Mobility management challenges in 3GPP heterogeneous networks," *IEEE Commun. Mag.*, vol. 50, no. 12, pp. 70–78, Dec. 2012.
- [310] A. Damjanovic et al., "A survey on 3GPP heterogeneous networks," *IEEE Wireless Commun.*, vol. 18, no. 3, pp. 10–21, Mar. 2011.
- [311] M. Kassar, B. Kervella, and G. Pujolle, "An overview of vertical handover decision strategies in heterogeneous wireless networks," *Comput. Commun.*, vol. 31, no. 10, pp. 2607–2620, Oct. 2008.
- [312] T. Taleb and A. Ksentini, "An analytical model for follow me cloud," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Atlanta, GA, USA, Dec. 2013, pp. 1291–1296.
- [313] A. Ksentini, T. Taleb, and M. Chen, "A Markov decision process-based service migration procedure for follow me cloud," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Sydney, NSW, Australia, Jun. 2014, pp. 1350–1354.
- [314] Z. Zhao et al., "Mobility management with transferable reinforcement learning trajectory prediction," *IEEE Trans. Netw. Service Manag.*, vol. 17, no. 4, pp. 2102–2116, Dec. 2020.
- [315] M. Palas et al., "Multi-criteria handover mobility management in 5G cellular network," *Comput. Commun.*, vol. 174, pp. 81–91, Jun. 2021.
- [316] Y. Chen, X. Lin, T. Khan, and M. Mozaffari, "Efficient drone mobility support using reinforcement learning," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Seoul, South Korea, May 2020, pp. 1–6.
- [317] H. Zhang, R. Wang, and J. Liu, "Mobility management for ultra-dense edge computing: A reinforcement learning approach," in *Proc. IEEE 90th Veh. Technol. Conf. (VTC-Fall)*, Honolulu, HI, USA, Sep. 2019, pp. 1–5.
- [318] Z. Zhang et al., "Reinforcement learning based computation-aware mobility management in ultra dense networks," *J. Int. Technol.*, vol. 21, no. 6, pp. 1785–1794, Nov. 2020.
- [319] Z. Han, T. Lei, Z. Lu, X. Wen, W. Zheng, and L. Guo, "Artificial intelligence-based handoff management for dense WLANs: A deep reinforcement learning approach," *IEEE Access*, vol. 7, pp. 31688–31701, 2019.
- [320] M. Mollel, S. Kaijage, and K. Michael, "Deep reinforcement learning based handover management for millimeter wave communication," *Int. J. Adv. Comput. Sci. Appl.*, vol. 12, no. 2, p. 21, 2021.
- [321] W. Liu, P. Si, E. Sun, M. Li, C. Fang, and Y. Zhang, "Green mobility management in UAV-assisted IoT based on dueling DQN," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Shanghai, China, May 2019, pp. 1–8.
- [322] S. Mollel et al., "Intelligent handover decision scheme using double deep reinforcement learning," *Physica Commun.*, vol. 42, Oct. 2020, Art. no. 101133.
- [323] F. Yang, W. Wu, X. Wang, Y. Zhang, and P. Si, "Deep reinforcement learning based handoff algorithm in end-to-end network slicing enabling HetNets," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Nanjing, China, Mar. 2021, pp. 1–7.
- [324] H. Zhang, R. Wang, W. Sun, and H. Zhao, "Mobility management for blockchain-based ultra-dense edge computing: A deep reinforcement learning approach," *IEEE Trans. Wireless Commun.*, vol. 20, no. 11, pp. 7346–7359, Nov. 2021.
- [325] Y. Cui, D. Zhang, J. Zhang, T. Zhang, L. Cao, and L. Chen, "Distributed task migration optimization in MEC by deep reinforcement learning strategy," in *Proc. IEEE 46th Conf. Local Comput. Netw. (LCN)*, Edmonton, AB, Canada, Oct. 2021, pp. 411–414.
- [326] S. Li, F. Wu, S. Luo, Z. Fan, J. Chen, and S. Fu, "Dynamic online trajectory planning for a UAV-enabled data collection system," *IEEE Trans. Veh. Technol.*, vol. 71, no. 12, pp. 13332–13343, Dec. 2022.
- [327] R. F. Atallah, C. M. Assi, and J. Y. Yu, "A reinforcement learning technique for optimizing downlink scheduling in an energy-limited vehicular network," *IEEE Trans. Veh. Technol.*, vol. 66, no. 6, pp. 4592–4601, Jun. 2017.
- [328] M. Yan, G. Feng, J. Zhou, and S. Qin, "Smart multi-RAT access based on multiagent reinforcement learning," *IEEE Trans. Veh. Technol.*, vol. 67, no. 5, pp. 4539–4551, May 2018.
- [329] Z. Ning, P. Dong, X. Wang, J. J. P. C. Rodrigues, and F. Xia, "Deep reinforcement learning for vehicular edge computing: An intelligent offloading system," *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 6, pp. 1–24, Oct. 2019.
- [330] Z. Ning et al., "When deep reinforcement learning meets 5G-enabled vehicular networks: A distributed offloading framework for traffic big data," *IEEE Trans. Ind. Informat.*, vol. 16, no. 2, pp. 1352–1361, Feb. 2020.
- [331] M. Khayyat, I. A. Elgendy, A. Muthanna, A. S. Alshahrani, S. Alharbi, and A. Koucheryavy, "Advanced deep learning-based computational offloading for multilevel vehicular edge-cloud computing networks," *IEEE Access*, vol. 8, pp. 137052–137062, 2020.
- [332] Z. Ning et al., "Joint computing and caching in 5G-envisioned Internet of Vehicles: A deep reinforcement learning-based traffic control system," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 8, pp. 5201–5212, Aug. 2021.
- [333] H. Ke, J. Wang, L. Deng, Y. Ge, and H. Wang, "Deep reinforcement learning-based adaptive computation offloading for MEC in heterogeneous vehicular networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 7, pp. 7916–7929, Jul. 2020.
- [334] M. Li, J. Gao, L. Zhao, and X. Shen, "Deep reinforcement learning for collaborative edge computing in vehicular networks," *IEEE Trans. Cogn. Commun. Netw.*, vol. 6, no. 4, pp. 1122–1135, Dec. 2020.
- [335] H. Peng and X. Shen, "Deep reinforcement learning based resource management for multi-access edge computing in vehicular networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 7, no. 4, pp. 2416–2428, Oct.–Dec. 2020.
- [336] Z. Gao, L. Yang, and Y. Dai, "Fast adaptive task offloading and resource allocation via multi-agent reinforcement learning in heterogeneous vehicular fog computing," *IEEE Internet Things J.*, vol. 10, no. 8, pp. 6818–6835, Apr. 2023.
- [337] S. S. Hassan, Y. M. Park, Y. K. Tun, W. Saad, Z. Han, and C. S. Hong, "Satellite-based ITS data offloading & computation in 6G networks: A cooperative multi-agent proximal policy optimization DRL with attention approach," *IEEE Trans. Mobile Comput.*, early access, Aug. 1, 2023, doi: [10.1109/TMC.2023.3300314](https://doi.org/10.1109/TMC.2023.3300314).
- [338] Z. Wei, B. Li, R. Zhang, X. Cheng, and L. Yang, "Dynamic many-to-many task offloading in vehicular fog computing: A multi-agent DRL approach," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Dec. 2022, pp. 6301–6306.
- [339] S. Xia, Z. Yao, G. Wu, and Y. Li, "Distributed offloading for cooperative intelligent transportation under heterogeneous networks," *IEEE Trans. Intell. Transport. Syst.*, vol. 23, no. 9, pp. 16701–16714, Sep. 2022.
- [340] L. T. Tan, R. Q. Hu, and L. Hanzo, "Twin-timescale artificial intelligence aided mobility-aware edge caching and computing in vehicular networks," *IEEE Trans. Veh. Technol.*, vol. 68, no. 4, pp. 3086–3099, Apr. 2019.
- [341] L. Zhao, Y. Ran, H. Wang, J. Wang, and J. Luo, "Towards cooperative caching for vehicular networks with multi-level federated reinforcement learning," in *Proc. IEEE Int. Conf. Commun. (ICC)*, Montreal, QC, Canada, Jun. 2021, pp. 1–6.
- [342] S. Malektaji, A. Ebrahimzadeh, H. Elbiaze, R. H. Glitho, and S. Kianpisheh, "Deep reinforcement learning-based content migration for edge content delivery networks with vehicular nodes," *IEEE Trans. Netw. Services Manag.*, vol. 18, no. 3, pp. 3415–3431, Sep. 2021.

- [343] Y. Dai et al., "Deep reinforcement learning and permissioned blockchain for content caching in vehicular edge computing and networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 4, pp. 4312–4324, Apr. 2020.
- [344] G. Qiao, S. Leng, S. Maharjan, Y. Zhang, and N. Ansari, "Deep reinforcement learning for cooperative content caching in vehicular edge computing and networks," *IEEE Internet Things J.*, vol. 7, no. 1, pp. 247–257, Jan. 2020.
- [345] Z. Qin, S. Leng, J. Zhou, and S. Mao, "Collaborative edge computing and caching in vehicular networks," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Seoul, South Korea, May 2020, pp. 1–6.
- [346] Y. Wu, Z. Zhang, P. Zheng, Y. Hu, and A. Schmeink, "V2E association and resource allocation via deep reinforcement learning in MEC-based HetVNs," in *Proc. IEEE 95th Veh. Technol. Conf. (VTC-Spring)*, Helsinki, Finland, Jun. 2022, pp. 2390–2397.
- [347] K. Tan, D. Bremner, J. L. Kerne, Y. Sambo, L. Zhang, and M. A. Imran, "Intelligent handover algorithm for vehicle-to-network communications with double-deep Q-learning," *IEEE Trans. Veh. Technol.*, vol. 71, no. 7, pp. 7848–7862, Jul. 2022.
- [348] Y. Lin, Z. Zhang, Y. Huang, J. Li, F. Shu, and L. Hanzo, "Heterogeneous user-centric cluster migration improves the connectivity-handover trade-off in vehicular networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 12, pp. 16027–16043, Dec. 2020.
- [349] H. Khan, A. Elgabli, S. Samarakoon, M. Bennis, and C. S. Hong, "Reinforcement learning-based vehicle-cell association algorithm for highly mobile millimeter wave communication," *IEEE Trans. Cogn. Commun. Netw.*, vol. 5, no. 4, pp. 1073–1085, Dec. 2019.
- [350] J. Wang, J. Hu, G. Min, Q. Ni, and T. El-Ghazawi, "Online service migration in mobile edge with incomplete system information: A deep recurrent actor-critic learning approach," *IEEE Trans. Mobile Comput.*, vol. 22, no. 11, pp. 6663–6675, Nov. 2023.
- [351] L. Zhou, "A vehicle service migration strategy algorithm in 5G NR-V2X," *Front. Comput. Intell. Syst.*, vol. 2, no. 3, pp. 48–55, 2022.
- [352] J. Xu, B. Yang, Y. Liu, C. Chen, and X. Guan, "Joint task offloading and resource allocation for multihop Industrial Internet of Things," *IEEE Internet Things J.*, vol. 9, no. 21, pp. 22022–22033, Nov. 2022.
- [353] L. Zhang et al., "Digital twin-assisted edge computation offloading in Industrial Internet of Things with NOMA," *IEEE Trans. Veh. Technol.*, vol. 72, no. 9, pp. 11935–11950, Sep. 2023.
- [354] Z. Hong, W. Chen, H. Huang, S. Guo, and Z. Zheng, "Multi-hop cooperative computation offloading for Industrial IoT-edge-cloud computing environments," *IEEE Trans. Parallel Distrib. Syst.*, vol. 30, no. 12, pp. 2759–2774, Dec. 2019.
- [355] S. Boyd and L. Vandenberghe, *Convex Optimization*. England, U.K: Cambridge Univ. Press, 2009.
- [356] F. Drew and J. Tirole, *Game Theory*. Cambridge, MA, USA: MIT Press, 1991.
- [357] Y. Ren, Y. Sun, and M. Peng, "Deep reinforcement learning based computation offloading in fog enabled Industrial Internet of Things," *IEEE Trans. Ind. Informat.*, vol. 17, no. 7, pp. 4978–4987, Jul. 2021.
- [358] N. Zhu, X. Xu, S. Han, and S. Lv, "Sleep-scheduling and joint computation-communication resource allocation in MEC networks for 5G IIoT," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Nanjing, China, Mar. 2021, pp. 1–7.
- [359] J. Wang et al., "Deep reinforcement learning based task offloading in SDN-enabled industrial Internet of Things," in *Proc. Int. Conf. Artif. Intell. Commun. Netw. (ICAICN)*, Jul. 2019, pp. 425–437.
- [360] Y. Chen, Z. Liu, Y. Zhang, Y. Wu, X. Chen, and L. Zhao, "Deep reinforcement learning-based dynamic resource management for mobile edge computing in Industrial Internet of Things," *IEEE Trans. Ind. Informat.*, vol. 17, no. 7, pp. 4925–4934, Jul. 2021.
- [361] P. Yu et al., "Intelligent-driven green resource allocation for Industrial Internet of Things in 5G heterogeneous networks," *IEEE Trans. Ind. Informat.*, vol. 18, no. 1, pp. 520–530, Jan. 2022.
- [362] L. Lin, W. Zhou, Z. Yang, and J. Liu, "Deep reinforcement learning-based task scheduling and resource allocation for NOMA-MEC in Industrial Internet of Things," *Peer-to-Peer Netw. Appl.*, vol. 16, pp. 170–188, Sep. 2022.
- [363] Y. Lyu, Z. Liu, R. Fan, C. Zhan, H. Hu, and J. An, "Optimal computation offloading in collaborative LEO-IoT enabled MEC: A multi-agent deep reinforcement learning approach," *IEEE Trans. Green Commun. Netw.*, vol. 7, no. 2, pp. 996–1011, Jun. 2023.
- [364] Y. Xiao, Y. Song, and J. Liu, "Multi-agent deep reinforcement learning based resource allocation for ultra-reliable low-latency Internet of controllable Things," *IEEE Trans. Wireless Commun.*, vol. 22, no. 8, pp. 5414–5430, Aug. 2023.
- [365] Y. Wang, F. Shang, J. Lei, X. Zhu, H. Qin, and J. Wen, "Dual-attention assisted deep reinforcement learning algorithm for energy-efficient resource allocation in Industrial Internet of Things," *Future Gener. Comput. Syst.*, vol. 142, pp. 150–164, May 2023.
- [366] Q. Tang et al., "Distributed task scheduling in serverless edge computing networks for the Internet of Things: A learning approach," *IEEE Internet Things J.*, vol. 9, no. 20, pp. 19634–19648, Oct. 2022.
- [367] S. Xu, Y. Li, S. Guo, C. Lei, D. Liu, and X. Qiu, "CloudCEdge collaborative SFC mapping for Industrial IoT using deep reinforcement learning," *IEEE Trans. Ind. Informat.*, vol. 18, no. 6, pp. 4158–4168, Jun. 2022.
- [368] A. Hazra, M. Adhikari, T. Amgoth, and S. N. Srirama, "Intelligent service deployment policy for next-generation industrial edge networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 5, pp. 3057–3066, Sep./Oct. 2022.
- [369] H. Chen, S. Wang, G. Li, L. Nie, X. Wang, and Z. Ning, "Distributed orchestration of service function chains for edge intelligence in the Industrial Internet of Things," *IEEE Trans. Ind. Informat.*, vol. 18, no. 9, pp. 6244–6254, Sep. 2022.
- [370] J. Li, R. Wang, and K. Wang, "Service function chaining in Industrial Internet of Things with edge intelligence: A natural actor-critic approach," *IEEE Trans. Ind. Informat.*, vol. 19, no. 1, pp. 491–502, Jan. 2023.
- [371] S. Wang, C. Yuen, W. Ni, Y. L. Guan, and T. Lv, "Multi-agent deep reinforcement learning for cost-and delay-sensitive virtual network function placement and routing," *IEEE Trans. Commun.*, vol. 70, no. 8, pp. 5208–5224, Aug. 2022.
- [372] D. Ren, X. Gui, and K. Zhang, "Adaptive request scheduling and service caching for MEC-assisted IoT networks: An online learning approach," *IEEE Internet Things J.*, vol. 9, no. 18, pp. 17372–17386, Sep. 2022.
- [373] Y. Lee, A. Masood, W. Noh, and S. Cho, "DQN based user association control in hierarchical mobile edge computing systems for mobile IoT services," *Future Gener. Comput. Syst.*, vol. 137, pp. 53–69, Dec. 2022.
- [374] T. M. Ho and K.-K. Nguyen, "Joint server selection, cooperative offloading and handover in multi-access edge computing wireless network: A deep reinforcement learning approach," *IEEE Trans. Mobile Comput.*, vol. 21, no. 7, pp. 2421–2435, Jul. 2022.
- [375] C. F. Kwong, C. Shi, Q. Liu, S. Yang, D. Chieng, and P. Kar, "Autonomous handover parameter optimisation for 5G cellular networks using deep deterministic policy gradient," in *Proc. SSRN* vol. 9, Mar. 2023, pp. 1–28, doi: [10.2139/ssrn.4374897](https://doi.org/10.2139/ssrn.4374897).
- [376] M. Chiputa et al., "Enhancing handover for 5G mmWave mobile networks using jump Markov linear system and deep reinforcement learning," *Sensors*, vol. 22, no. 3, p. 746, Jan. 2022.
- [377] Z. Cheng, X. Ji, W. You, Y. Zhao, and Z. Guo, "Secho: A deep reinforcement learning-based scheme for secure handover in mobile edge computing," *Computer Netw.*, vol. 229, Jun. 2023, Art. no. 109769.
- [378] S. S. Oyewobi, G. P. Hancke, A. M. Abu-Mahfouz, and A. J. Onumanyi, "An effective spectrum handoff based on reinforcement learning for target channel selection in the Industrial Internet of Things," *Sensors*, vol. 19, no. 6, p. 1395, Mar. 2019.
- [379] Y. Lin, J. Bao, Y. Zhang, J. Li, F. Shu, and L. Hanzo, "Privacy-preserving joint edge association and power optimization for the Internet of Vehicles via federated multi-agent reinforcement learning," *IEEE Trans. Veh. Technol.*, vol. 72, no. 6, pp. 8256–8261, Jun. 2023.
- [380] A. A. Abdellatif, N. Mhaisen, A. Mohamed, A. Erbad, and M. Guizani, "Reinforcement learning for intelligent healthcare systems: A review of challenges, applications, and open research issues," *IEEE Internet Things J.*, vol. 10, no. 24, pp. 21982–22007, Dec. 2023, doi: [10.1109/JIOT.2023.3288050](https://doi.org/10.1109/JIOT.2023.3288050).
- [381] S. Kharche and J. Kharche, "6G intelligent healthcare framework: A review on role of technologies, challenges and future directions," *J. Mobile Multimedia*, vol. 19, no. 3, pp. 603–644, Feb. 2023.
- [382] S. Beborra, S. S. Tripathy, S. Basheer, and C. L. Chowdhary, "DeepMist: Toward deep learning assisted mist computing framework for managing healthcare big data," *IEEE Access*, vol. 11, pp. 42485–42496, 2023.
- [383] P. Tiwari, A. Lakhan, R. H. Jhaveri, and T.-M. Gronli, "Consumer-centric Internet of Medical Things for cyborg applications based on federated reinforcement learning," *IEEE Trans. Consumer Electron.*, early access, Feb. 7, 2023, doi: [10.1109/TCE.2023.3242375](https://doi.org/10.1109/TCE.2023.3242375).
- [384] X. Yuan et al., "A DQN-based frame aggregation and task offloading approach for edge-enabled IoMT," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 3, pp. 1339–1351, May/Jun. 2023.

- [385] B. Wu, Z. Cai, W. Wu, and X. Yin, "AoI-aware resource management for smart health via deep reinforcement learning," *IEEE Access*, vol. 11, pp. 81180–81195, 2023.
- [386] P. Lin, Q. Song, F. R. Yu, D. Wang, and L. Guo, "Task offloading for wireless VR-enabled medical treatment with blockchain security using collective reinforcement learning," *IEEE Internet Things J.*, vol. 8, no. 21, pp. 15749–15761, Nov. 2021.
- [387] Y. Wang, H. Wu, R. H. Jhaveri, and Y. Djenouri, "DRL-based URLLC-constraint and energy-efficient task offloading for Internet of Health Things," *IEEE J. Biomed. Health Inform.*, early access, Jul. 24, 2023, doi: [10.1109/JBHI.2023.3297525](https://doi.org/10.1109/JBHI.2023.3297525).
- [388] J. Jiang, J. Guo, M. Khan, Y. Cui, and W. Lin, "Energy-saving service offloading for the Internet of Medical Things using deep reinforcement learning," *ACM Trans. Sensor Netw.*, vol. 19, no. 3, pp. 1–20, Mar. 2023.
- [389] Q. He et al., "A blockchain-based scheme for secure data offloading in healthcare with deep reinforcement learning," *IEEE/ACM Trans. Netw.*, early access, Jun. 5, 2023, doi: [10.1109/TNET.2023.3274631](https://doi.org/10.1109/TNET.2023.3274631).
- [390] A. M. Seid, A. Erbad, H. N. Abishu, A. Albaser, M. Abdallah, and M. Guizani, "Multi-agent federated reinforcement learning for resource allocation in UAV-enabled Internet of Medical Things networks," *IEEE Internet Things J.*, vol. 10, no. 22, pp. 19695–19711, Nov. 2023.
- [391] J. Liu, L. Shu, X. Lu, and Y. Liu, "Survey of intelligent agricultural IoT based on 5G," *Electronics*, vol. 12, no. 10, p. 2336, May 2023.
- [392] S. Santosh and R. Raghavendra, "IoT-enabled technologies for sustainable smart agriculture and their comprehensive survey," in *Proc. Int. Conf. Artif. Intell. Smart Commun. (AISC)*, Apr. 2023, pp. 467–472.
- [393] G. G. Devarajan, S. M. Nagarajan, T. V. Ramana, T. Vignesh, U. Ghosh, and W. Alnumay, "DDNSAS: Deep reinforcement learning based deep Q-learning network for smart agriculture system," *Sustain. Comput. Informat. Syst.*, vol. 39, Sep. 2023, Art. no. 100890.
- [394] F. Khoramnejad, *AI-Enabled System Optimization With Carrier Aggregation and Task Offloading in 5G and 6G*, Univ. Ottawa, Ottawa, ON, Canada, 2023.
- [395] Y. Li et al., "Research on winter wheat growth stages recognition based on mobile edge computing," *Agriculture*, vol. 13, no. 3, p. 534, Feb. 2023.
- [396] F. Bu and X. Wang, "A smart agriculture IoT system based on deep reinforcement learning," *Future Gener. Comput. Syst.*, vol. 99, pp. 500–507, Oct. 2019.
- [397] C. Dwork, K. Kenthapadi, F. McSherry, I. Mironov, and M. Naor, "Our data, ourselves: Privacy via distributed noise generation," in *Proc. 24th Annu. Int. Conf. Theory Apps. Cryptol. Techn.*, St. Petersburg, Russia, May 2006, pp. 486–503.
- [398] M. Yang et al., "Local differential privacy and its applications: A comprehensive survey," 2020, *arXiv:2008.03686*.
- [399] X. Pan et al., "How you act tells a lot: Privacy-leakage attack on deep reinforcement learning," 2019, *arXiv:1904.11082*.
- [400] H. Ono and T. Takahashi, "Locally private distributed reinforcement learning," 2020, *arXiv:2001.11718*.
- [401] P. Gajane, U. Tanguy, and K. Emilie, "Corrupt bandits for preserving local privacy," in *Proc. Algorithmic Learn. Theory*, vol. 83, 2018, pp. 387–412.
- [402] D. Basu, C. Dimitrakakis, and A. Tossou, "Differential privacy for multi-armed bandits: What is it and what is its cost?" 2019, *arXiv:1905.12298*.
- [403] J. Li et al., "A privacy-preserving online deep learning algorithm based on differential privacy," in *Proc. 26th Int. Conf. Comput. Supported Cooper. Work (CSCWD)*, Rio de Janeiro, Brazil, Jun. 2023, pp. 559–564.
- [404] D. Wei, N. Xi, X. Ma, M. Shojafar, S. Kumari, and J. Ma, "Personalized privacy-aware task offloading for edge-cloud-assisted Industrial Internet of Things in automated manufacturing," *IEEE Trans. Ind. Informat.*, vol. 18, no. 11, pp. 7935–7945, Nov. 2022.
- [405] X. Pang, Z. Wang, J. Li, R. Zhou, J. Ren, and Z. Li, "Towards online privacy-preserving computation offloading in mobile edge computing," in *Proc. IEEE Conf. Comput. Commun. (INFOCOM)*, London, U.K., Jun. 2022, pp. 1179–1188.
- [406] S. Gan, M. Siew, C. Xu, and T. Q. Quek, "Differentially private deep Q-learning for pattern privacy preservation in MEC offloading," 2020, *arXiv:2302.04608*.
- [407] H. Gao, W. Huang, T. Liu, Y. Yin, and Y. Li, "PPO2: Location privacy-oriented task offloading to edge computing using reinforcement learning for intelligent autonomous transport systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 7, pp. 7599–7612, Jul. 2023.
- [408] W. Lan, K. Chen, Y. Li, J. Cao, and Y. Sahni, "Deep reinforcement learning for privacy-preserving task offloading in integrated satellite-terrestrial networks," 2023, *arXiv:2306.17183*.
- [409] D. Wei, J. Zhang, M. Shojafar, S. Kumari, N. Xi, and J. Ma, "Privacy-aware multiagent deep reinforcement learning for task offloading in VANET," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 11, pp. 13108–13122, Nov. 2022, doi: [10.1109/TITS.2022.3202196](https://doi.org/10.1109/TITS.2022.3202196).
- [410] Y. Zeng, Y. Huang, J. Liu, and Y. Yang, "Privacy-preserving distributed edge caching for mobile data offloading in 5G networks," in *Proc. IEEE 40th Int. Conf. Distrib. Comput. Syst. (ICDCS)*, Singapore, Feb. 2020, pp. 541–551.
- [411] X. Zhang et al., "Data-driven caching with users content preference privacy in information-centric networks," *IEEE Trans. Wireless Commun.*, vol. 20, no. 9, pp. 5744–5753, Sep. 2021.
- [412] Z. Zhang, T. Cao, X. Wang, H. Xiao, and J. Guan, "VC-PPQ: Privacy-preserving Q-learning based video caching optimization in mobile edge networks," *IEEE Trans. Netw. Sci. Eng.*, vol. 9, no. 6, pp. 4129–4144, Nov./Dec. 2022.
- [413] X. Sun et al., "Synthesizing realistic trajectory data with differential privacy," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 5, pp. 5502–5515, May 2023.
- [414] M. Tsao, K. Yang, S. Zoepf, and M. Pavone, "Trust but verify: Cryptographic data privacy for mobility management," *IEEE Trans. Control Netw. Syst.*, vol. 9, no. 1, pp. 50–61, Mar. 2022.
- [415] L. Wang, D. Yang, X. Han, D. Zhang, and X. Ma, "Mobile crowdsourcing task allocation with differential-and-distortion geo-obfuscation," *IEEE Trans. Depend. Secure Comput.*, vol. 18, no. 2, pp. 967–981, Mar./Apr. 2021.
- [416] F. Tian, S. Zhang, L. Lu, H. Liu, and X. Gui, "A novel personalized differential privacy mechanism for trajectory data publication," in *Proc. Int. Conf. Netw. Netw. Apps. (NaNA)*, Kathmandu, Nepal, Jan. 2017, pp. 61–68.
- [417] IEC. "Edge intelligence (white paper)." Accessed: 2017. [Online]. Available: <https://basecamp.iec.ch/download/iec-white-paper-edge-intelligence-en/>
- [418] "ITU-T Y.3000-series-artificial intelligence standardization roadmap." Accessed: 2022. [Online]. Available: <https://handle.itu.int/11.1002/1000/15249>
- [419] ITU-T Q.5001. "Signalling requirements and architecture of intelligent edge computing." Accessed: Oct. 2018. [Online]. Available: <https://handle.itu.int/11.1002/1000/13701>
- [420] ITU-T. "Unified architecture for machine learning in 5G and future networks." Accessed: Feb. 2019. [Online]. Available: <https://www.itu.int/en/ITU/focusgroups/ml5g/Documents/ML5G-deliverables.pdf>
- [421] ITU-T Q. 5028. "Data management interfaces for intelligent edge computing-based smart agriculture service." Accessed: Dec. 2023. [Online]. Available: <https://www.itu.int/rec/T-REC-Q.5028/en>
- [422] ITU-T Q.IEC-PRO. "Signalling architecture for microservices based intelligent edge computing." Accessed: Oct. 2023. [Online]. Available: <https://www.itu.int/md/T22-SG11-231010-TD-GEN-0738/en>
- [423] 3GPP. "Study on 5G system support for AI/ML-based services; stage 2 (release 18)." Oct. 2022. [Online]. Available: <https://www.3gpp.org/news-events/3gpp-news/rel18-sa2>
- [424] *IEEE Standard for Adoption of OpenFog Reference Architecture for Fog Computing*, OpenFog Consortium, Scottsdale, AZ, USA, 2018, pp. 1–176.
- [425] *IEEE Standard for Enabling Mobile Device Platforms to Be Used as High-Performance Computing Clients*, IEEE Standard 802.1, 2016, pp. 1–27.
- [426] *Standard for the Adoption of IEEE 1918.1, Recommended Practice for the Definition of the Internet of Things (IoT)*, IEEE Standard 1918.1, 2019.
- [427] *Standard for Local and Metropolitan Area Networks—Time-Sensitive Networking for Fronthaul*, IEEE Standard P802.1CM, 2015.
- [428] "Requirements for the deployment of edge computing." 2020. [Online]. Available: <https://handle.itu.int/11.1002/1000/14133>
- [429] "Architecture of edge computing." 2020. [Online]. Available: <https://handle.itu.int/11.1002/1000/14134>
- [430] J. Chen, H. Xing, Z. Xiao, L. Xu, and T. Tao, "A DRL agent for jointly optimizing computation offloading and resource allocation in MEC," *IEEE Internet Things J.*, vol. 8, no. 24, pp. 17508–17524, Dec. 2021.
- [431] X. Wang, Y. Han, C. Wang, Q. Zhao, X. Chen, and M. Chen, "In-edge AI: Intelligentizing mobile edge computing, caching and communication by federated learning," *IEEE Netw.*, vol. 33, no. 5, pp. 156–165, Sep./Oct. 2019.

- [432] C. Pan et al., "Reconfigurable intelligent surfaces for 6G systems: Principles, applications, and research directions," *IEEE Commun. Mag.*, vol. 59, no. 6, pp. 14–20, Jun. 2021.
- [433] H. Ye, L. Liang, G. Y. Li, J. Kim, L. Lu, and M. Wu, "Machine learning for vehicular networks: Recent advances and application examples," *IEEE Veh. Technol. Mag.*, vol. 13, no. 2, pp. 94–101, Jun. 2018.
- [434] X. Guo et al., "Learning mean-field games," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, p. 19.
- [435] B. He et al., "Deephop on edge: Hop-by-hop routing by distributed learning with semantic attention," in *Proc. 49th Int. Conf. Parallel Process.*, New York, NY, USA, Aug. 2020, pp. 1–11.
- [436] H. Mao, Z. Gong, and Z. Xiao, "Reward design in cooperative multi-agent reinforcement learning for packet routing," 2020, *arXiv:2003.03433*.
- [437] Z. He, L. Wang, H. Ye, G. Y. Li, and B.-H. F. Juang, "Resource allocation based on graph neural networks in vehicular communications," in *Proc. IEEE Global Commun. Conf. (GLOBECOM)*, Taipei, Taiwan, Dec. 2020, pp. 1–5.
- [438] S. Kaviani et al., "Robust and scalable routing with multi-agent deep reinforcement learning for MANETs," 2021, *arXiv:2101.03273*.
- [439] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd, "Quantum machine learning," *Nature*, vol. 549, no. 7671, pp. 195–202, 2017.
- [440] K. Bharti et al., "Noisy intermediate-scale quantum (NISQ) algorithms," 2021, *arXiv:2101.08448*.
- [441] D. Herman et al., "A survey of quantum computing for finance," 2022, *arXiv:2201.02773*.
- [442] S. S. Gill, "Quantum and blockchain based serverless edge computing: A vision, model, new trends and future directions," *Int. Technol. Lett.*, Feb. 2021, Art. no. e 275.
- [443] C. Zhang, P. Patras, and H. Haddadi, "Deep learning in mobile and wireless networking: A survey," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2224–2287, 3rd Quart., 2019.
- [444] M. Chen, U. Challita, W. Saad, C. Yin, and M. Debbah, "Artificial neural networks-based machine learning for wireless networks: A tutorial," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3039–3071, 4th Quart., 2019.
- [445] W. O'Quinn and S. Mao, "Quantum machine learning: Recent advances and outlook," *IEEE Wireless Commun.*, vol. 27, no. 3, pp. 126–131, Jun. 2020.
- [446] D. Shi, L. Li, T. Ohtsuki, M. Pan, Z. Han, and H. V. Poor, "Make smart decisions faster: Deciding D2D resource allocation via Stackelberg game guided multi-agent deep reinforcement learning," *IEEE Trans. Mobile Comput.*, vol. 21, no. 12, pp. 4426–4438, Dec. 2022.
- [447] R. Yang, F. R. Yu, P. Si, Z. Yang, and Y. Zhang, "Integrated blockchain and edge computing systems: A survey, some research issues and challenges," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 2, pp. 1508–1532, 2nd Quart., 2019.
- [448] L. Vasco and A. Luis, "An overview of blockchain integration with robotics and artificial intelligence," 2018, *arXiv:1810.00329*.
- [449] H. Wang, J. Ning, X. Huang, G. Wei, G. S. Poh, and X. Liu, "Secure fine-grained encrypted keyword search for E-healthcare cloud," *IEEE Trans. Depend. Secure Comput.*, vol. 18, no. 3, pp. 1307–1319, May/Jun. 2021.



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